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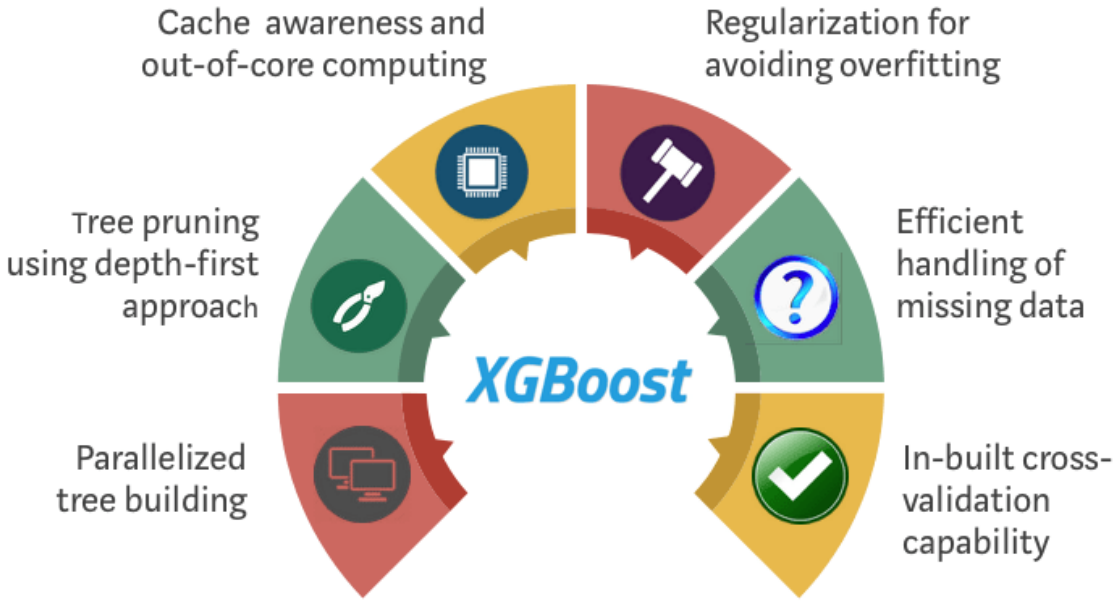
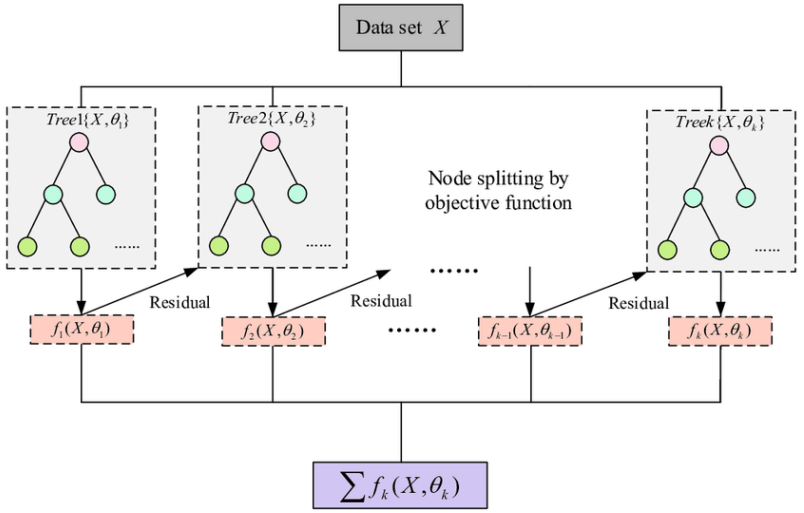
XGBoost

(Basic, Advanced Concepts and Its Applications)



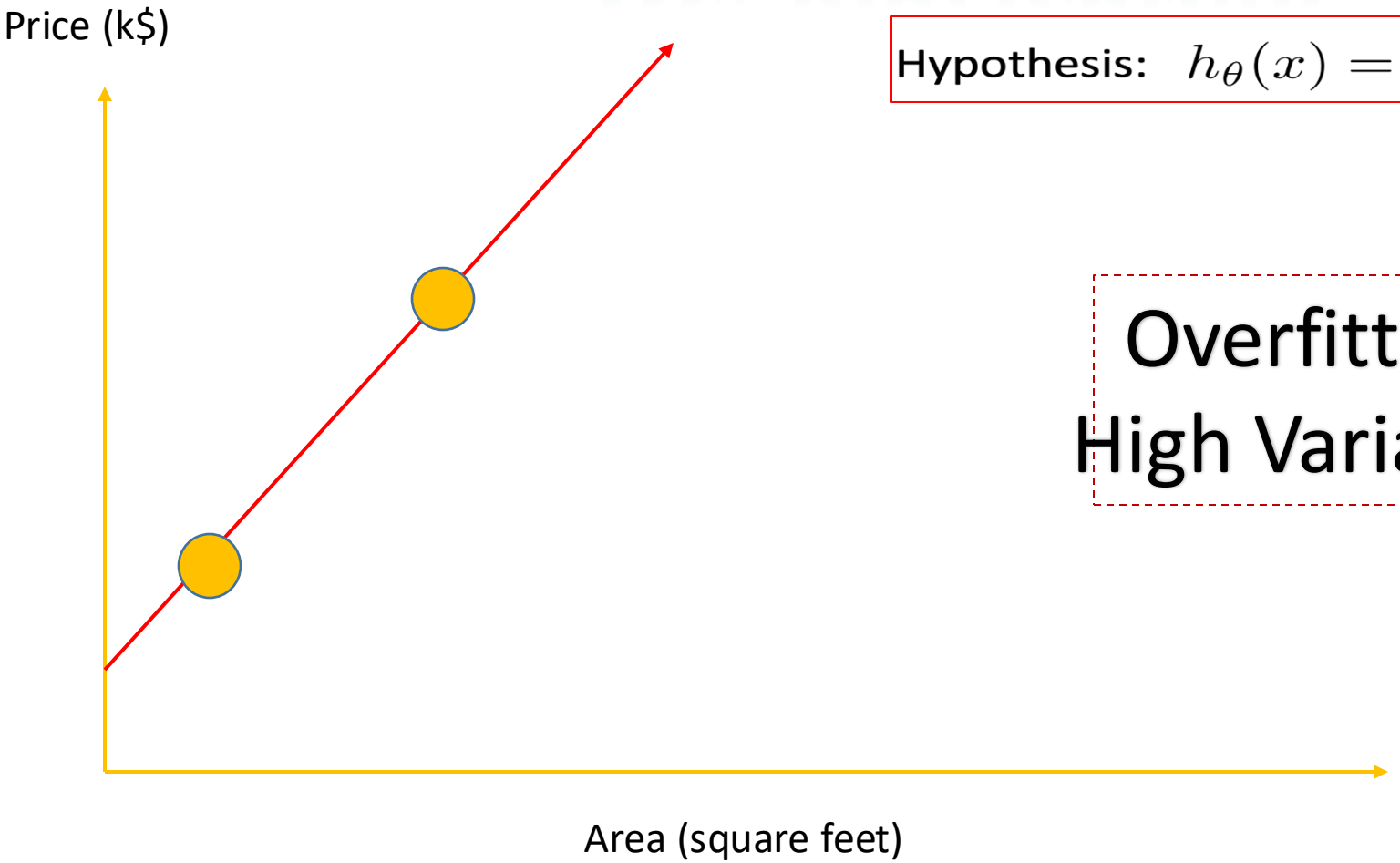


Vinh Dinh Nguyen
PhD in Computer Science

Outline

- Regularization
- Regression XGBoost
- Classification XGBoost
- XGBoost: Clearly Explain
- Time Series Example
- Summary

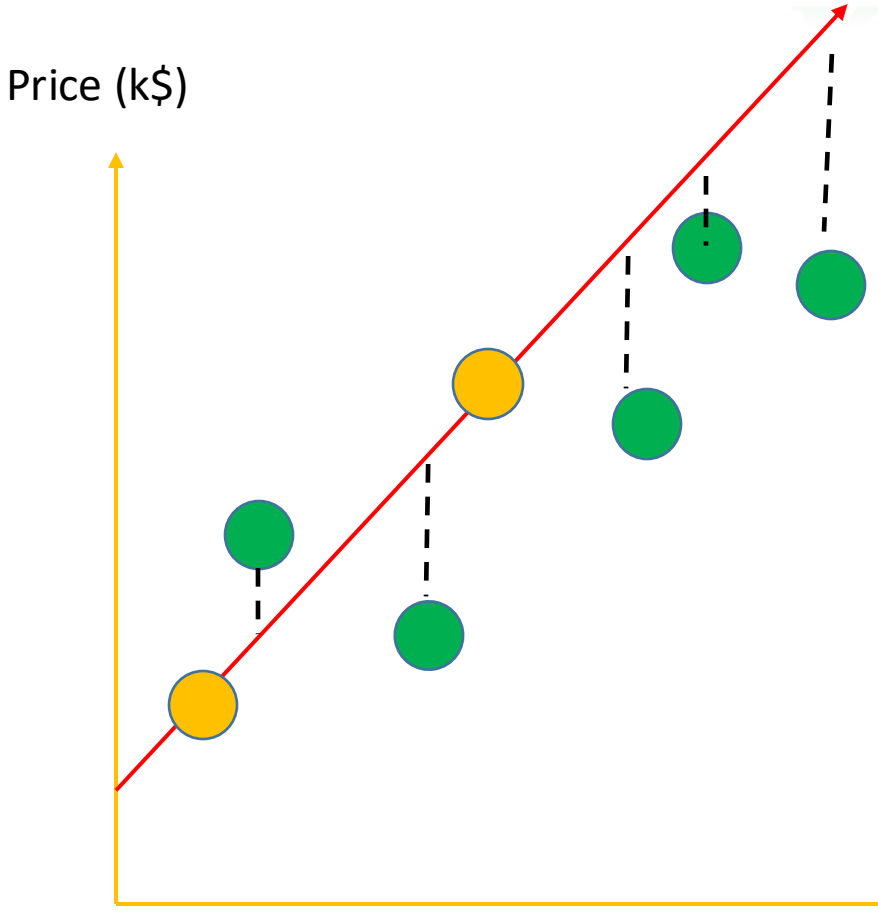
Regularization



Hypothesis: $h_\theta(x) = \theta_0 + \theta_1 x$

Overfitting
High Variance

Regularization



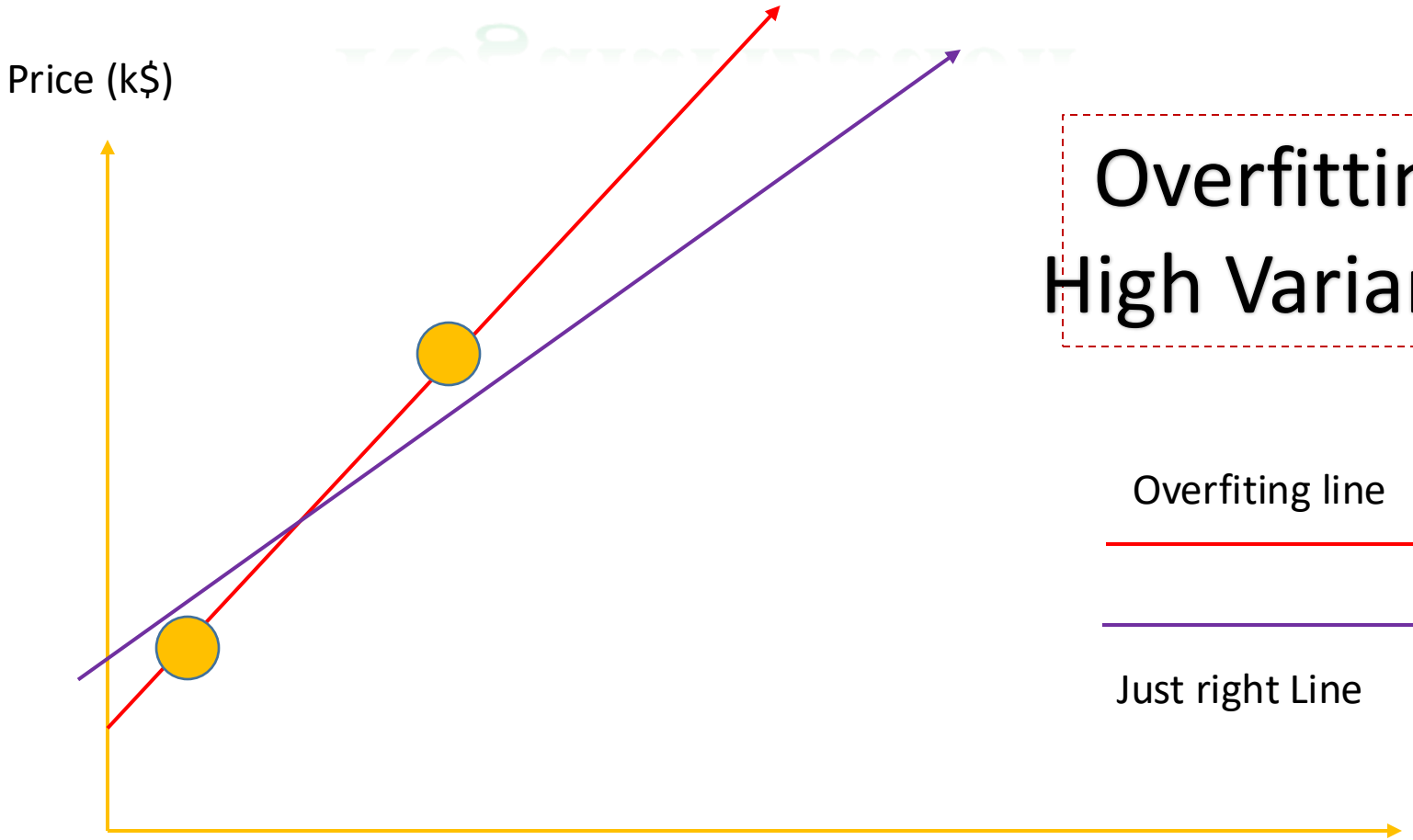
Overfitting
High Variance

Hypothesis: $h_\theta(x) = \theta_0 + \theta_1 x$

Train data

Test data

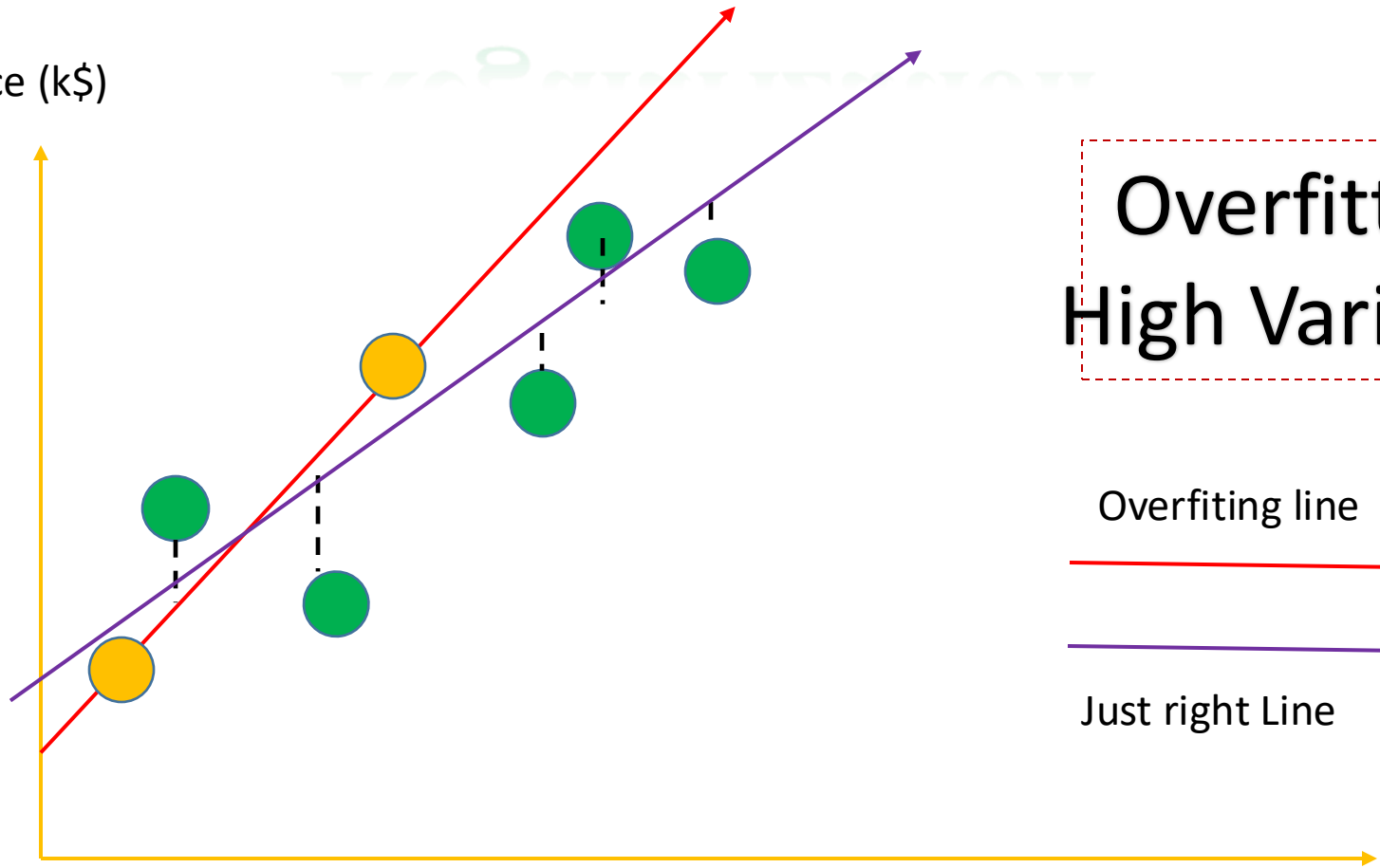
Regularization



Overfitting
High Variance

Hypothesis: $h_\theta(x) = \theta_0 + \theta_1 x$

Regularization



Overfitting
High Variance

Test data

Train data

Hypothesis: $h_\theta(x) = \theta_0 + \theta_1 x$

Price = Intercept + slope * area

Regularization

Hypothesis: $h_\theta(x) = \theta_0 + \theta_1 x$

Parameters: θ_0, θ_1

Cost Function: $J(\theta_0, \theta_1) = \frac{1}{2m} \sum_{i=1}^{m} \left(h_\theta(x^{(i)}) - y^{(i)}\right)^2$

Goal: $\underset{\theta_0, \theta_1}{\text{minimize}}\ J(\theta_0, \theta_1)$

Regularization.

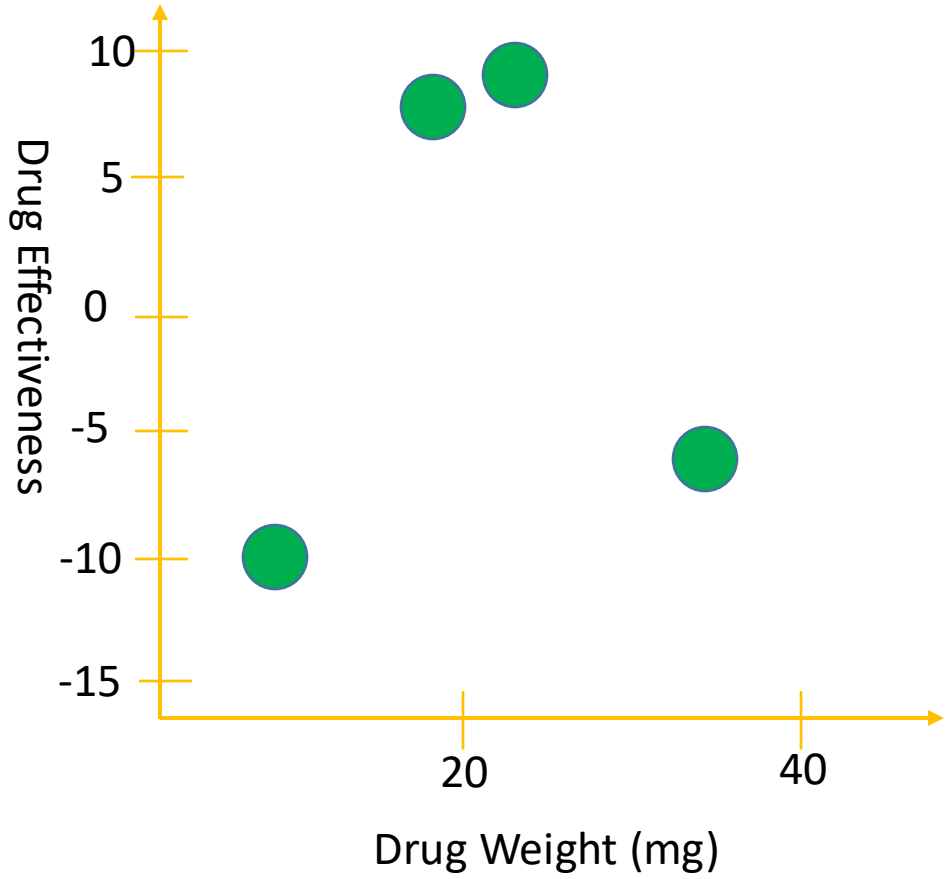
$$J(\theta) = \frac{1}{2m} \left[\sum_{i=1}^{m} (h_\theta(x^{(i)}) - y^{(i)})^2 + \lambda \sum_{j=1}^{n} \theta_j^2 \right]$$

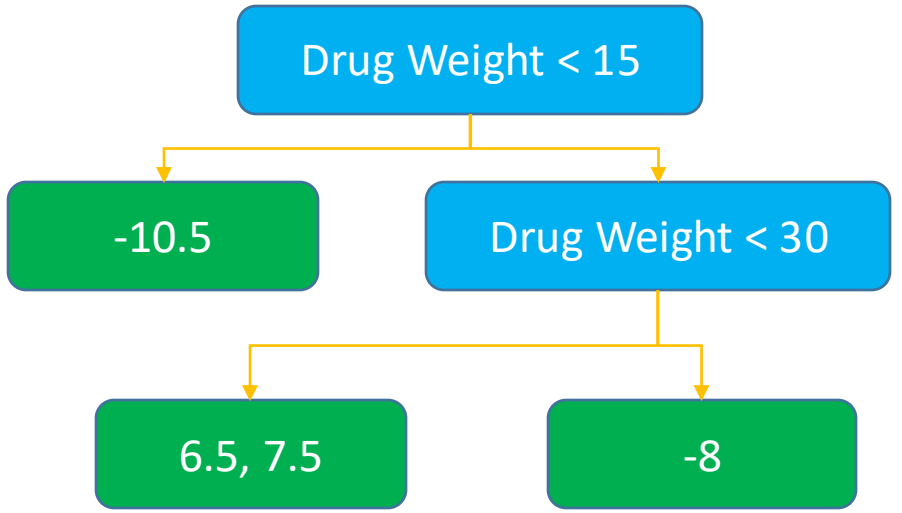
Price = [Intercept + slope * area] + λ*slope^2

Outline

- Regularization
- Regression XGBoost
- Classification XGBoost
- XGBoost: Clearly Explain
- Time Series Example
- Summary

XGBoost For Regression





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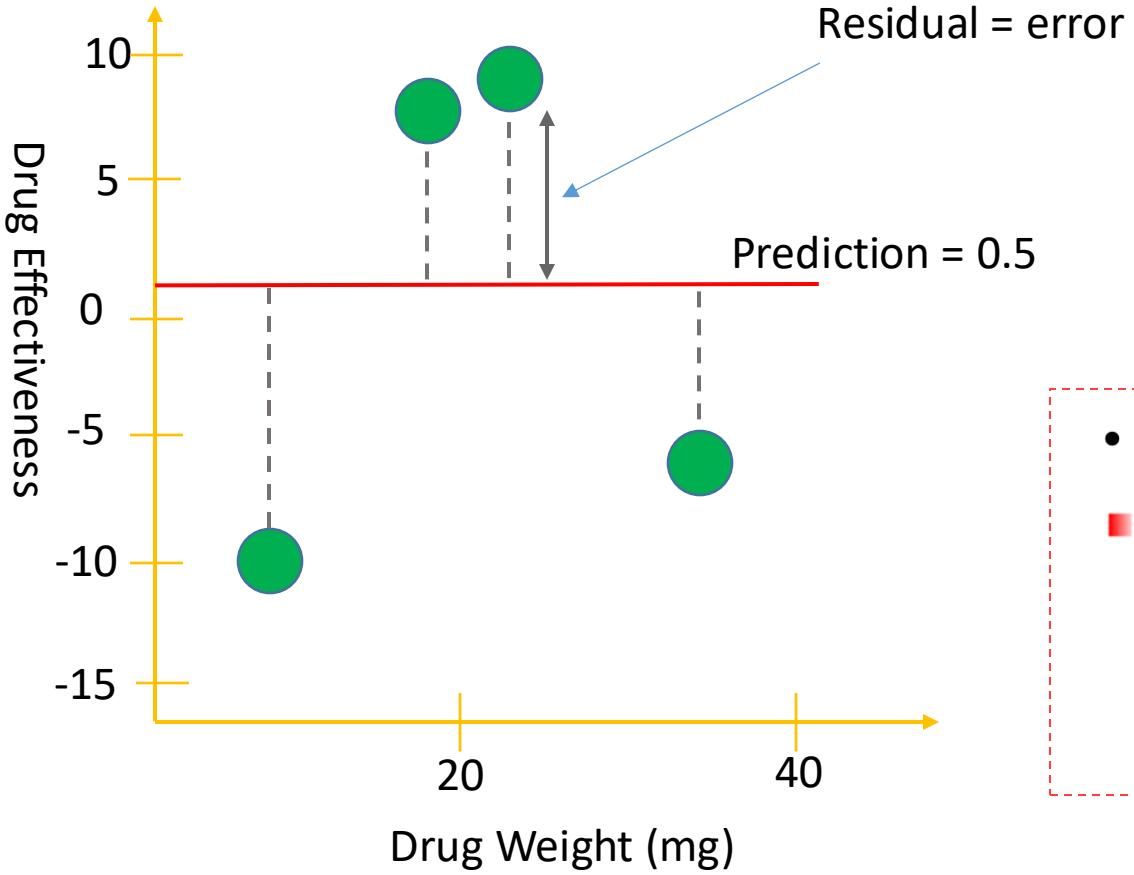
XGBoost For Regression

Step 1

- Initialize the first prediction for drug effectiveness
- Any number, for default, we set 1st prediction = 0.5



$$r_{im} = -\left[\frac{\partial L(y_i, F(x_i))}{\partial F(x_i)}\right]_{F(x)=F_{m-1}(x)}$$

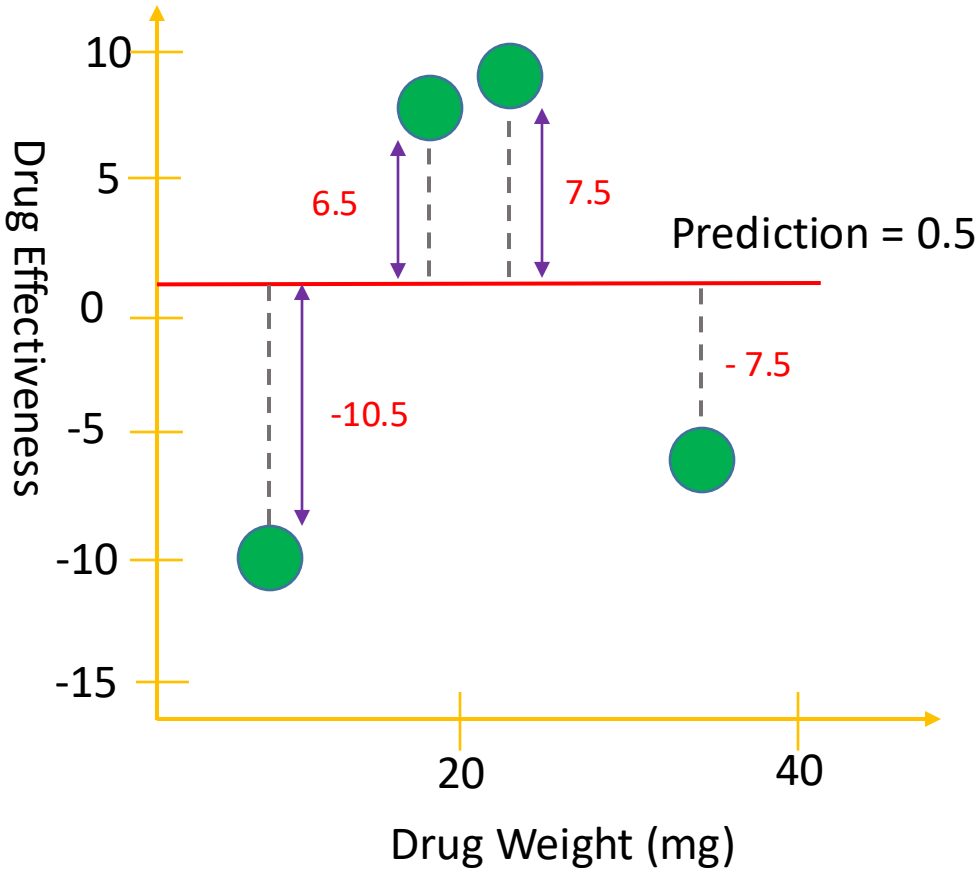
- $\{(x_i, y_i)\}^n_1$
- Loss function = $L(y_i, F(x)) = 1/2 * (\text{Output} - \text{Predicted})^2$

$$\frac{dL}{dPredicted} = 2/2\ (\text{Output} - \text{Predicted}) * -1 = -(\text{Output} - \text{Predicted})$$

Tricky implementation here

XGBoost For Regression

Step 1



Start with single Leaf of residuals

-10.5, 6.5, 7.5, -7.5

Compute Similarity Score

$$SC = \frac{[\sum (output - predicted)]^2}{m + \lambda}$$

m: number of samples

λ : *regularization parameters*

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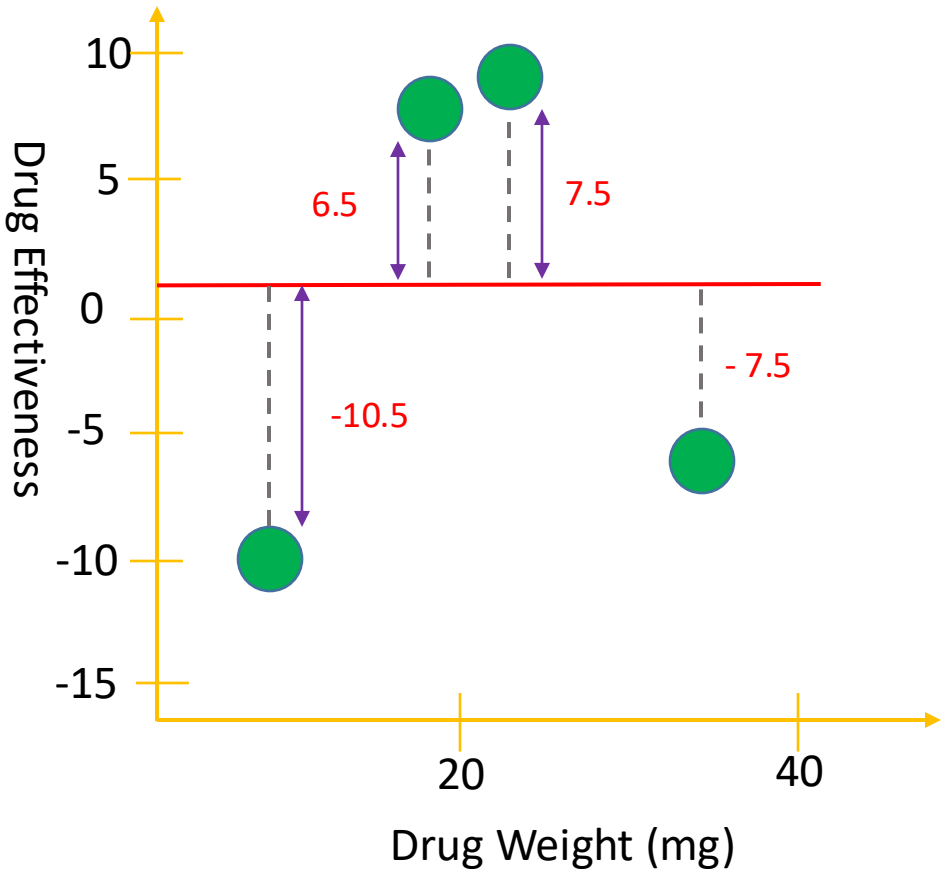
XGBoost For Regression

Step 1



Start with single Leaf of residuals

-10.5, 6.5, 7.5, -7.5

m = 4

$\lambda = 0$

Compute Similarity Score

$$SC = \frac{[\sum (output - predicted)]^2}{m + \lambda}$$

$$\textbf{SC} = \frac{[-10.5+7.5+6.5+(-7.5)]^2}{4} = \textbf{4}$$

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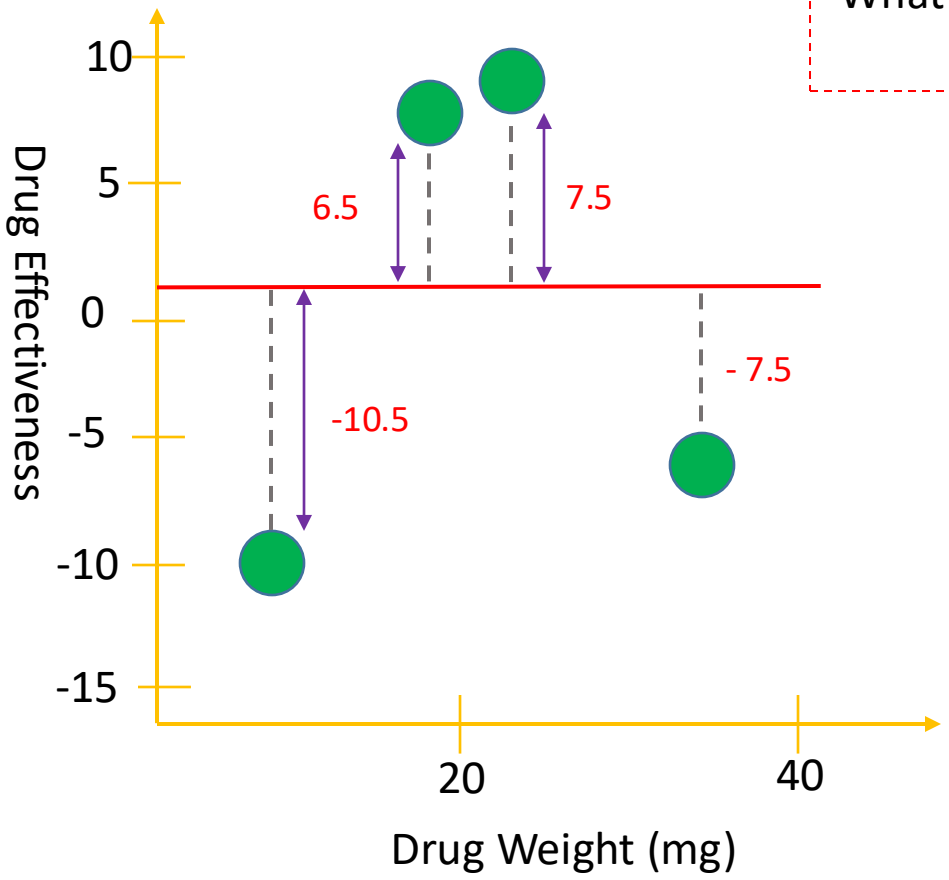
XGBoost For Regression

Step 1

SC = 4

-10.5, 6.5, 7.5, -7.5

What happens if we try to split residuals into two groups => measure the similarity score



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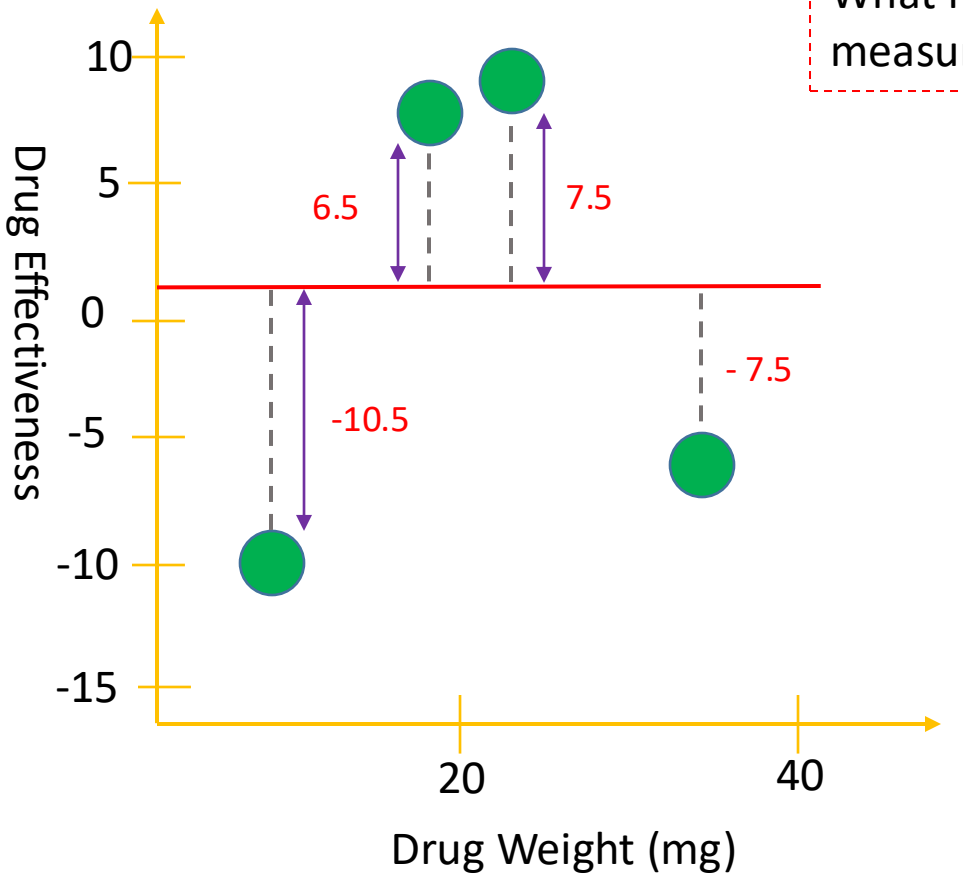
XGBoost For Regression

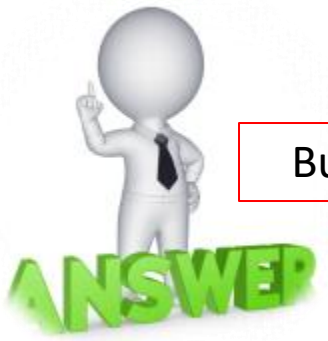
Step 1

SC = 4

-10.5, 6.5, 7.5, -7.5

What happens if we try to split residuals into two groups => measure the similarity score

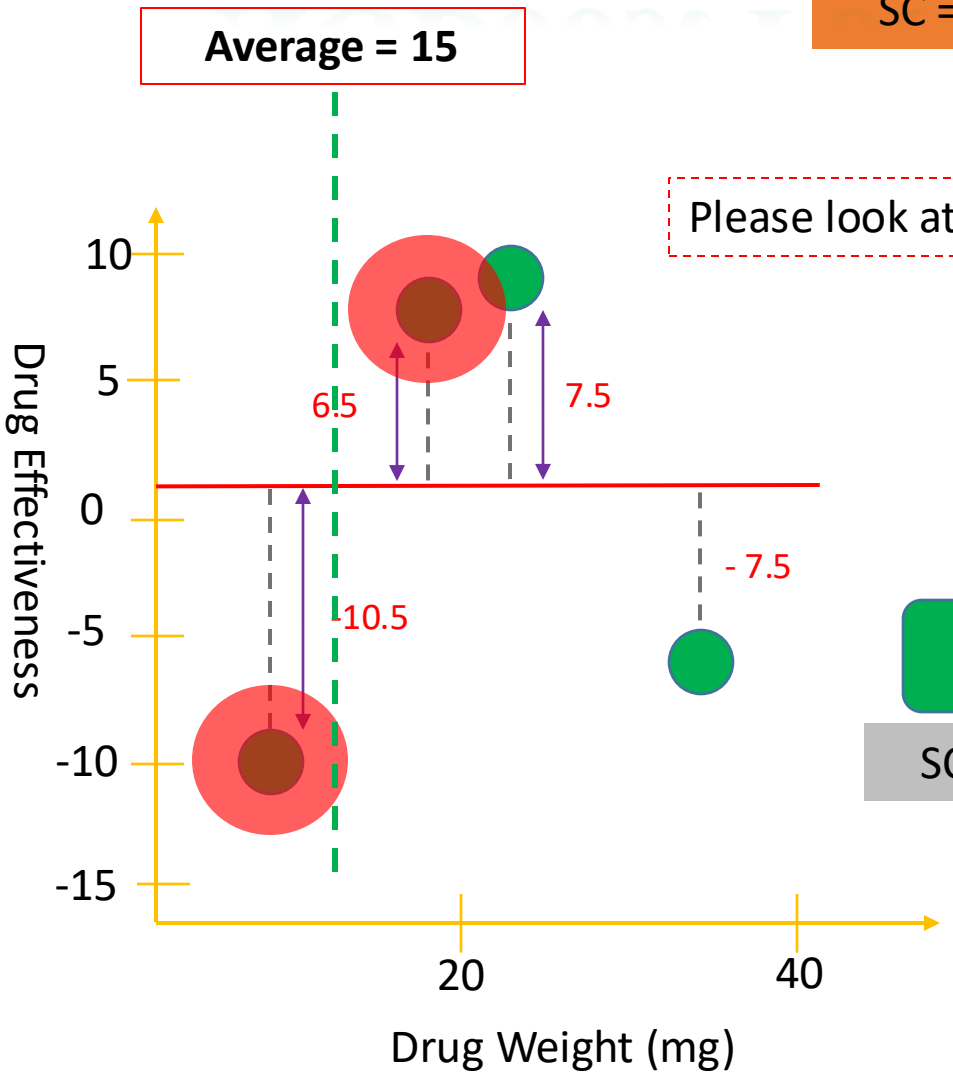




Build a tree on it

XGBoost For Regression

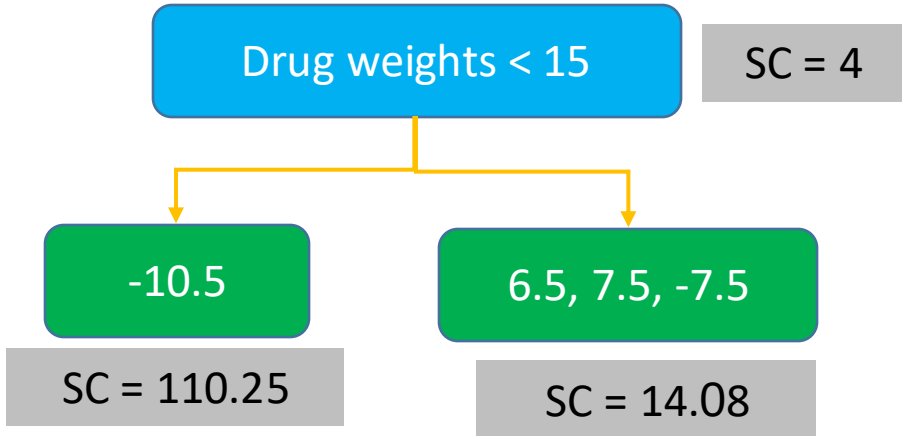
Step 1



SC = 4

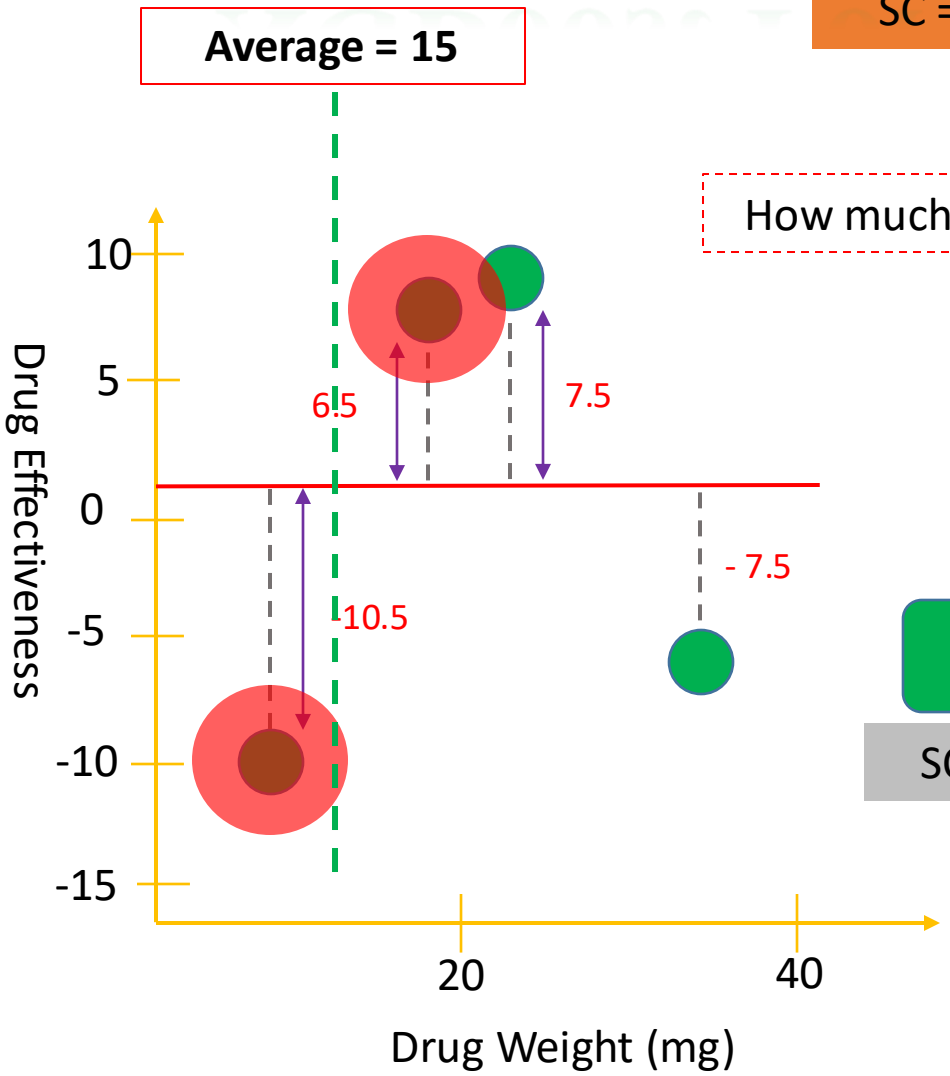
-10.5, 6.5, 7.5, -7.5

Please look at the two outputs with lowest drug weights



XGBoost For Regression

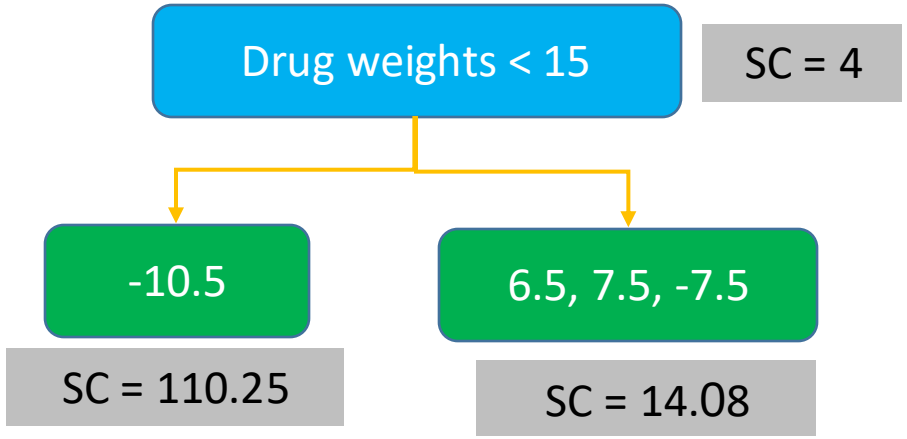
Step 1



SC = 4

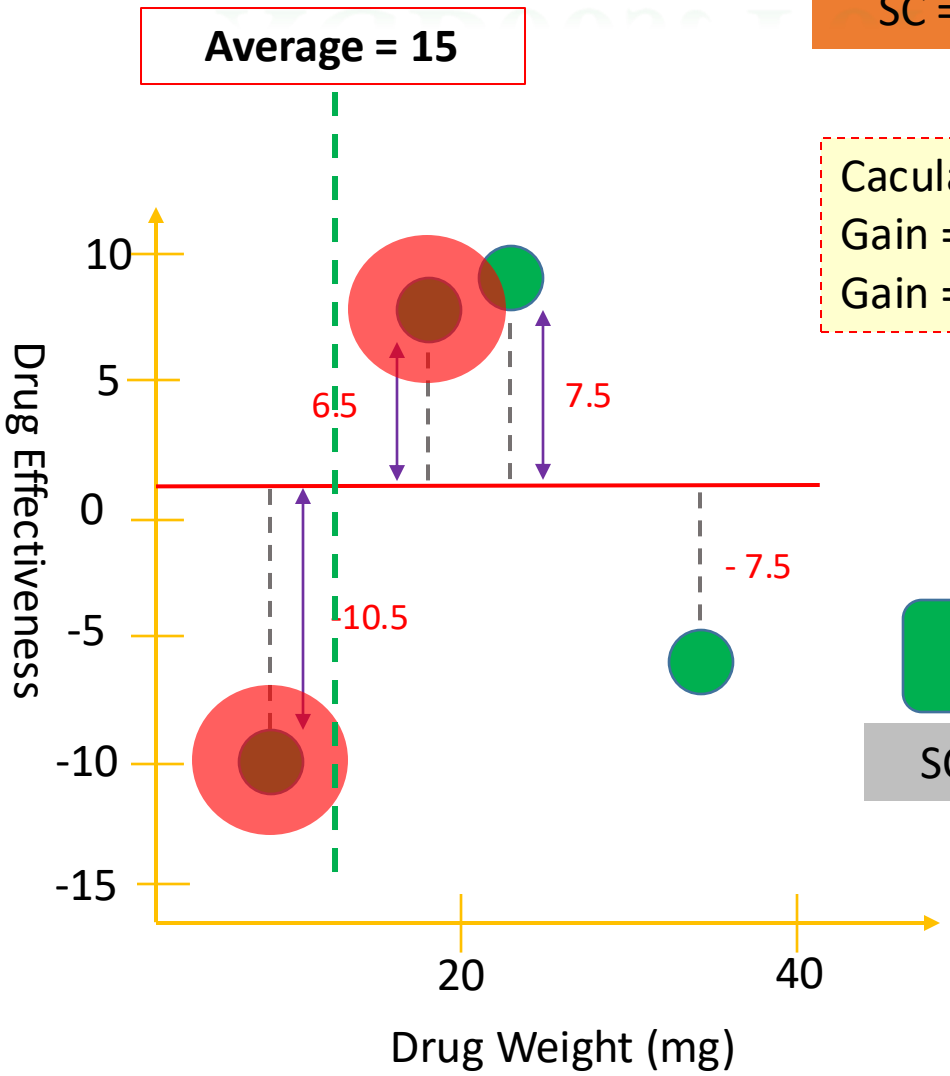
-10.5, 6.5, 7.5, -7.5

How much better leaves cluster similar Residual than the root?



XGBoost For Regression

❑ *Step 1*



SC = 4

-10.5, 6.5, 7.5, -7.5

Residual rất khác nhau, triệt tiêu lẫn nhau, nên SC nhỏ

Caculate the Gain.
Gain = Left SC + Right SC - Root SC
Gain = 120.33

Drug weights < 15

SC = 4

-10.5

SC = 110.25

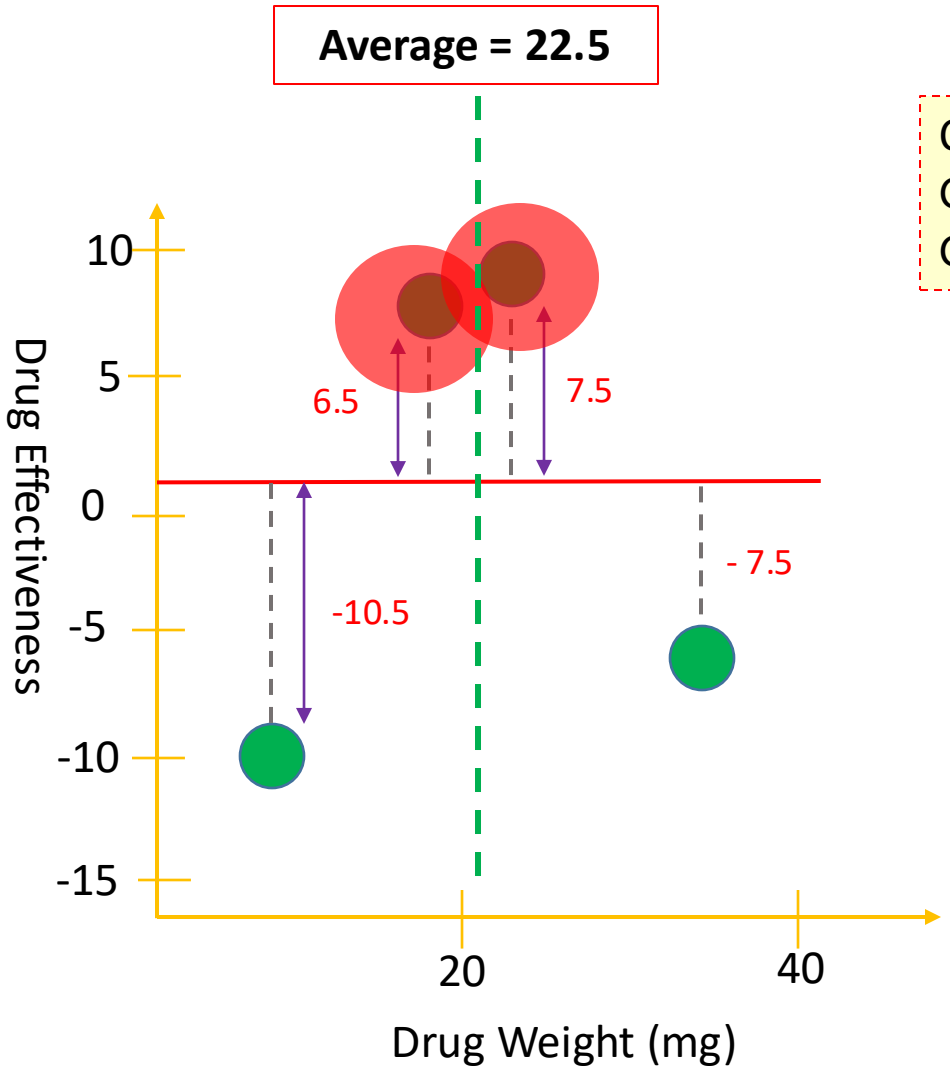
6.5, 7.5, -7.5

SC = 14.08

Residual giống nhau hoặc không triệt tiêu lẫn nhau, nên SC khá lớn

XGBoost For Regression

❑ *Step 1*



SC = 4

-10.5, 6.5, 7.5, -7.5

Residual rất khác nhau, triệt tiêu lẫn nhau, nên SC nhỏ

Caculate the Gain.
Gain = Left SC + Right SC - Root SC
Gain = 4.0

Drug weights < 22.5

SC = 4

-10.5, 6.5

SC = 8

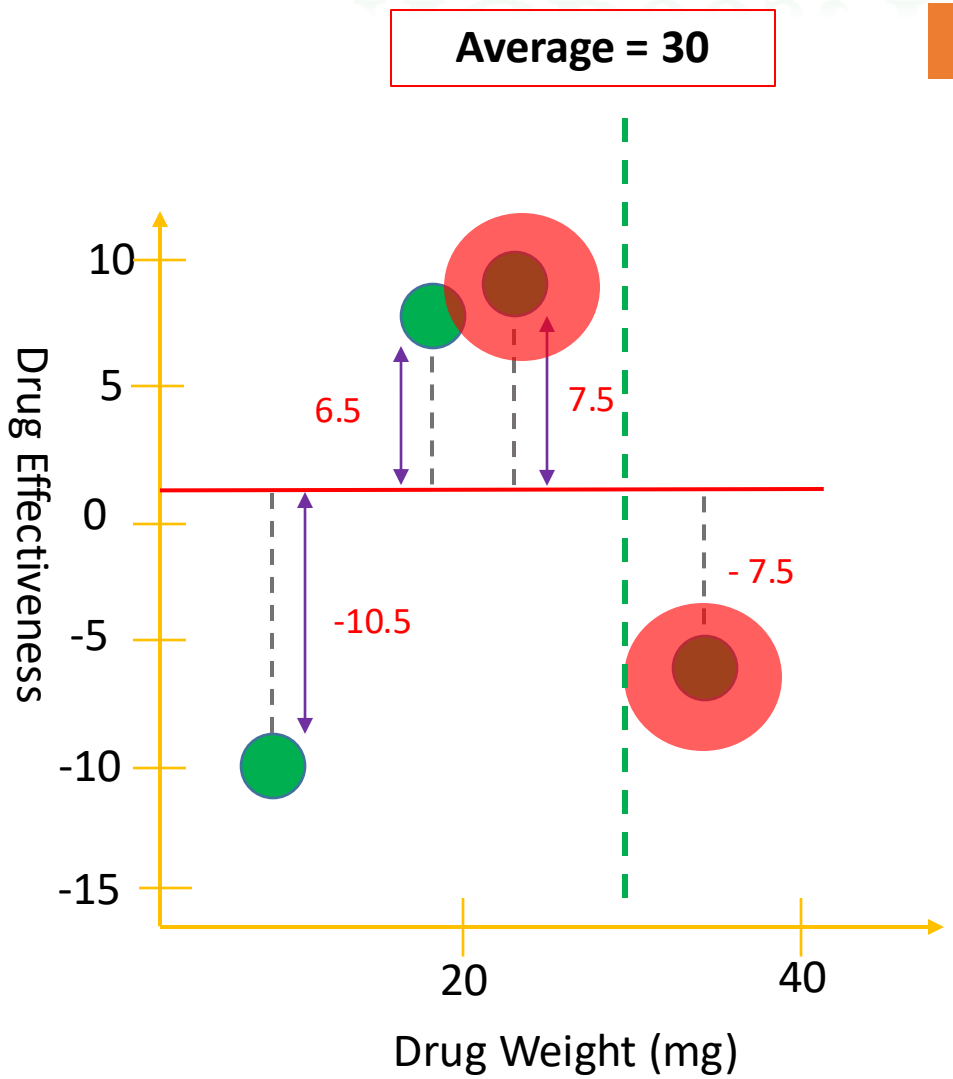
7.5, -7.5

SC = 0

Residual giống nhau hoặc không triệt tiêu lẫn nhau, nên SC khá lơn

XGBoost For Regression

❑ *Step 1*

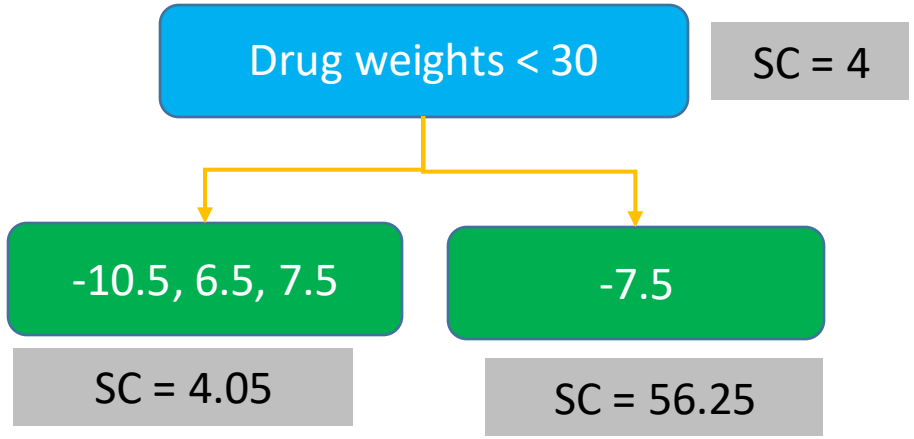


SC = 4

-10.5, 6.5, 7.5, -7.5

Residual rất khác nhau, triệt tiêu lẫn nhau, nên SC nhỏ

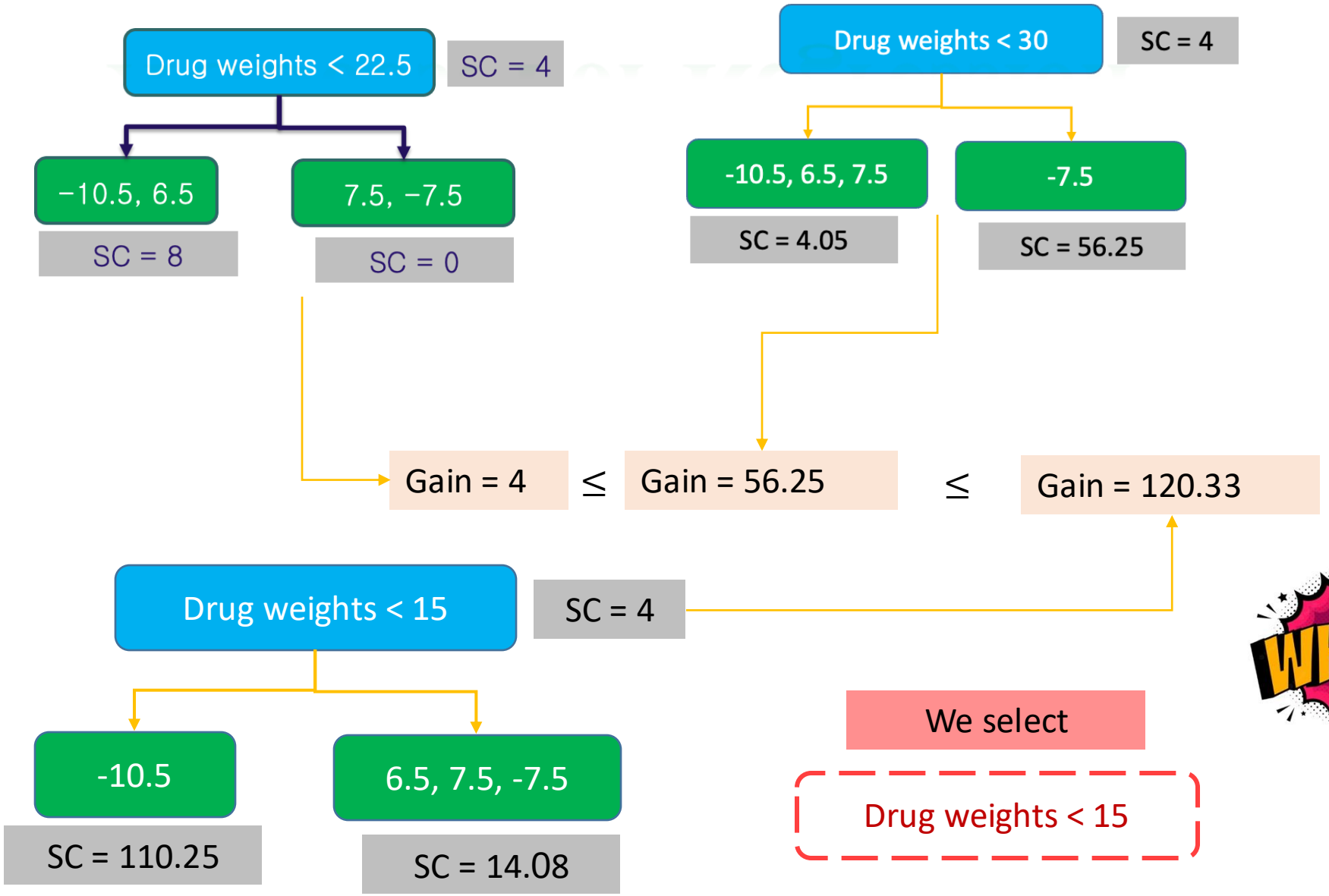
Caculate the Gain.
Gain = Left SC + Right SC - Root SC
Gain = 56.33



Residual giống nhau hoặc không triệt tiêu lẫn nhau, nên SC khá lơn

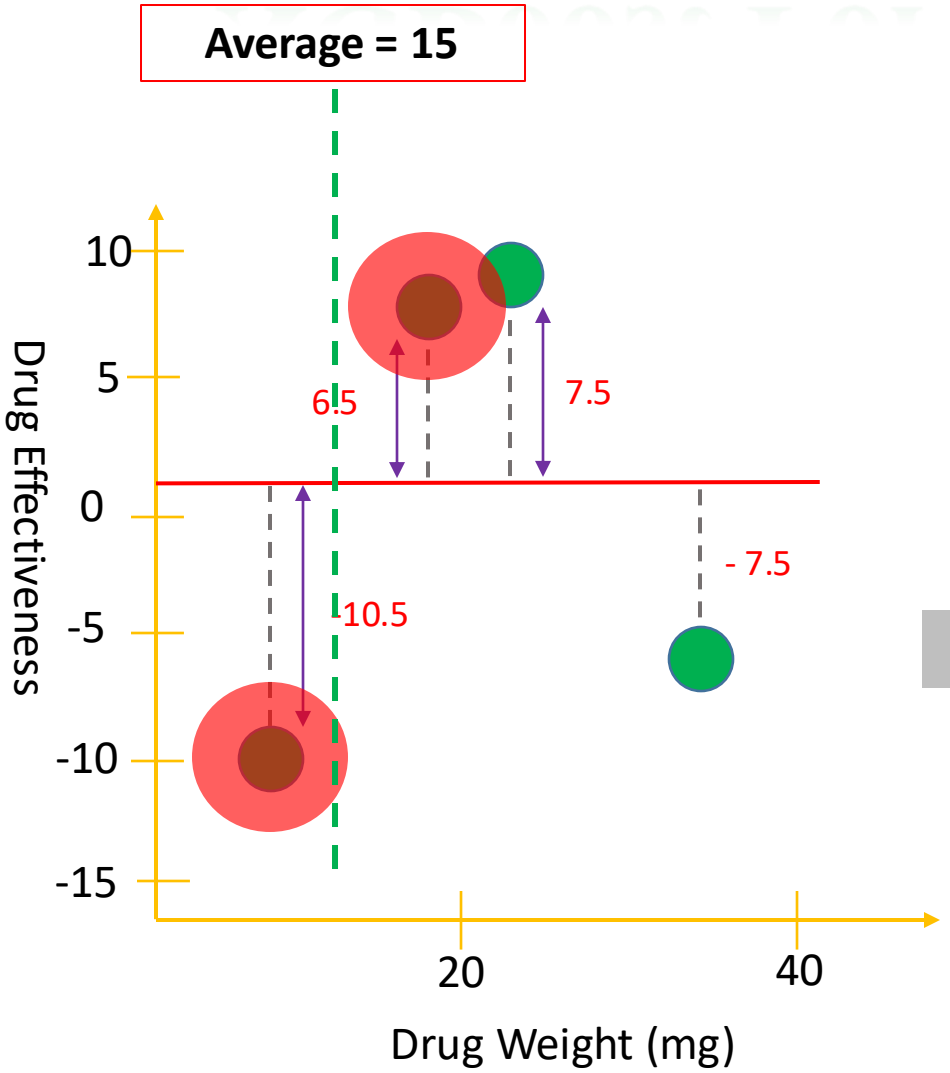
XGBoost For Regression

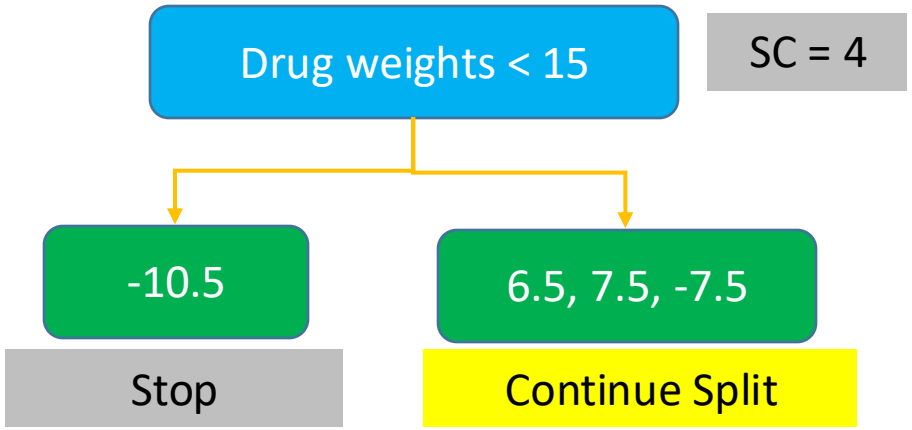
❑ *Step 1*



XGBoost For Regression

❑ *Step 1*





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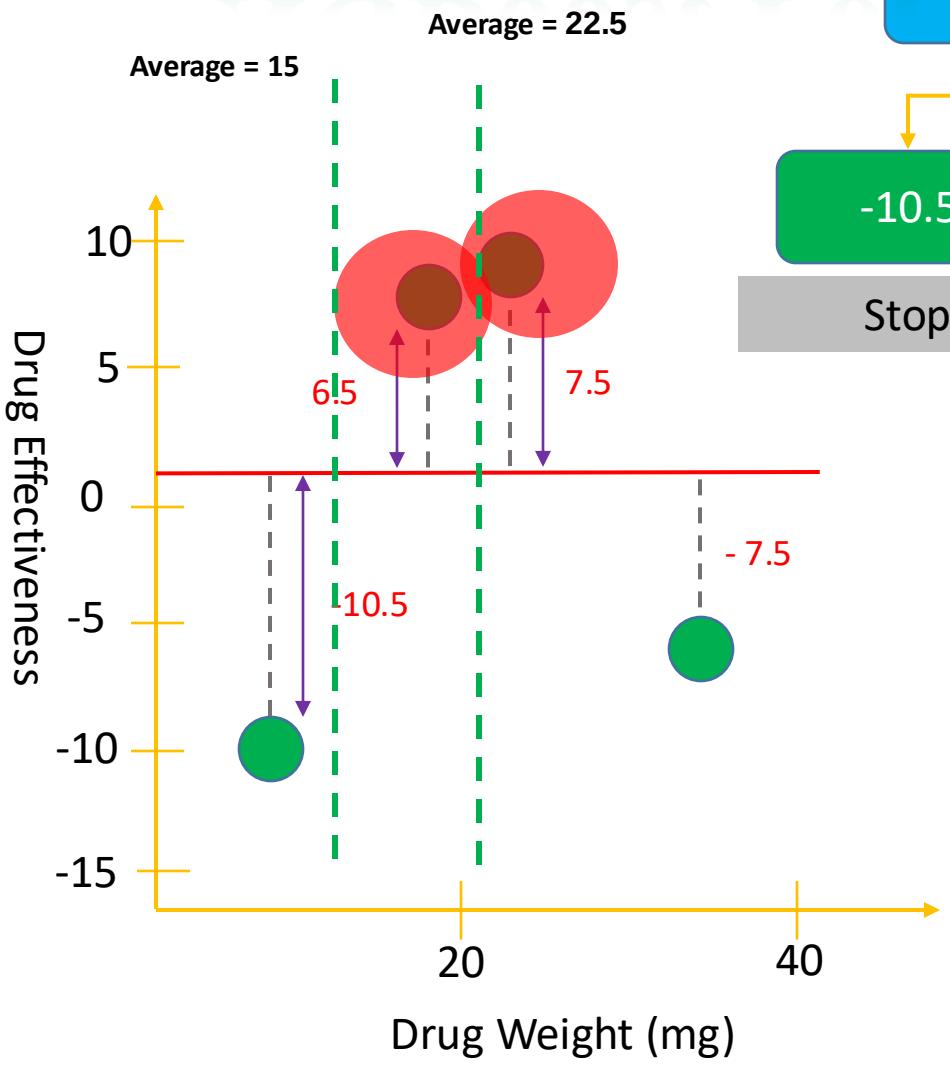
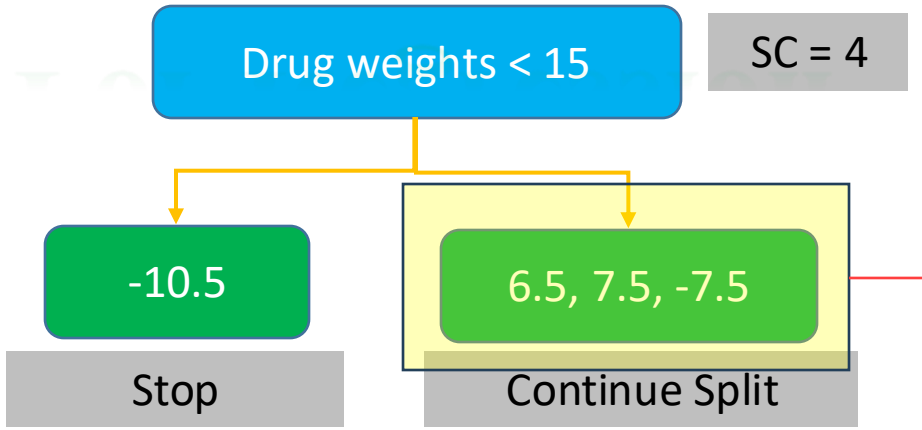
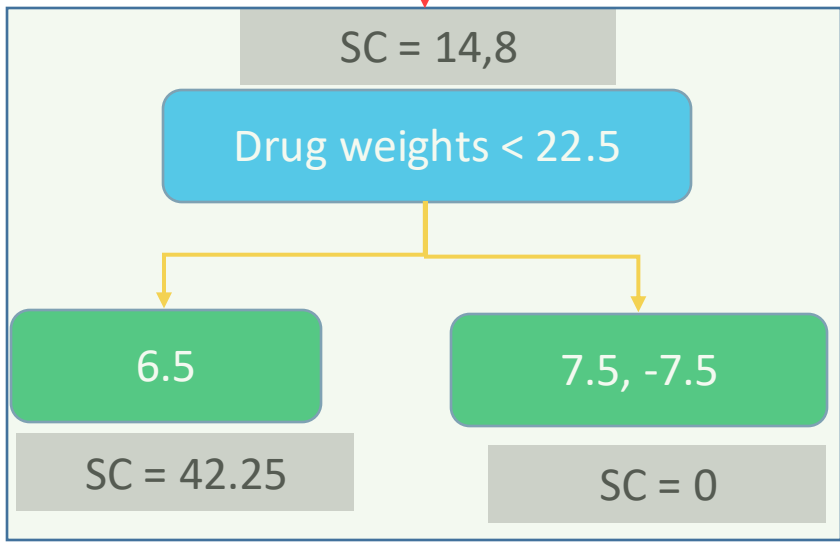
XGBoost For Regression

❑ *Step 1*







Caculate the Gain
Gain = Left SC + Right SC - Root SC
Gain = 28.17

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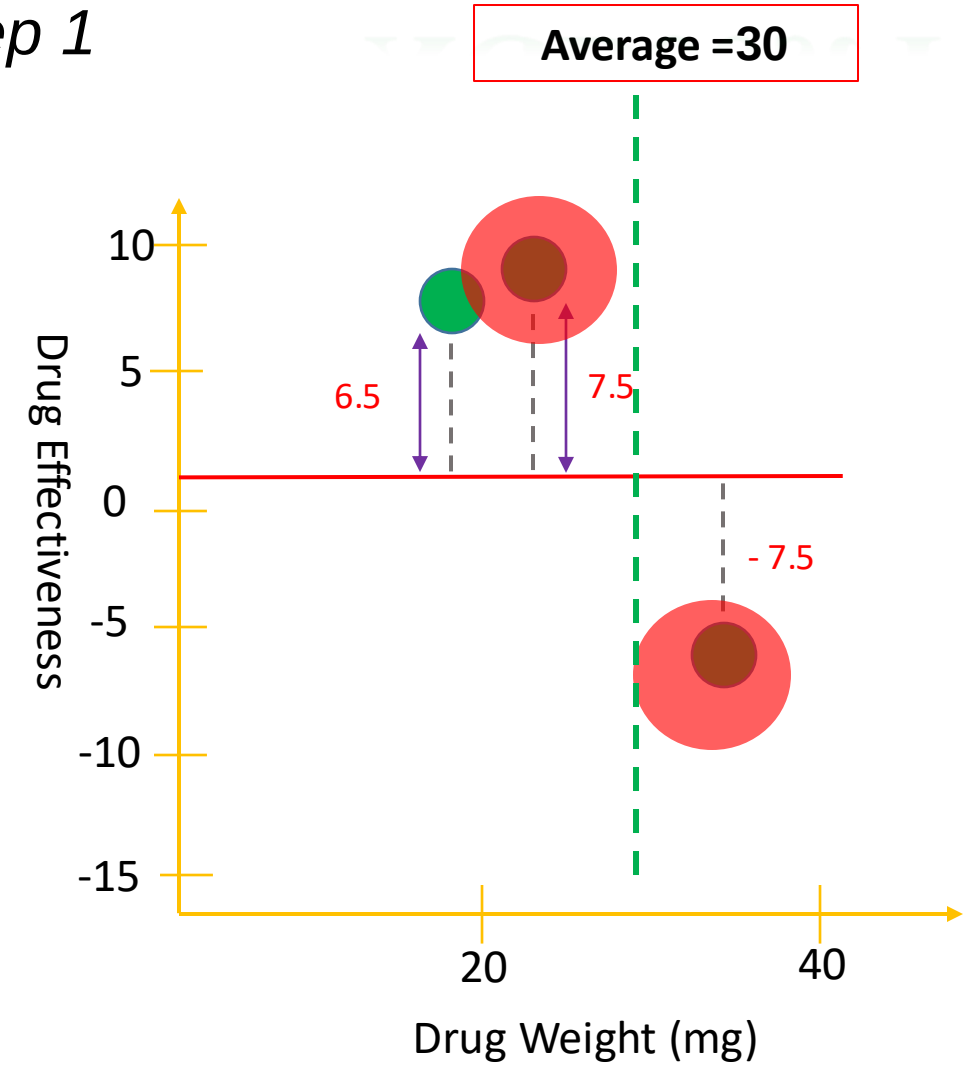
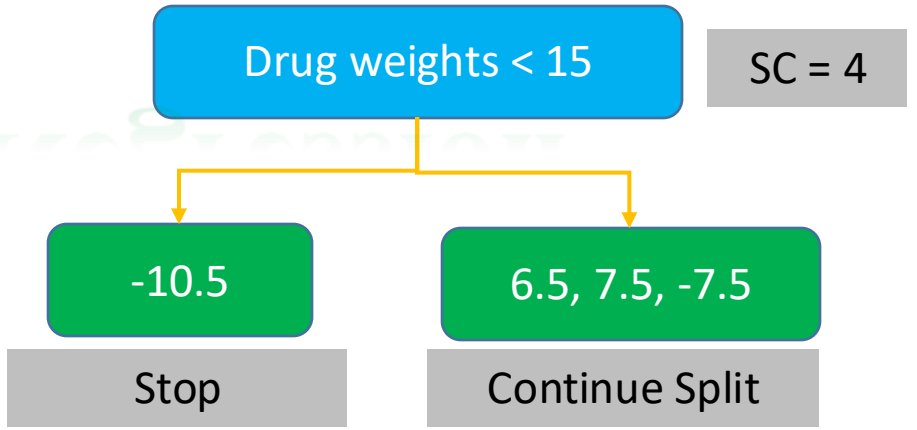
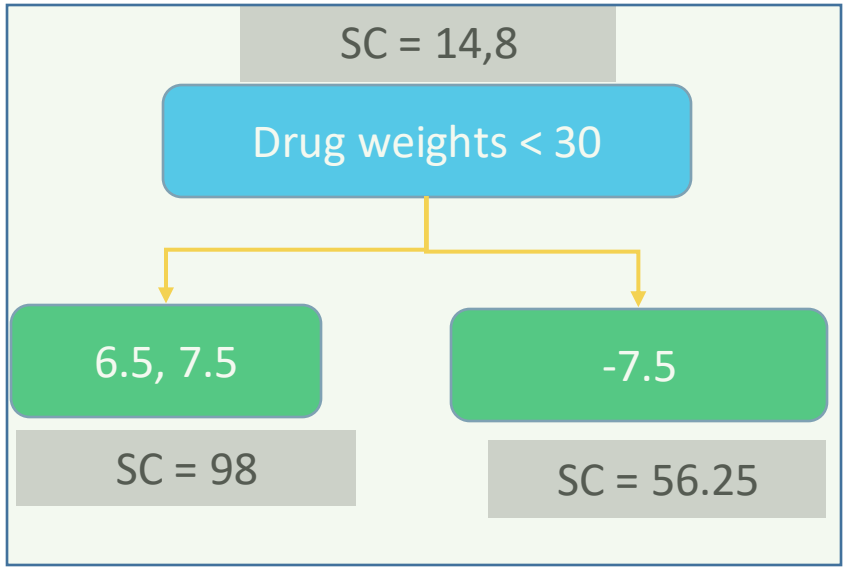
XGBoost For Regression

❑ *Step 1*



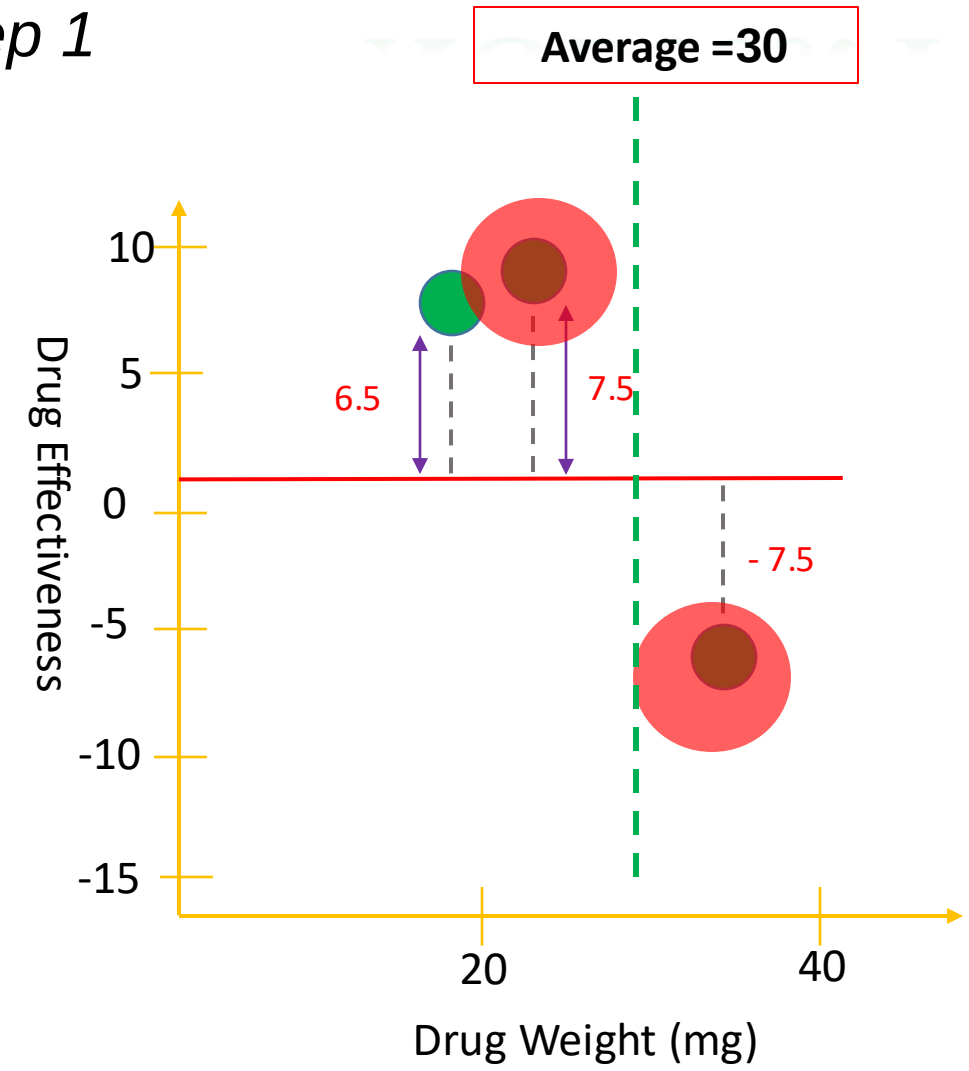


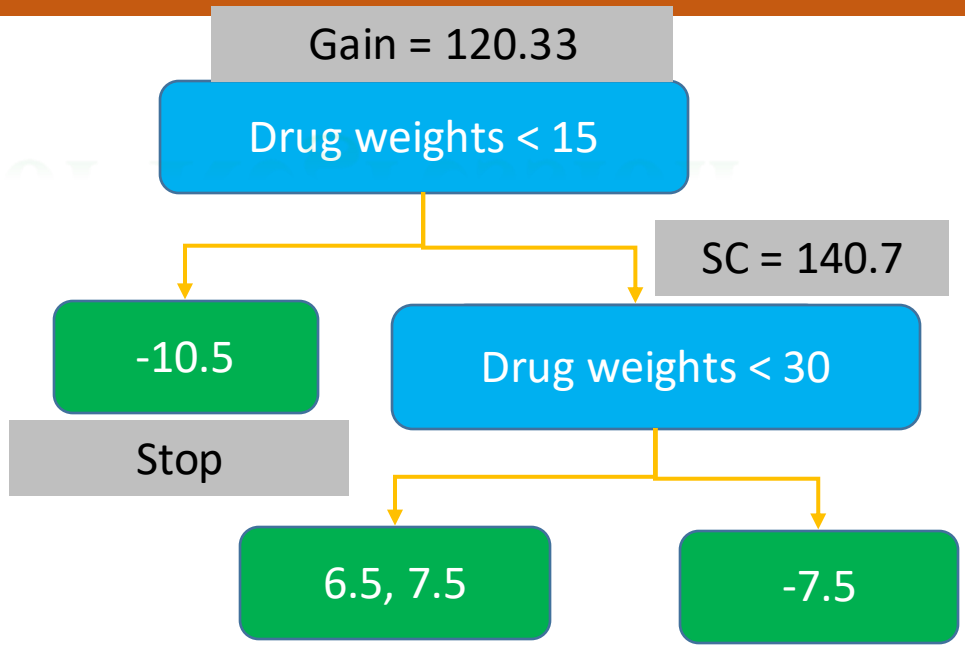


Caculate the Gain
Gain = Left SC + Right SC - Root SC
Gain = 140.17

XGBoost For Regression

❑ *Step 1*



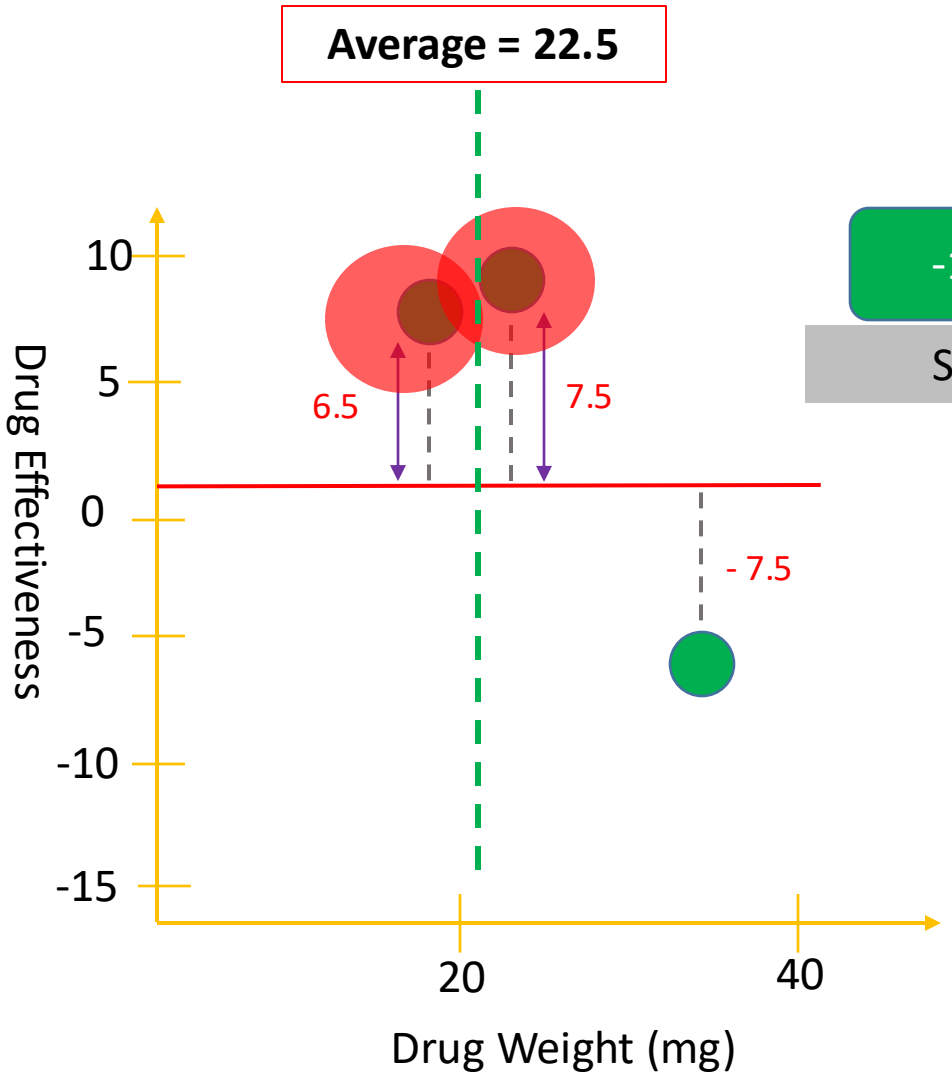


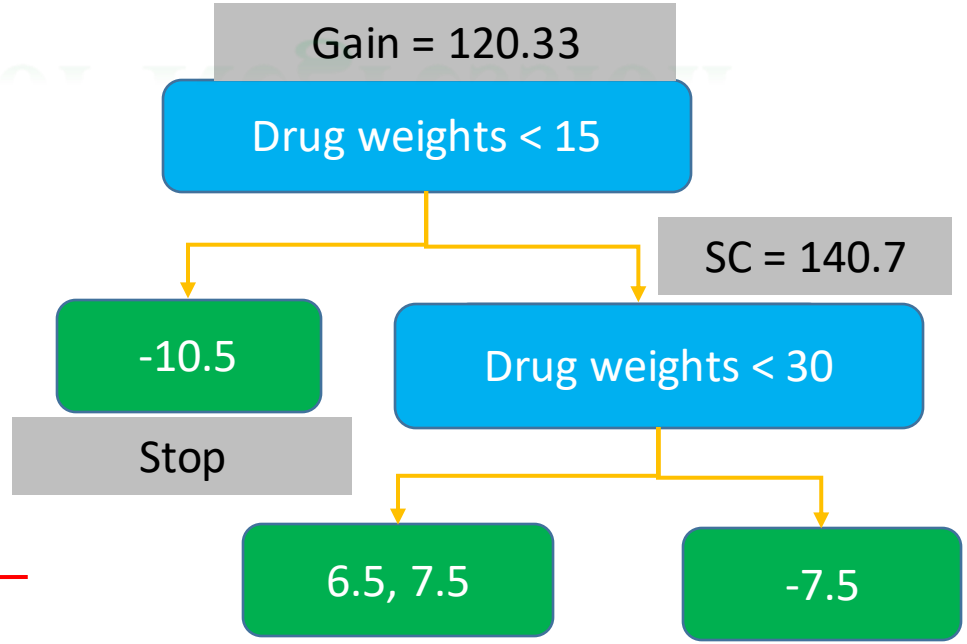
How to prune the tree to prevent Overfitting ? Gain information

$\gamma = 130$

XGBoost For Regression

❑ *Step 1*





How to prune the tree to prevent Overfitting ? Gain information

$$\gamma = 130$$

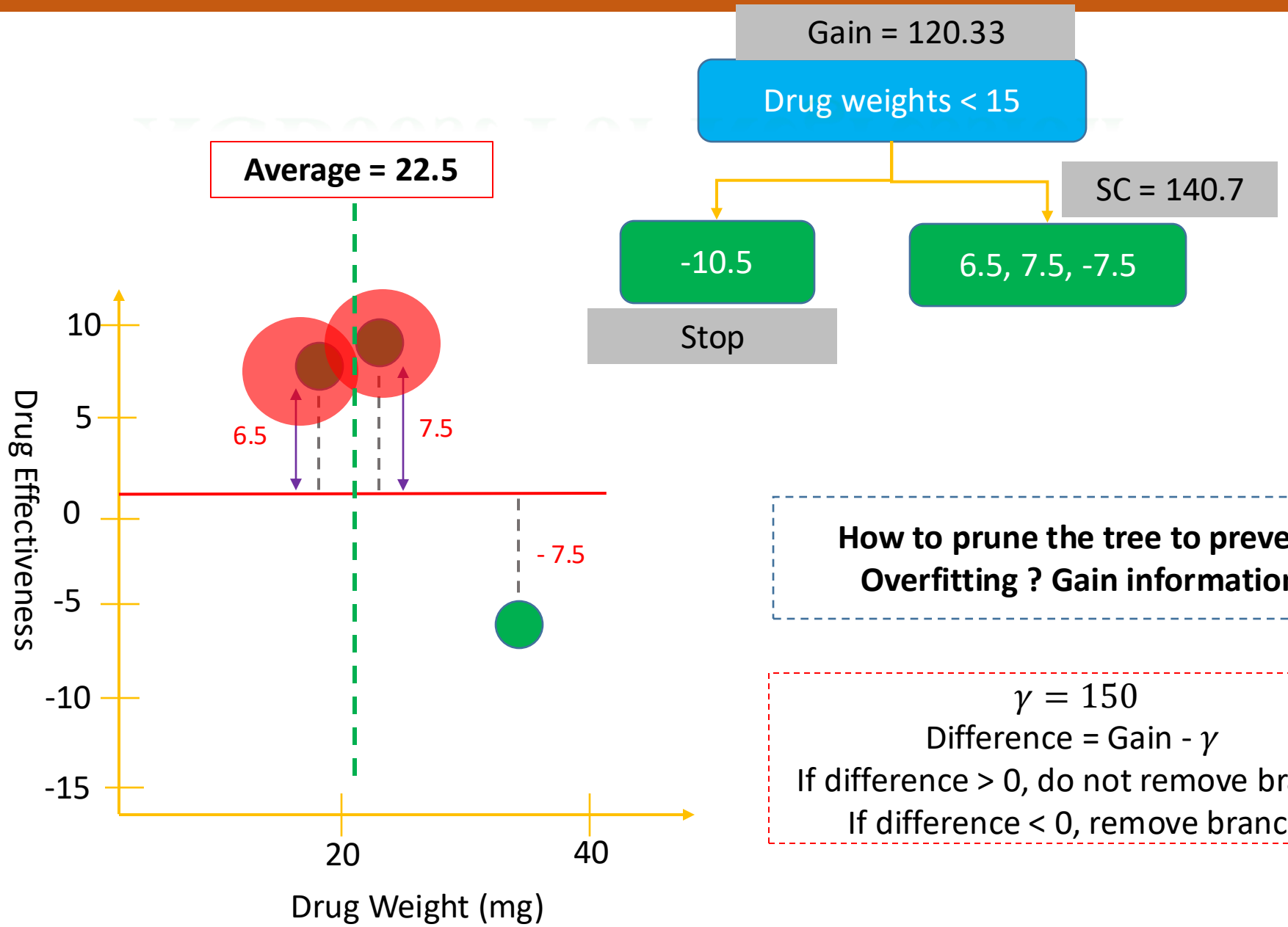
Difference = Gain - γ

If difference > 0, do not remove branch

If difference < 0, remove branch

XGBoost For Regression

❑ *Step 1*



How to prune the tree to prevent Overfitting ? Gain information

$\gamma = 150$

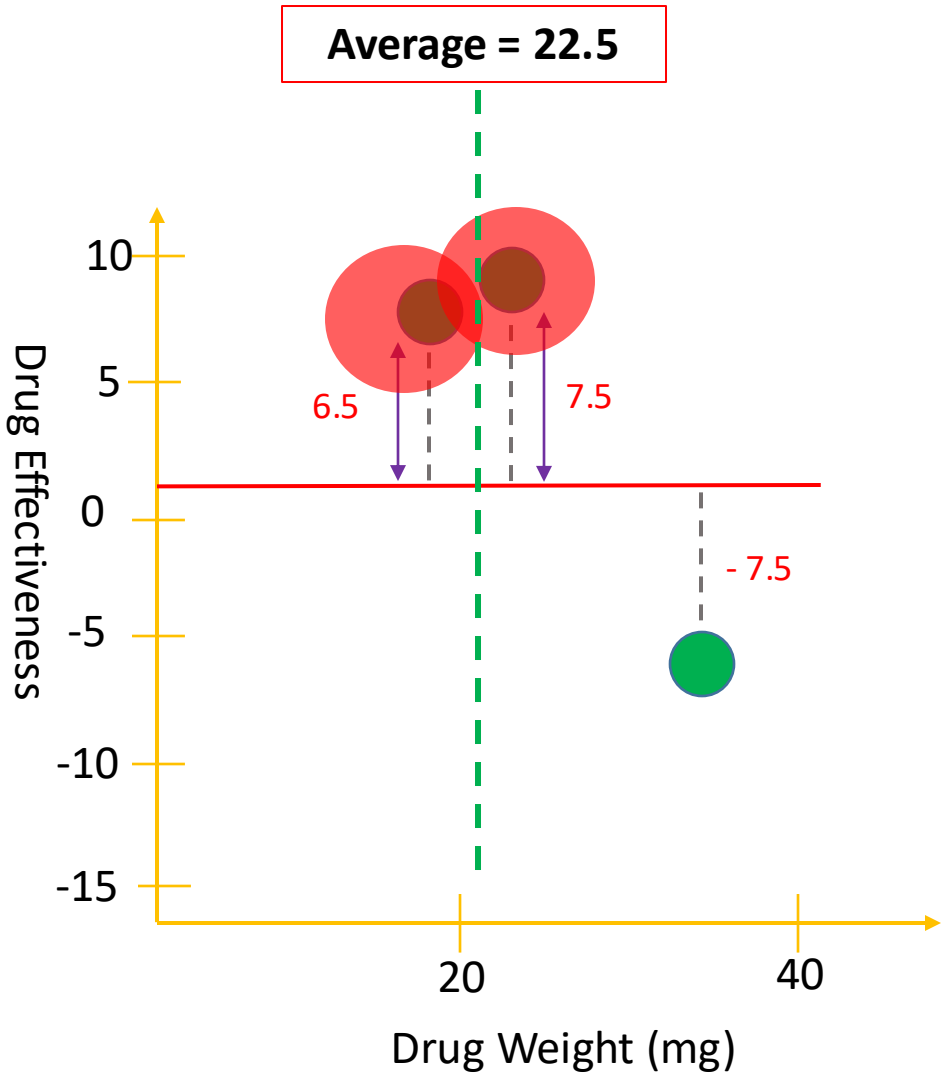
Difference = Gain - γ

If difference > 0, do not remove branch

If difference < 0, remove branch

XGBoost For Regression

❑ *Step 1*



0.5

How to prune the tree to prevent Overfitting ? Gain information

$\gamma = 150$

Difference = Gain - γ

If difference > 0, do not remove branch

If difference < 0, remove branch

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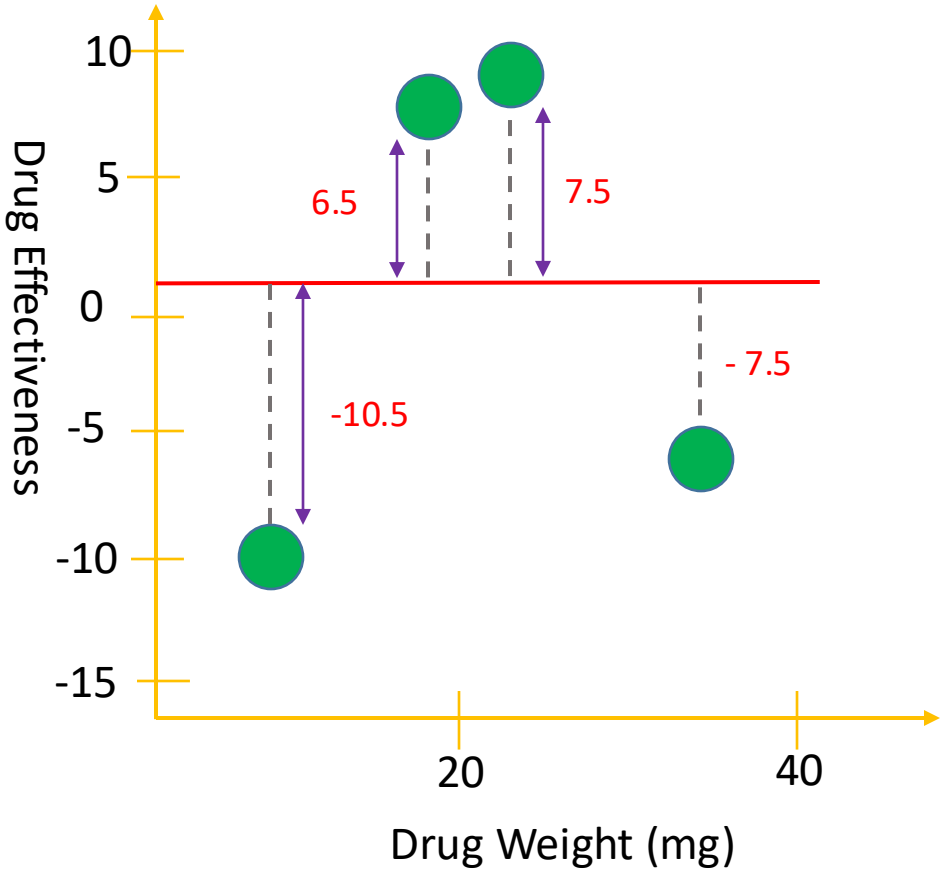
XGBoost For Regression

❑ *Step 1*



Start with single Leaf of residuals

-10.5, 6.5, 7.5, -7.5

m = 4
λ = 1

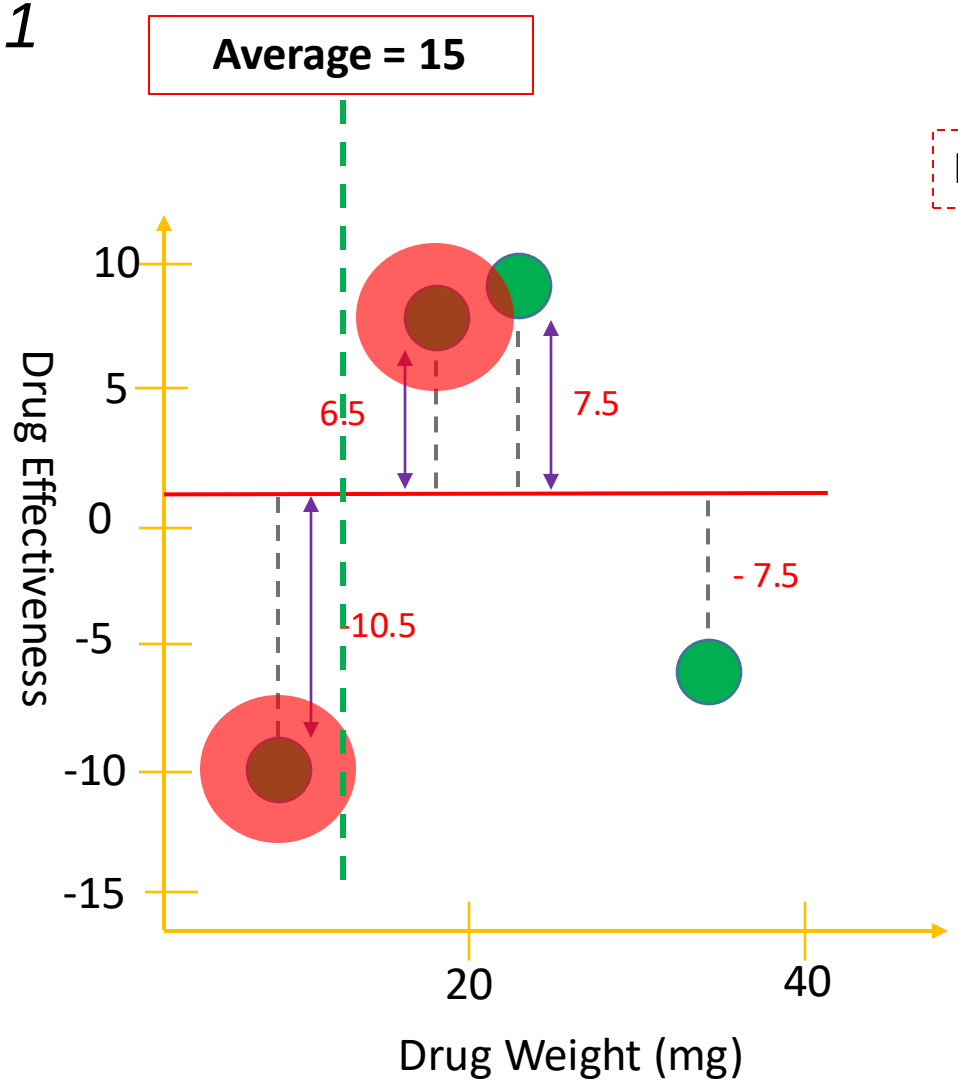
Compute Similarity Score

$$SC = \frac{\sum (output - predicted)^2}{m + \lambda}$$

$$SC = \frac{[-10.5 + 7.5 + 6.5 + (-7.5)]^2}{4+1} = 3.2$$

XGBoost For Regression

❑ *Step 1*



SC = 3.2

-10.5, 6.5, 7.5, -7.5

Please look at the two outputs with lowest drug weights

SC = 3.2

SC = 4

Drug weights < 15

$\lambda = 1$

$\lambda = 0$

-10.5

6.5, 7.5, -7.5

SC = 55.12

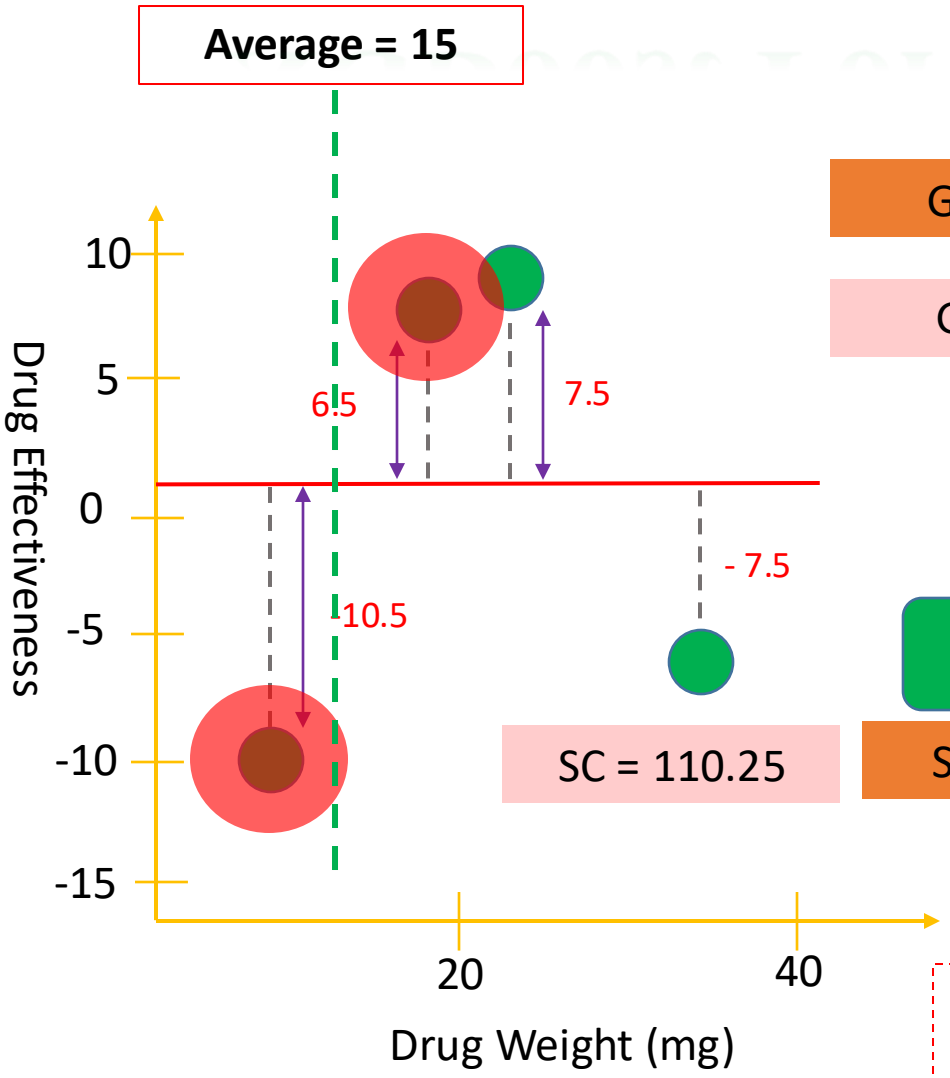
SC = 10.56

SC = 110.25

SC = 14.8

When $\lambda > 0$, the similarity score are smaller
Inversely proportional to the number of residuals

XGBoost For Regression



$\lambda = 1$ $\lambda = 0$

Gain = 55.12 + 10.56 – 3.2 = 62.48

Gain = 55.12 + 10.56 – 4 = 120.33

Drug weights < 15

Gain = 82.9

Gain = 140.17

-10.5

6.5, 7.5, -7.5

SC = 110.25

SC = 55.12

SC = 10.56

SC = 14.8

The amount of decrease is invertly propotional to the number of Residual in the nodes

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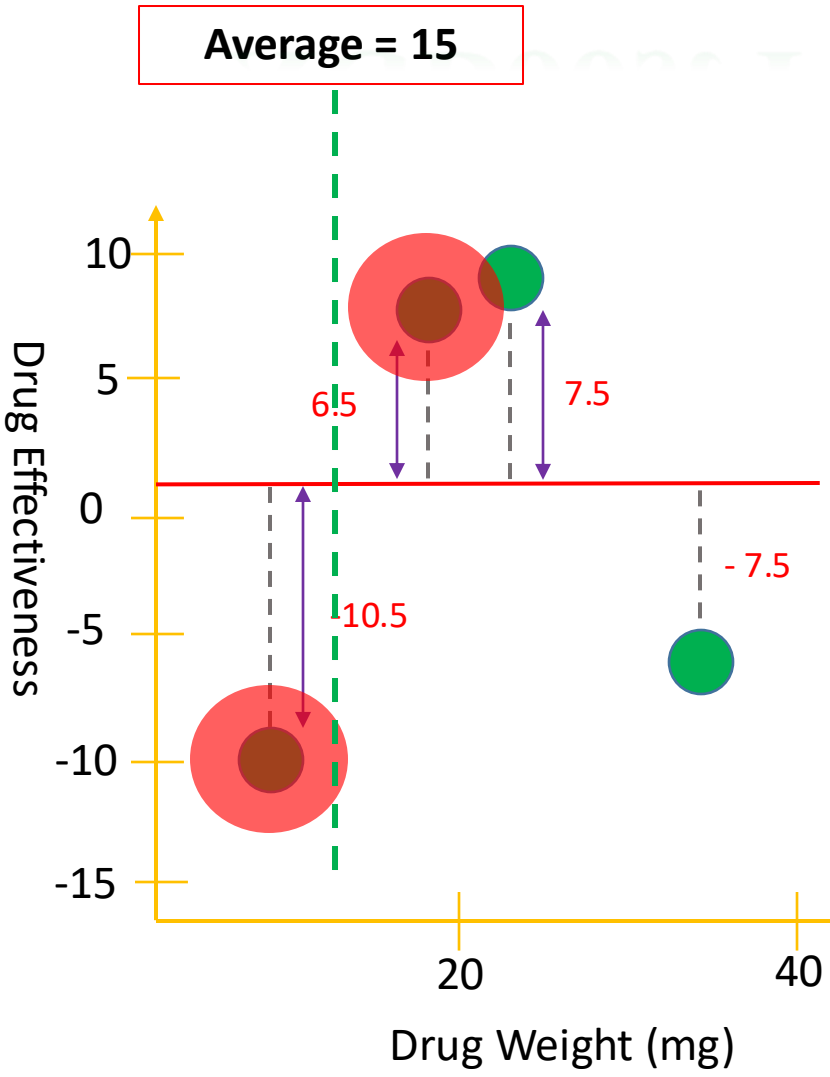
XGBoost For Regression



$\lambda = 0$

$\lambda = 1$

$\lambda > 0$: easy to prune the tree
Prevent overffiting

Gain = 55.12 + 10.56 – 4 = 62.48

Gain = 55.12 + 10.56 – 4 = 120.33

Prunning parameter: $\gamma = 130$

Drug weights < 15

Gain = 82.9

Gain = 140.17

-10.5

6.5, 7.5, -7.5

SC = 55.12

SC = 110.25

SC = 10.56

SC = 14.8

The amount of decrease is invertly propotional to the number of Residual in the nodes

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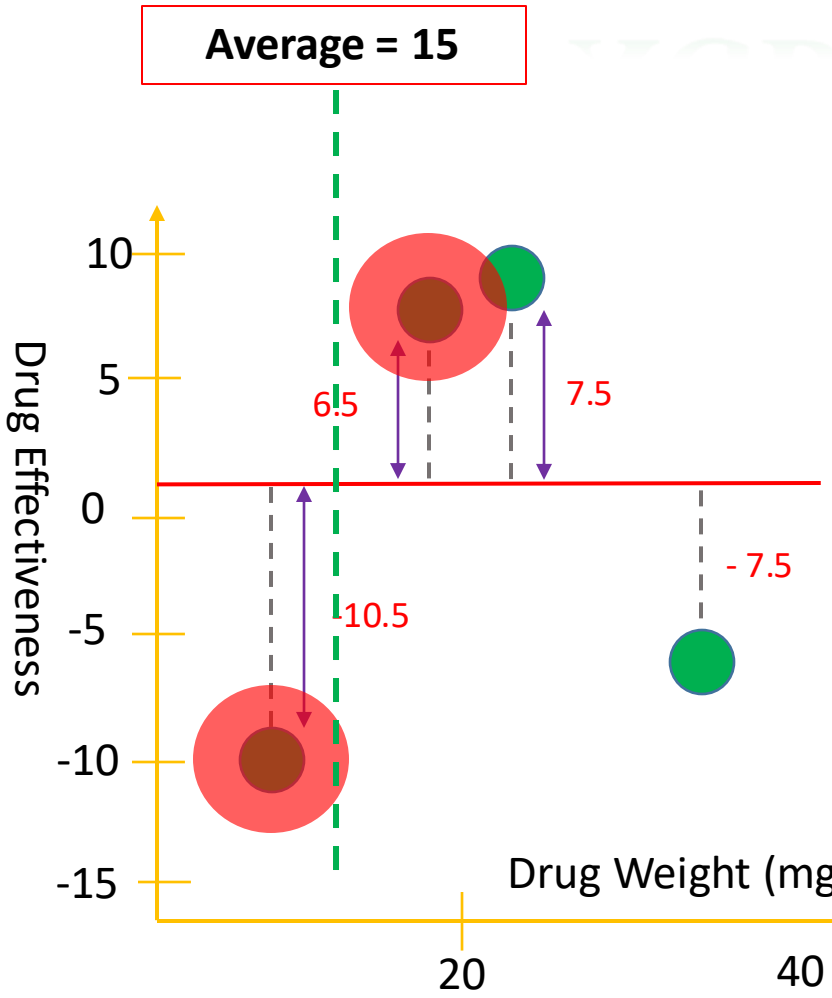
XGBoost For Regression



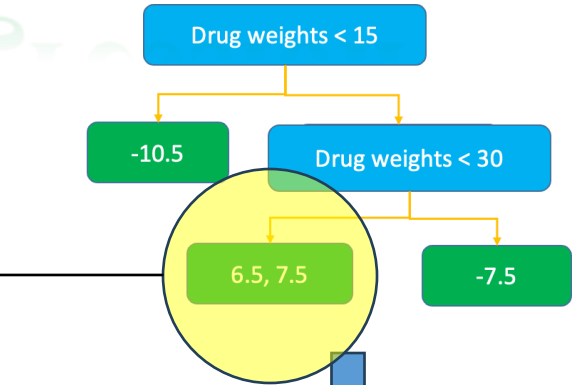
Setting $\gamma = 0$ do not turn off prunning

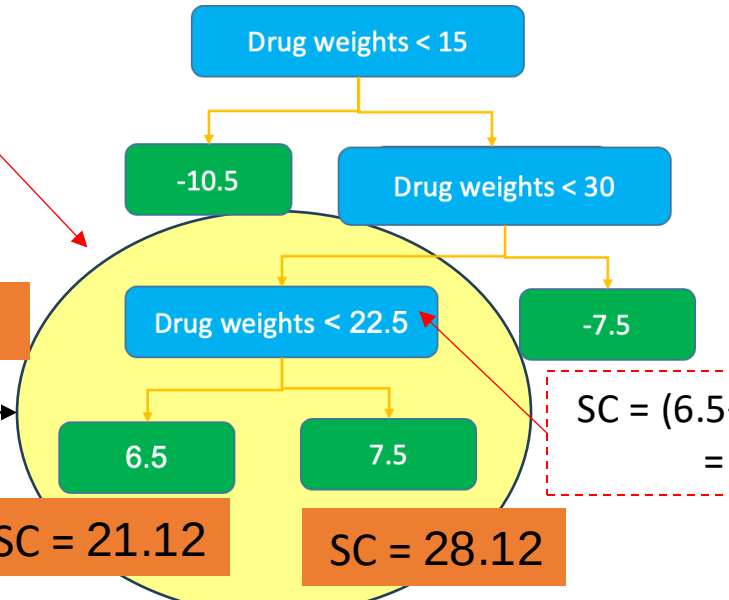
We will remove this branch: -16.06- γ < 0

Prunning parameter: $\gamma = 0$

$\lambda = 1$

SC = 65.3





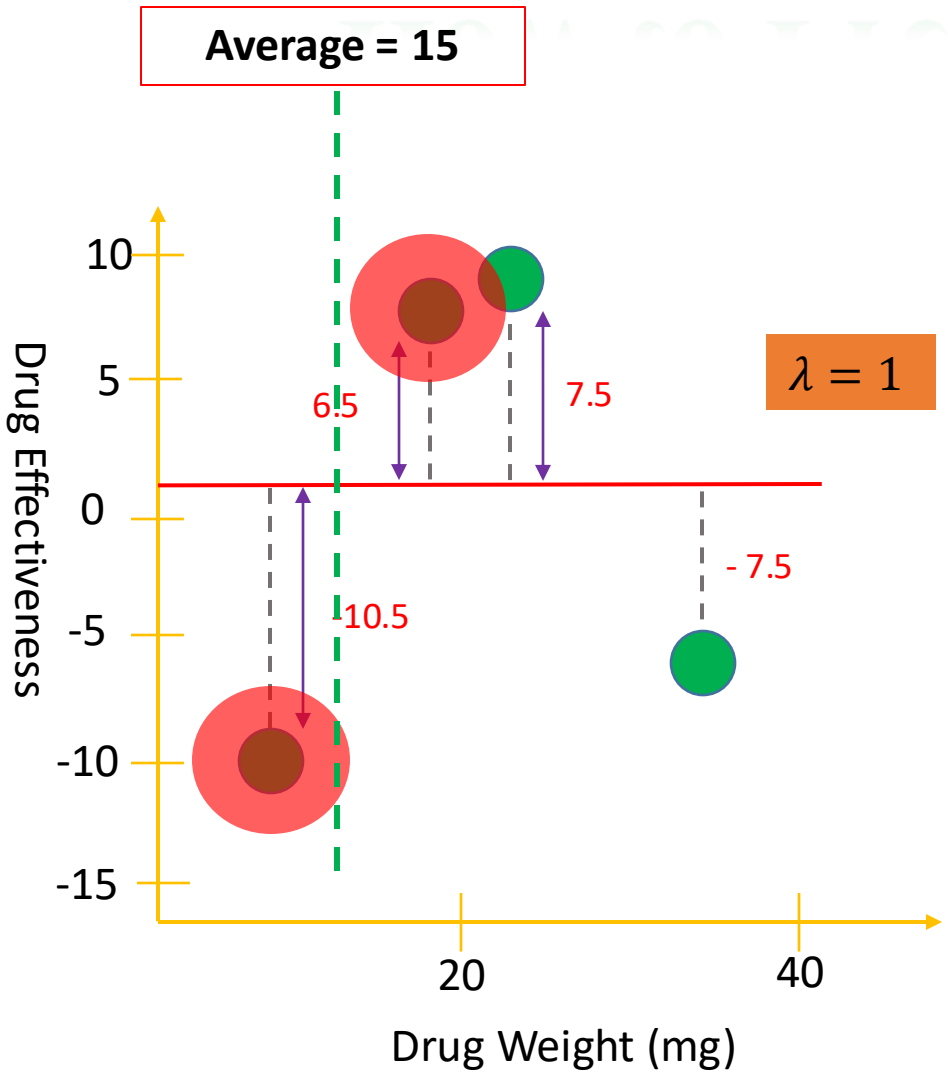
SC = $(6.5+7.5)^2/(2+1)$ = 65.3

SC = 21.12

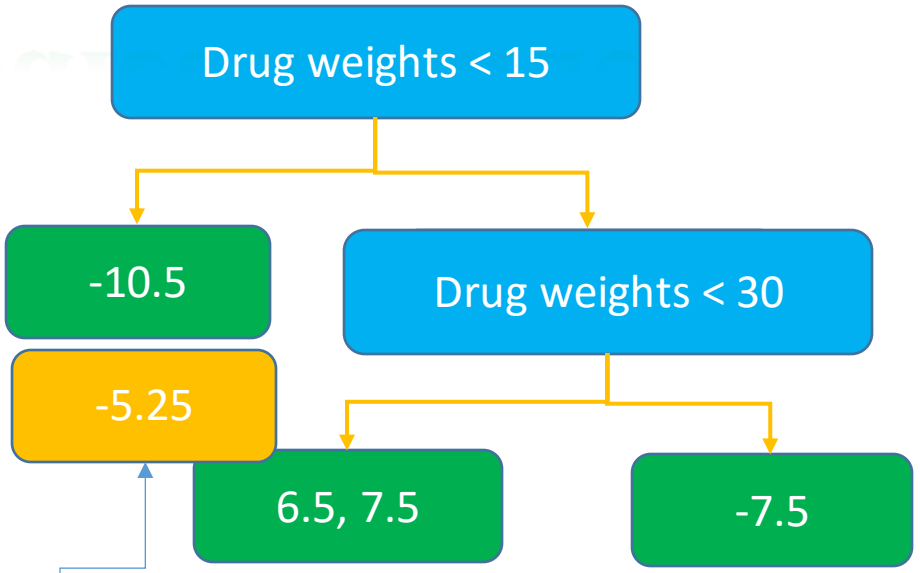
SC = 28.12

Gain = 21.12 + 28.12 – 65.3 = -16.06

How to Predict a Value



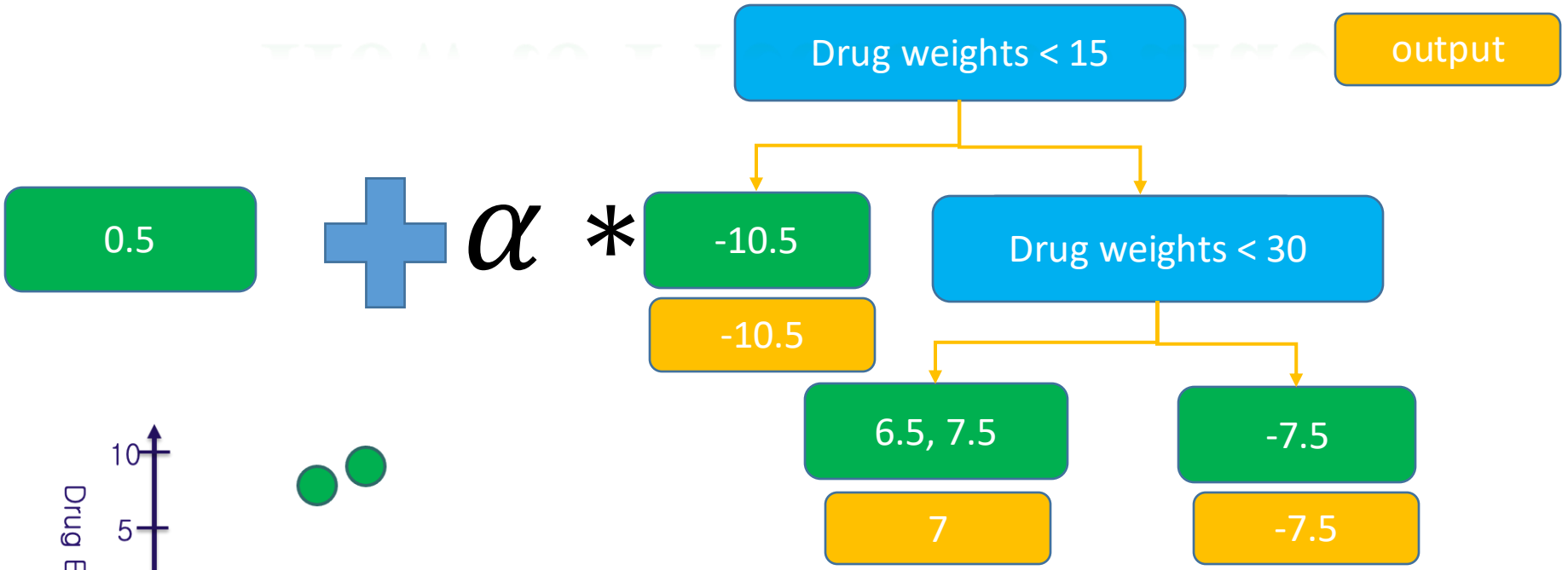
$\lambda = 1$

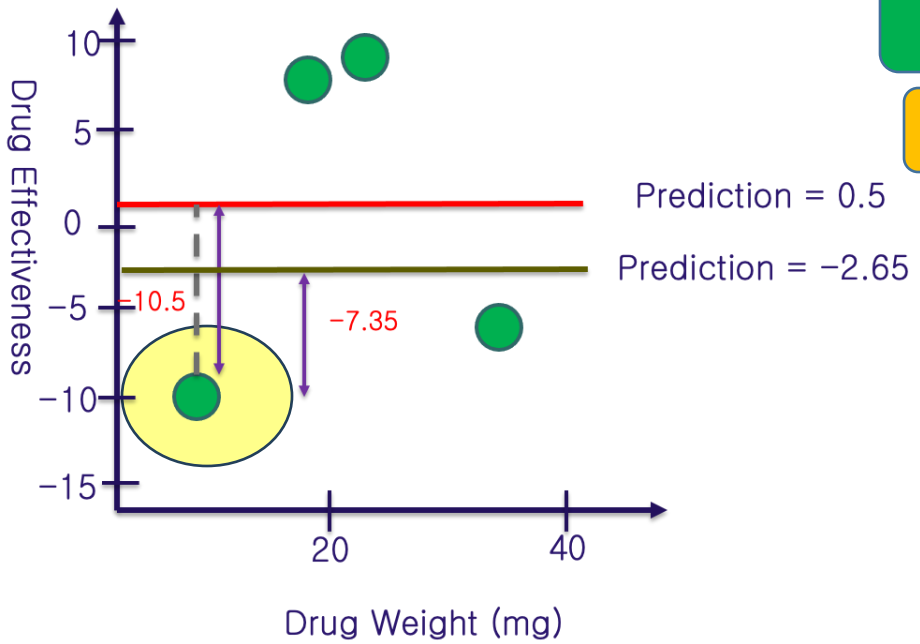


$$\text{Output Value} = \frac{\text{Sum of Residuals}}{\text{Number of Residuals} + \lambda}$$

When $\lambda > 0$, it will reduce the amount that this indiviual observation add to the overal prediction

How to Predict a Value





$\lambda = 0$ $\alpha = 0.3$

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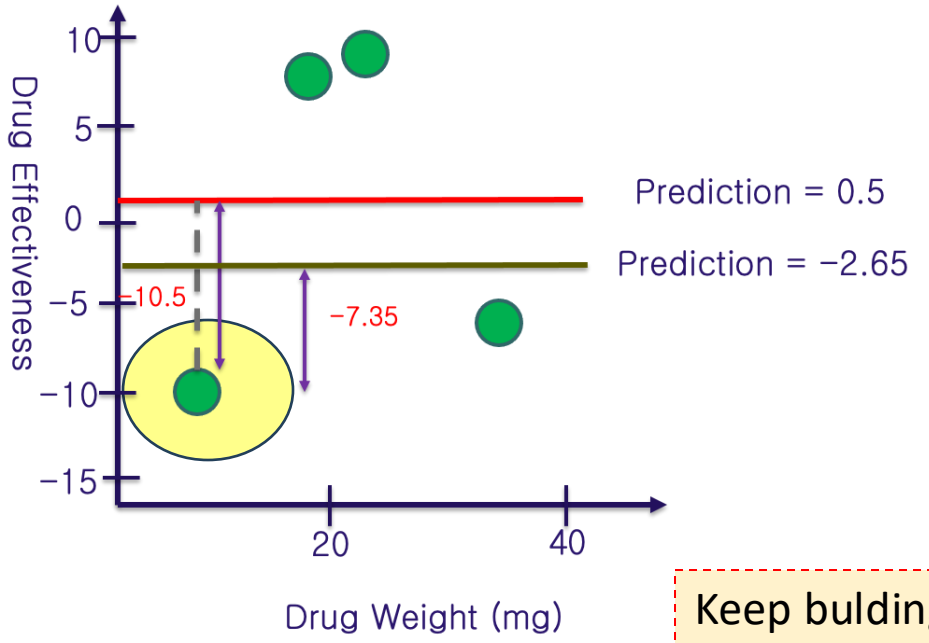
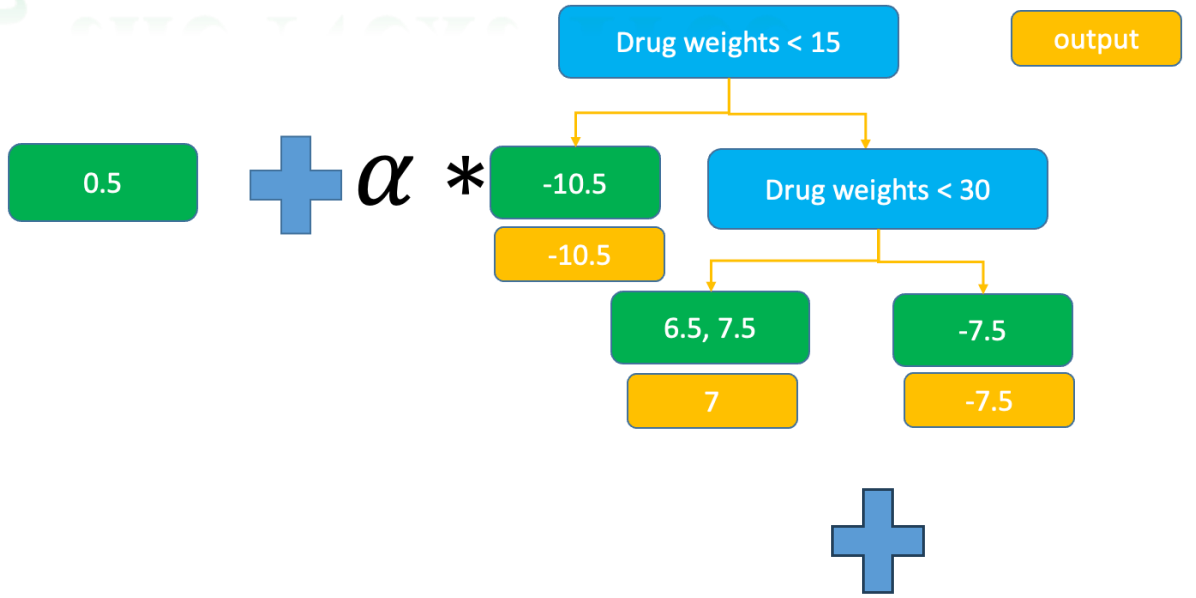
Building the Next Tree

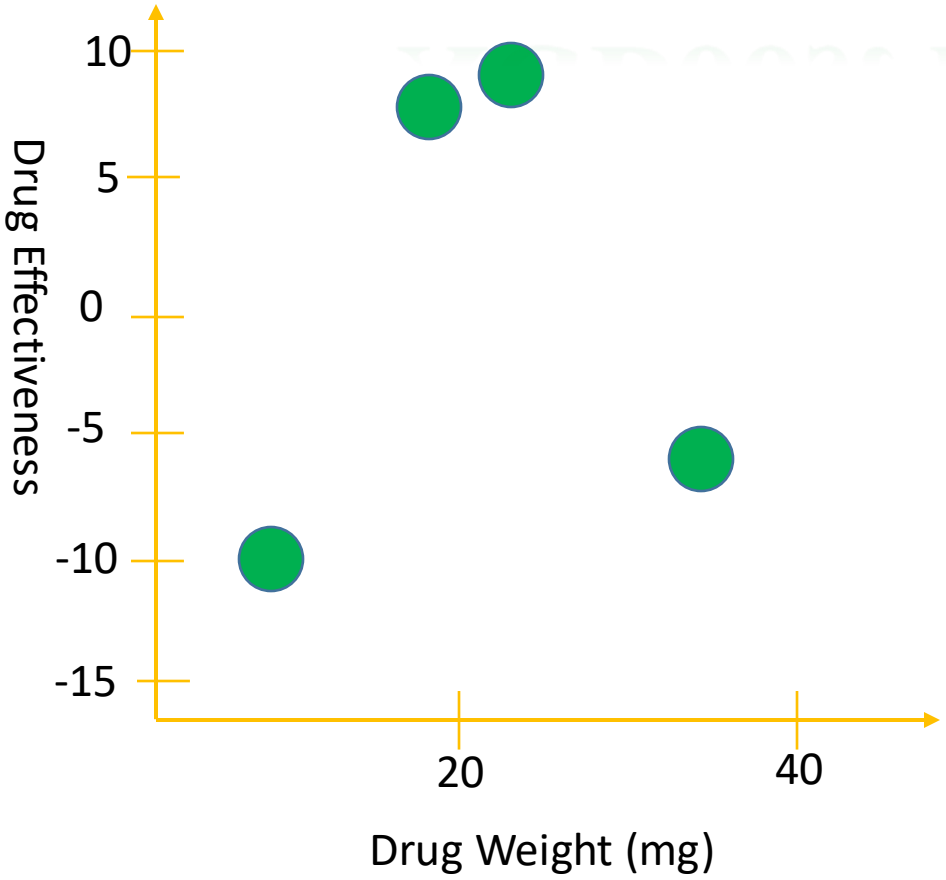


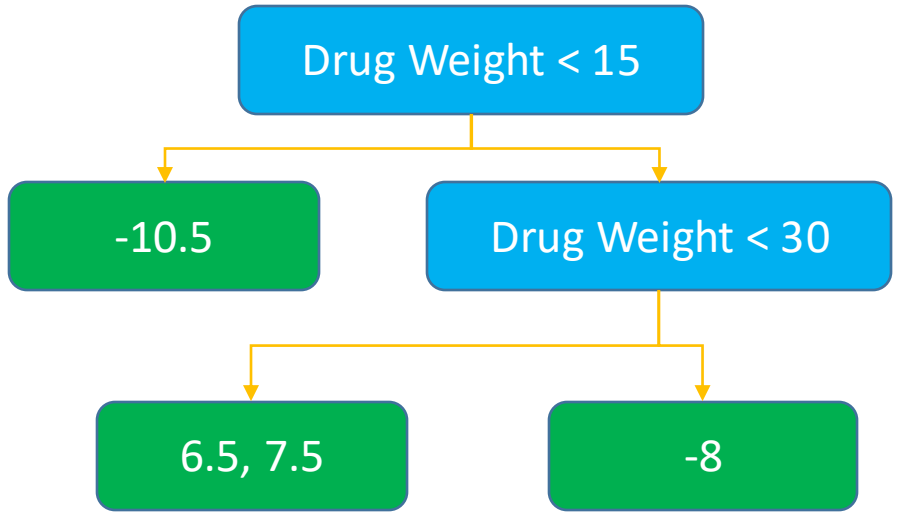


$\alpha *$ Next Tree Result

Keep bulding the Tree until the Residual are reach the predefined threshold. Or we reach to the maximum number of Tree

XGBoost for Regression



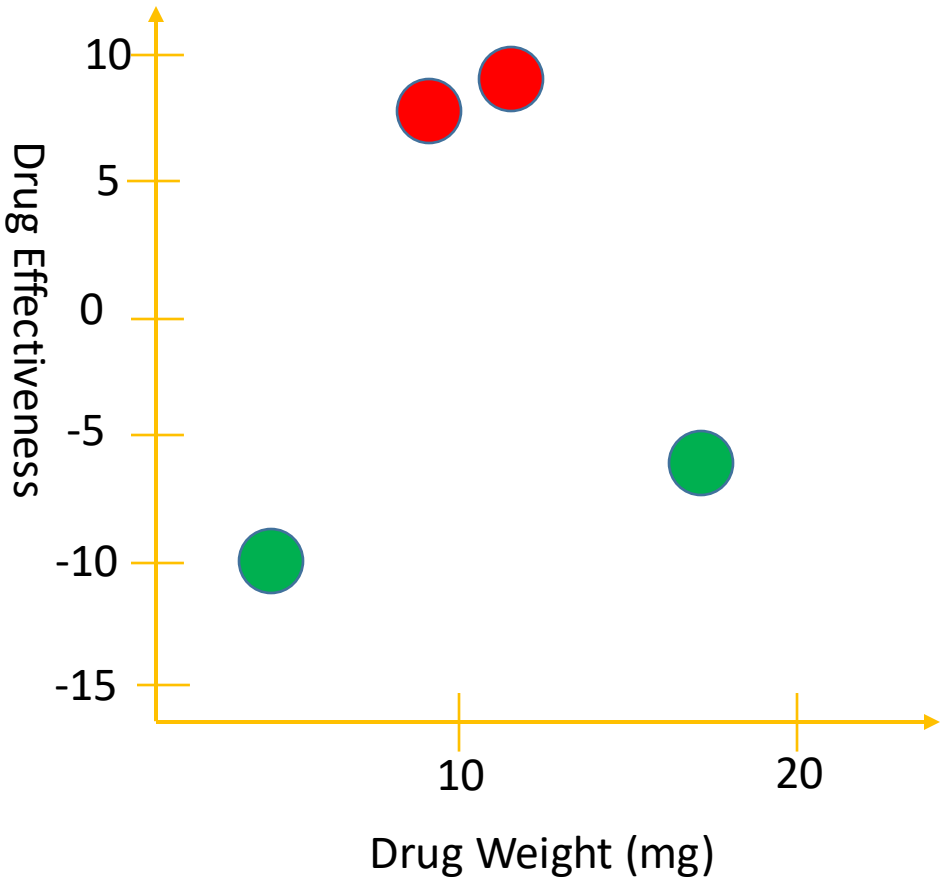


HOW TO FIND QUANTILES? => QUANTILE SKETCH APPROXIMATE SOLUTION

Outline

- Regularization
- Regression XGBoost
- Classification XGBoost
- XGBoost: Clearly Explain
- Time Series Example
- Summary

XGBoost For Classification



Not effectiviness

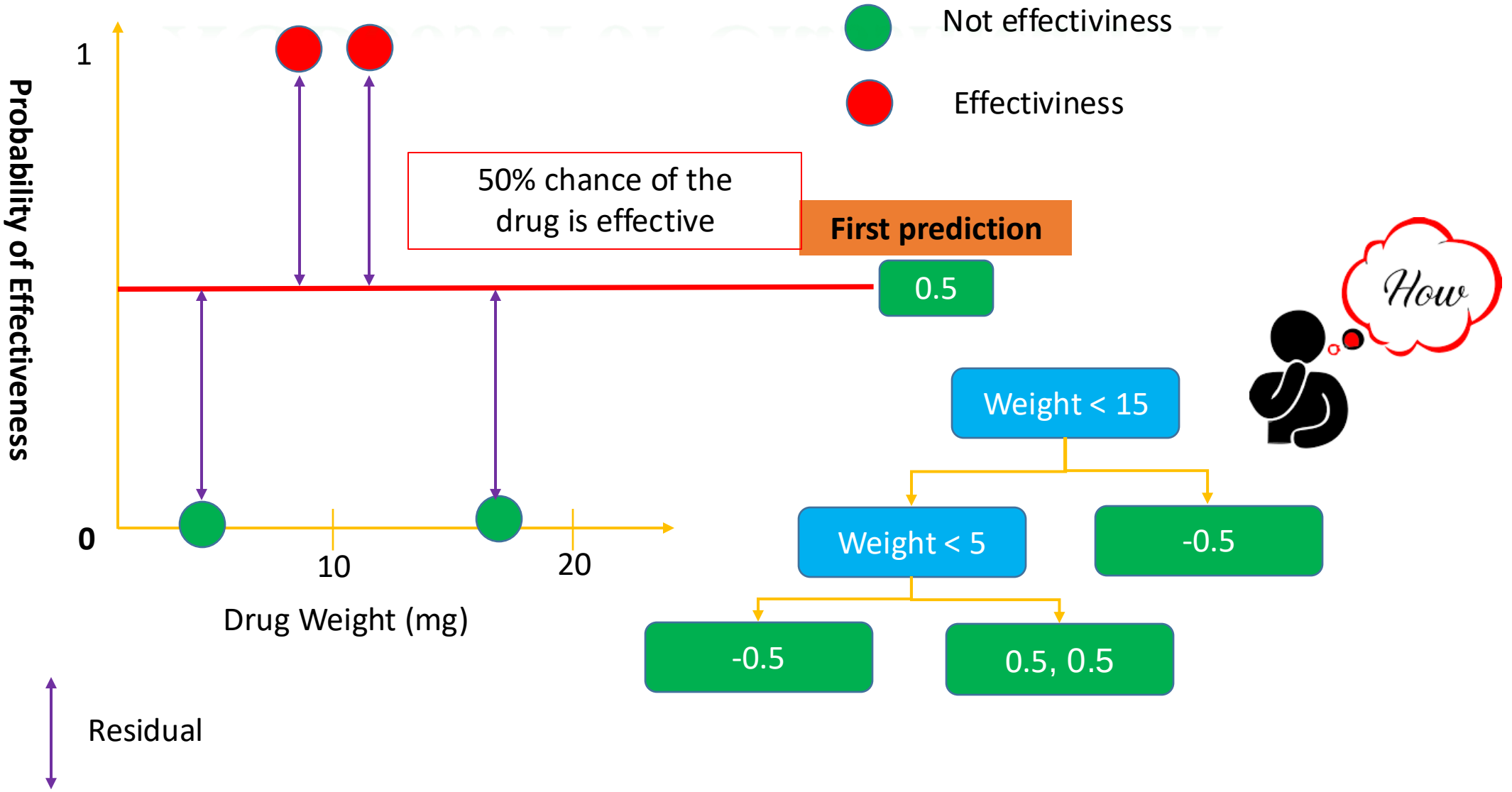
Effectiviness

First prediction

0.5

50% chance of the drug is effective

XGBoost For Classification



XGBoost For Classification

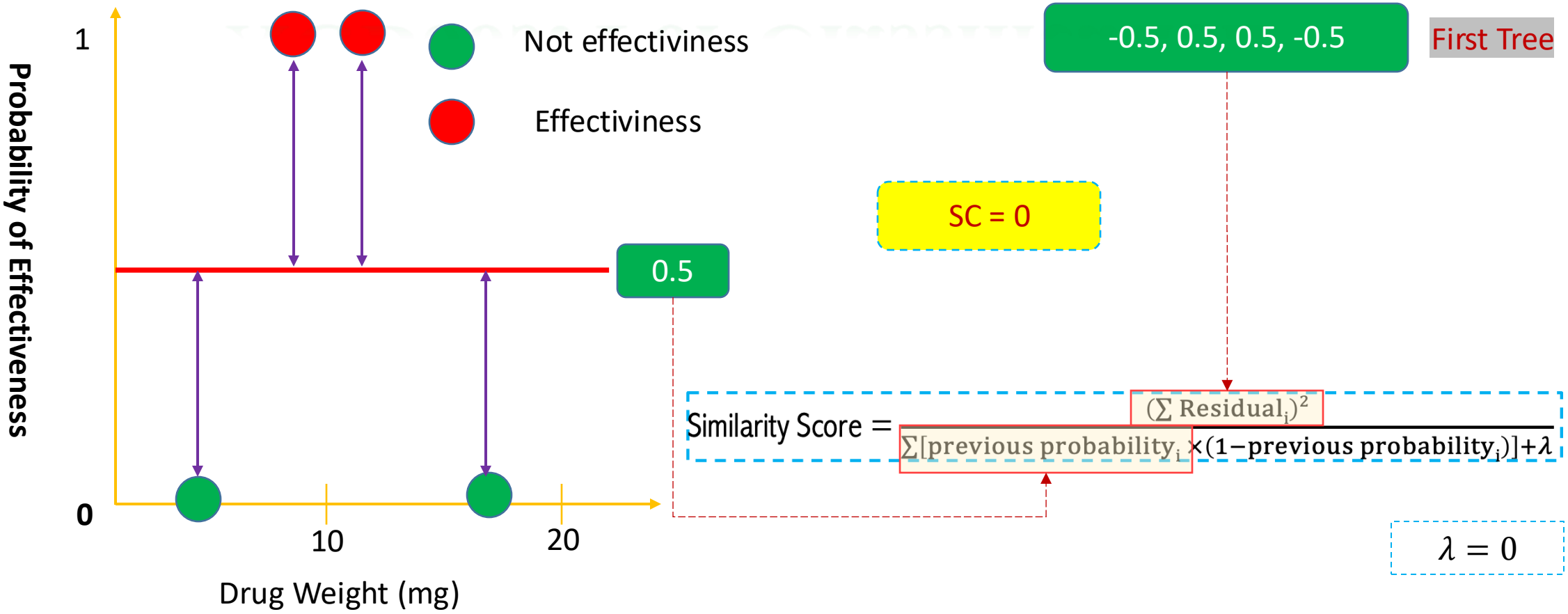
Similarity Score for Classification:

$$\text{Similarity Score} = \frac{(\sum \text{Residual}_i)^2}{\sum[\text{previous probability}_i \times (1 - \text{previous probability}_i)] + \lambda}$$

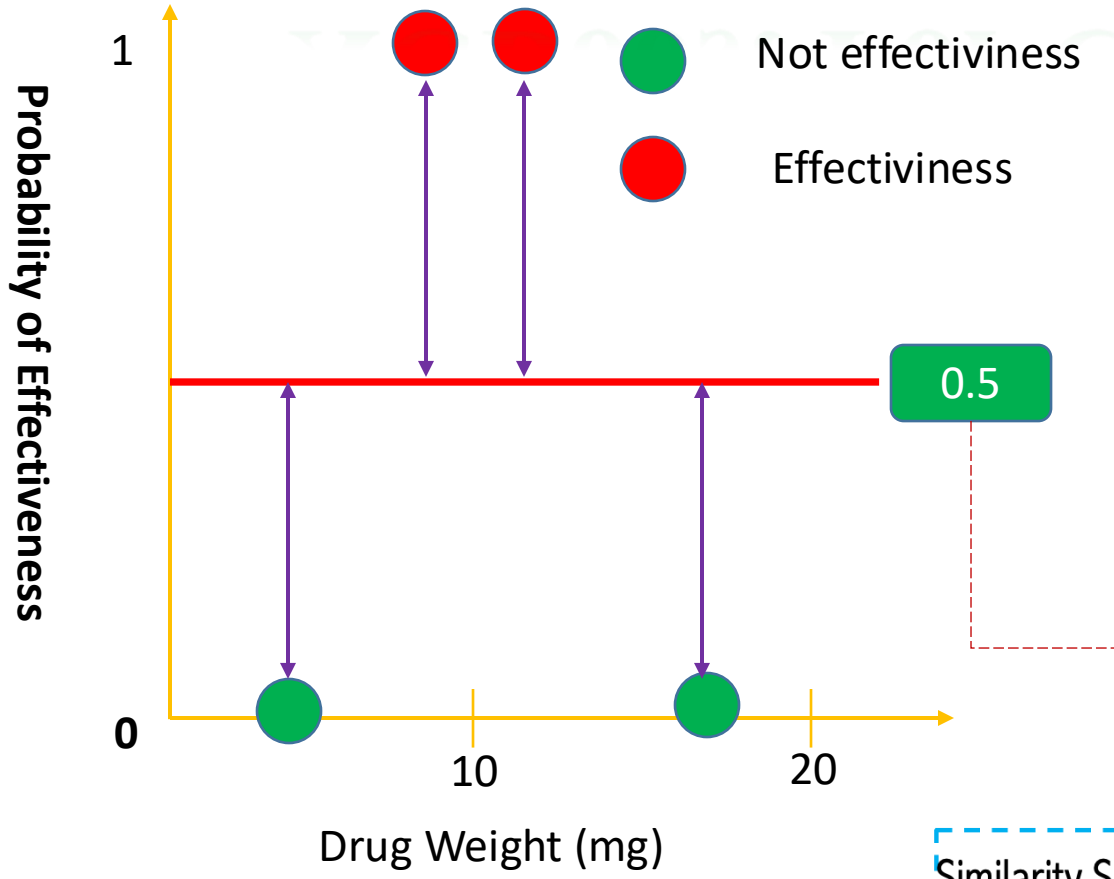
Similarity Score for Prediction (regression):

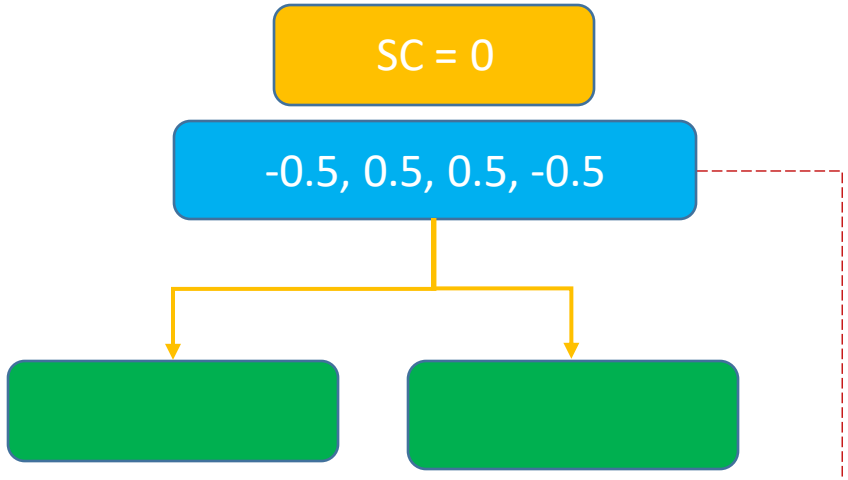
$$\text{Similarity Score} = \frac{(\sum \text{Residual}_i)^2}{\text{number of residual} + \lambda}$$

XGBoost For Classification



XGBoost For Classification





$$\text{Similarity Score} = \frac{(\sum \text{Residual}_i)^2}{\sum[\text{previous probability}_i \times (1 - \text{previous probability}_i)] + \lambda}$$

$\lambda = 0$

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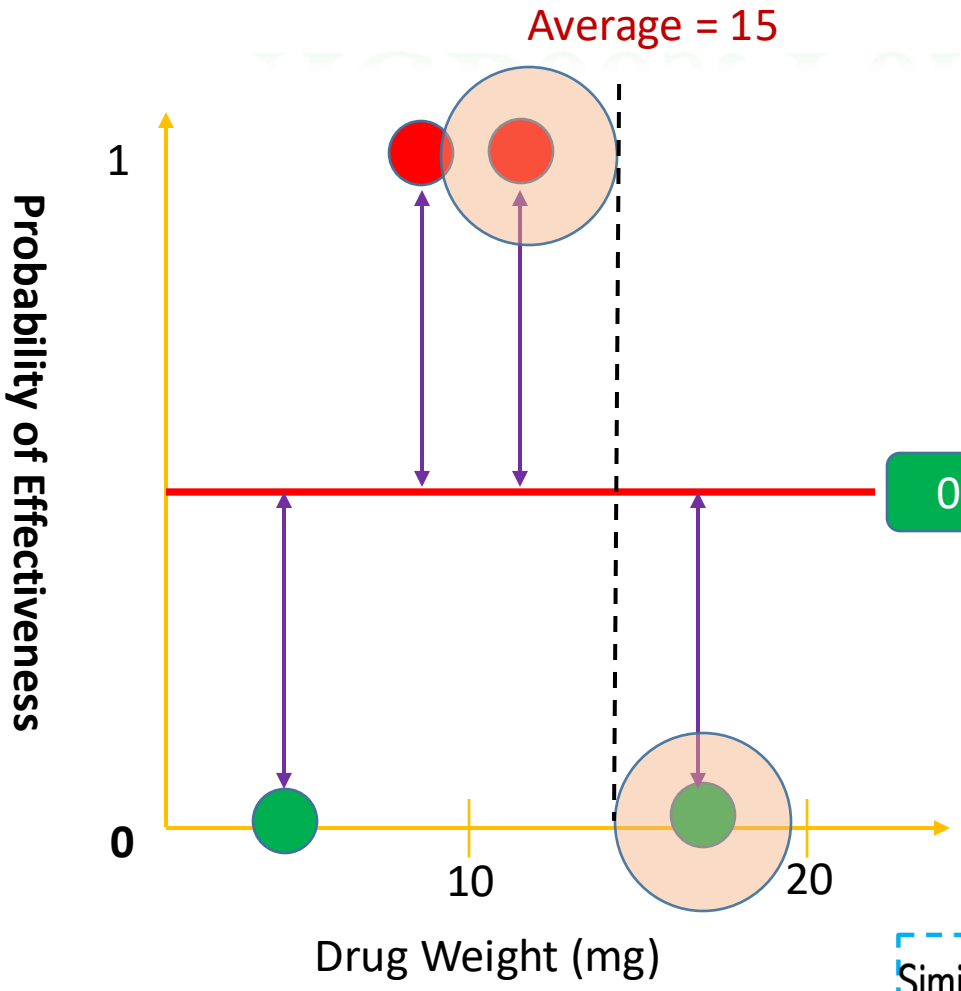
XGBoost For Classification



SC = 0

Weights < 15

-0.5, 0.5, 0.5

-0.5

SC = 0.33

SC = 1

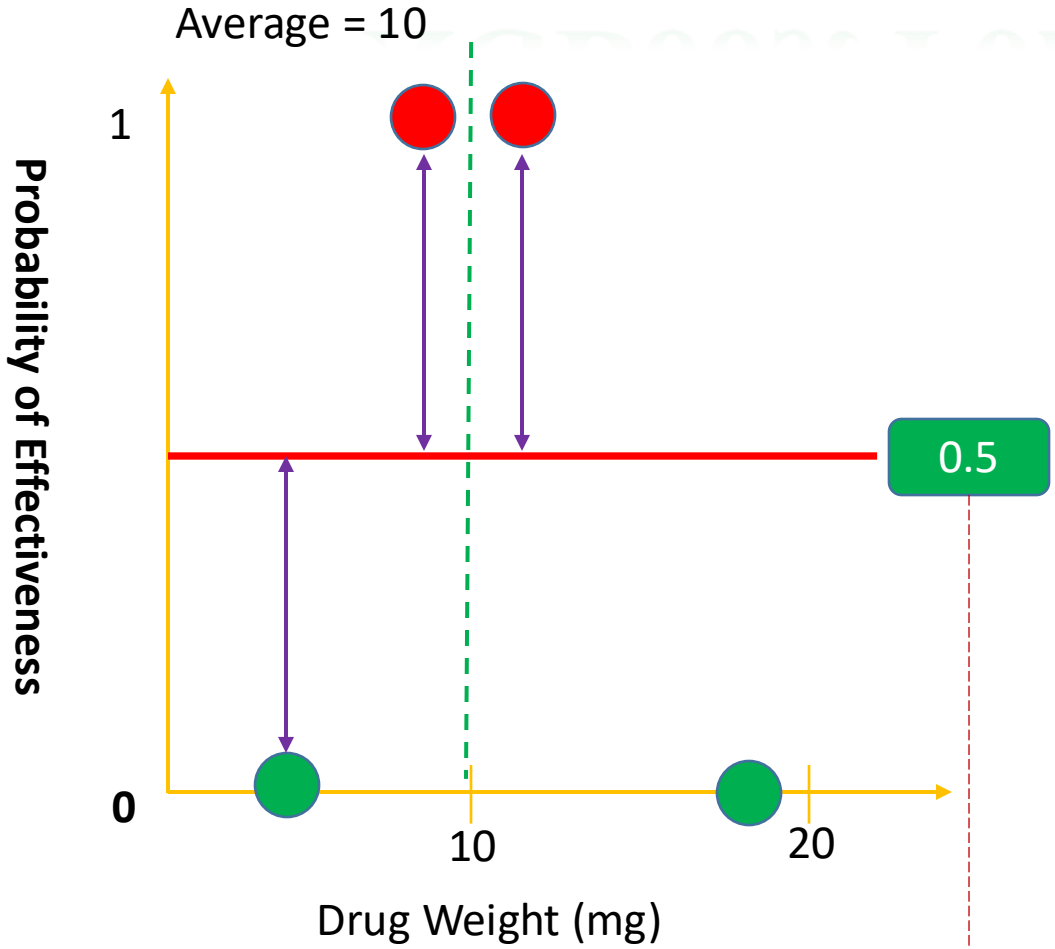
Gain = 0.33 + 1 – 0 =1.33

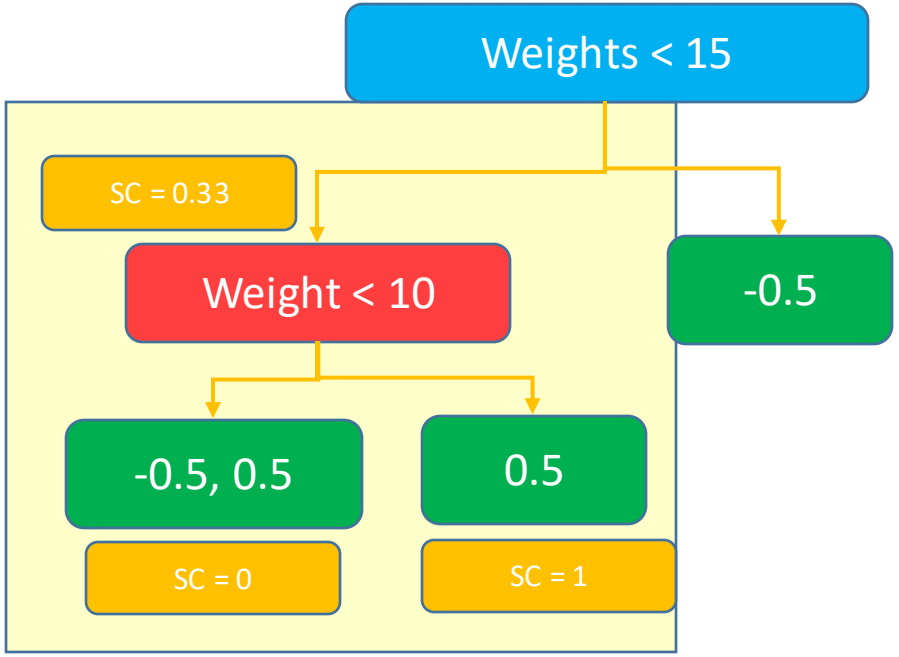
Supposing that weights < 15 is best threshold

$\lambda = 0$

$$\text{Similarity Score} = \frac{(\sum \text{Residual}_i)^2}{\sum[\text{previous probability}_i \times (1 - \text{previous probability}_i)] + \lambda}$$

XGBoost For Classification





Gain = 0 + 1 – 0.33 = 0.66

$\lambda = 0$

$$\text{Similarity Score} = \frac{(\sum \text{Residual}_i)^2}{\sum[\text{previous probability}_i \times (1 - \text{previous probability}_i)] + \lambda}$$

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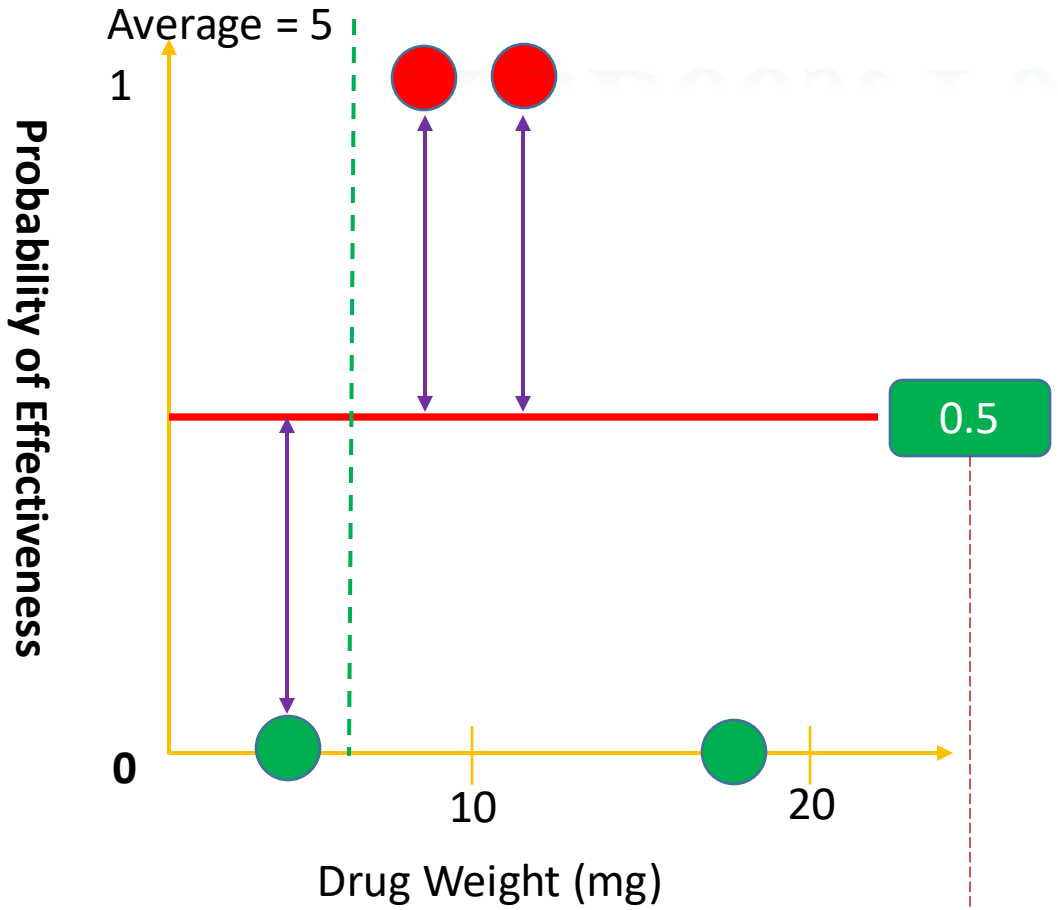
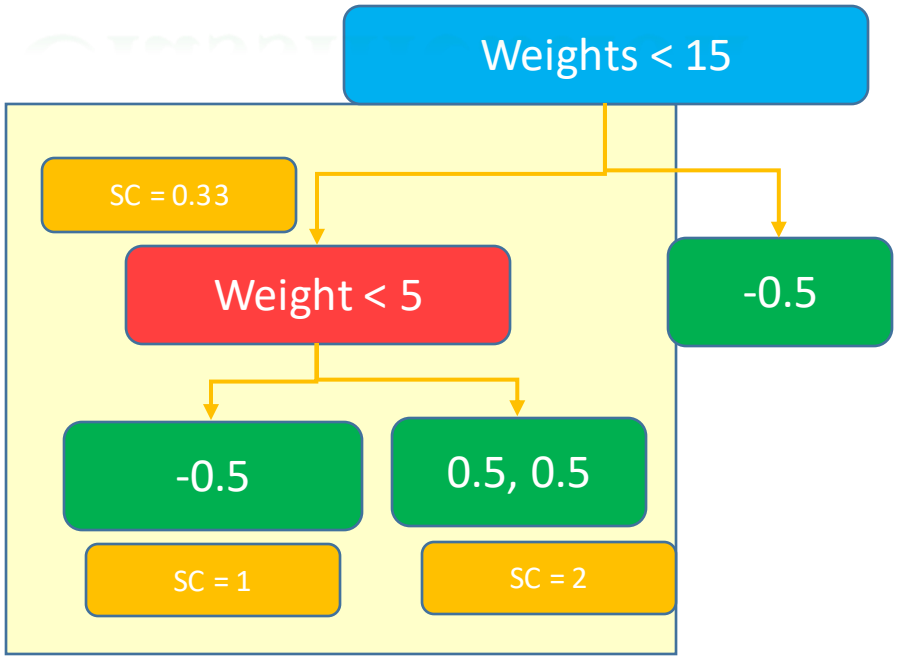
XGBoost For Classification

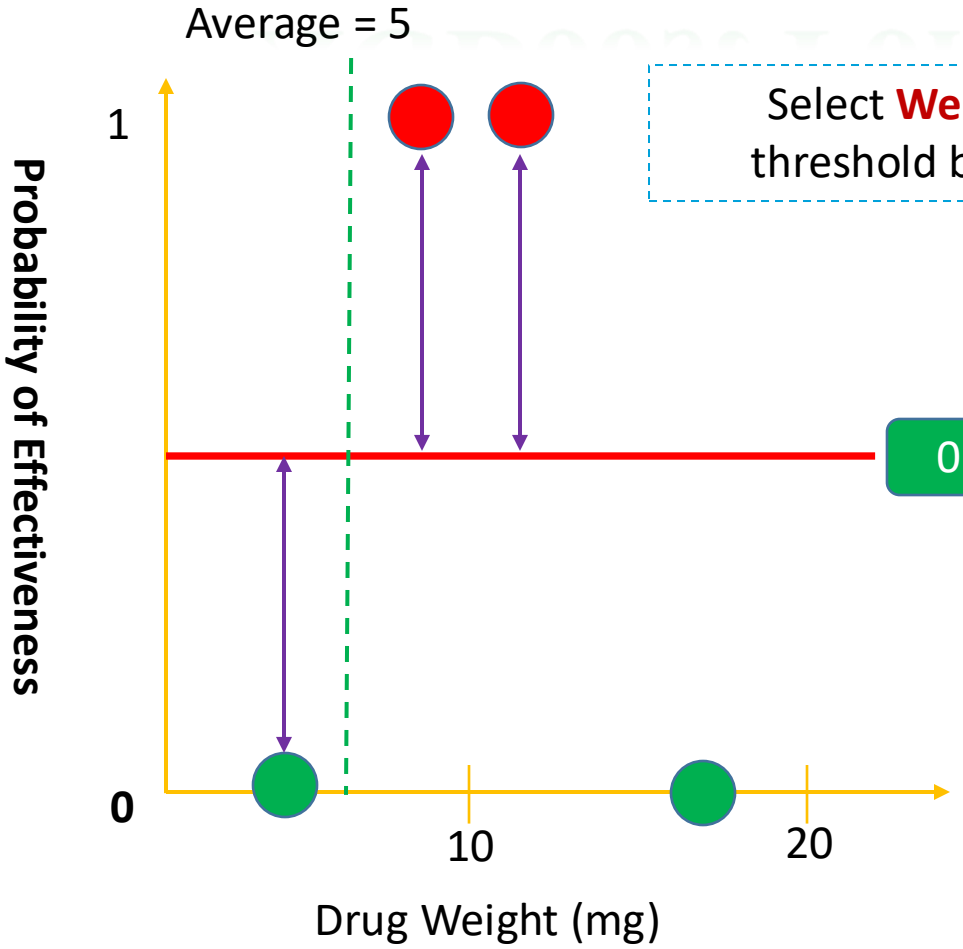




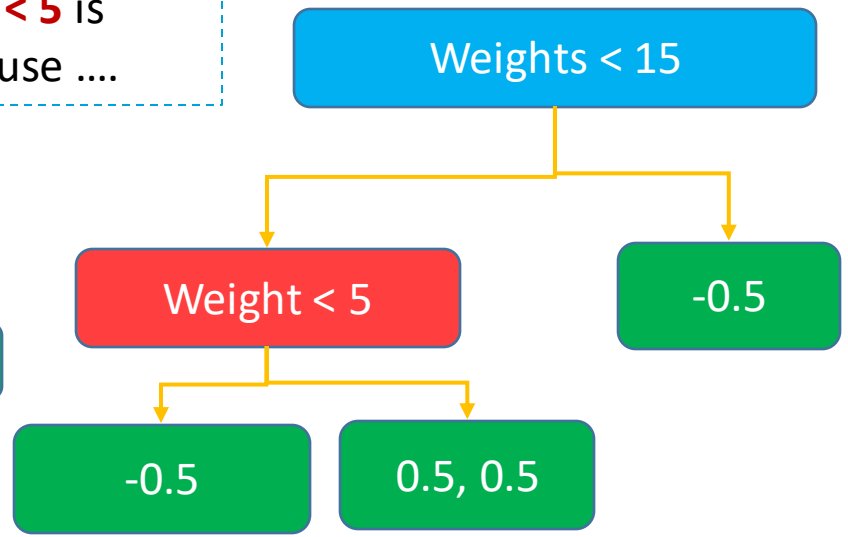
Gain = 1 + 2 – 0.33 = 2.66

$$\text{Similarity Score} = \frac{(\sum \text{Residual}_i)^2}{\sum[\text{previous probability}_i \times (1 - \text{previous probability}_i)] + \lambda}$$

XGBoost For Classification



Select **Weight < 5** is threshold because ....



Giả sử quy định depth level = 2, dừng xây dựng Tree

How to estimate the minimum number of Residuals in each leaf => **XGBoot Cover**

By default: Mininmum XGBoot Cover is set to 1

What is a Cover

Similarity Score for Classification:

$$\text{Similarity Score} = \frac{(\sum \text{Residual}_i)^2}{\sum[\text{previous probability}_i \times (1 - \text{previous probability}_i)] + \lambda}$$

Cover

Similarity Score for Prediction:

$$\text{Similarity Score} = \frac{(\sum \text{Residual}_i)^2}{\text{number of residual} + \lambda}$$

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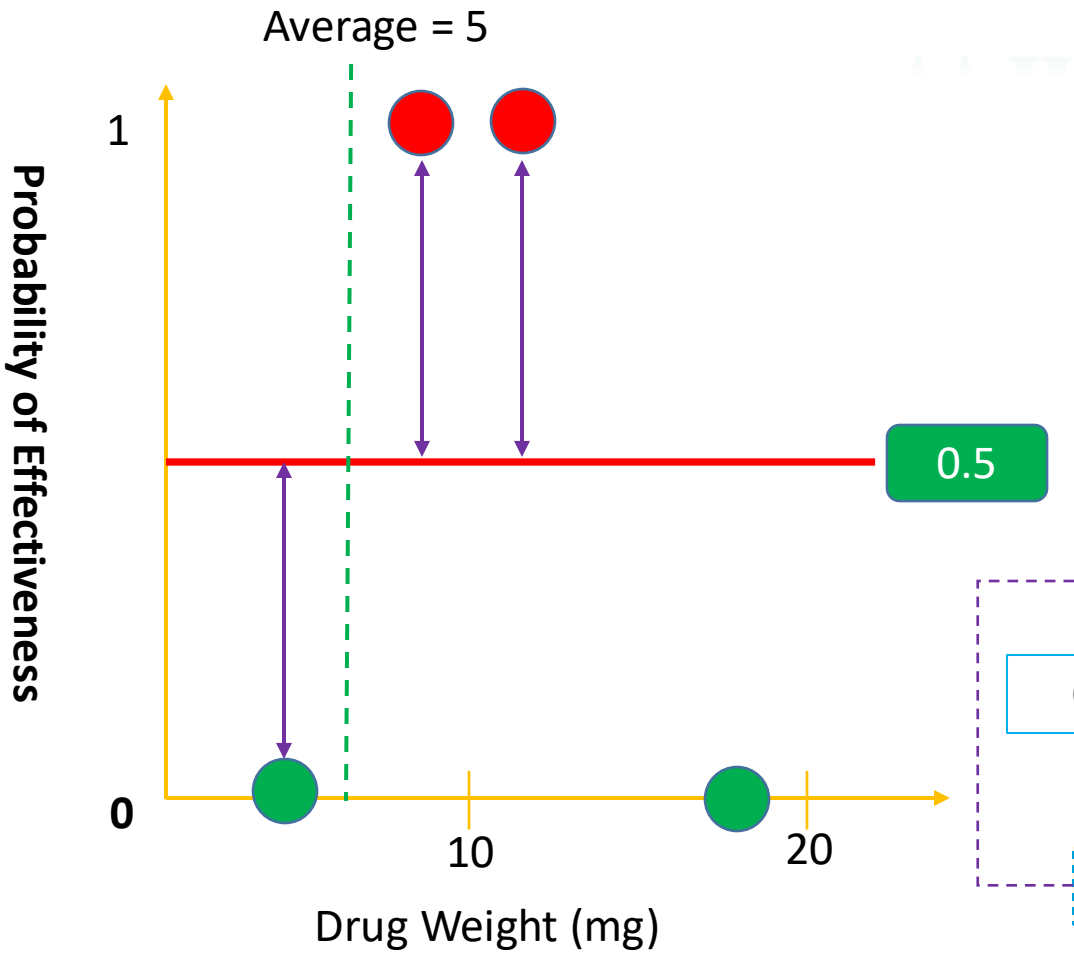
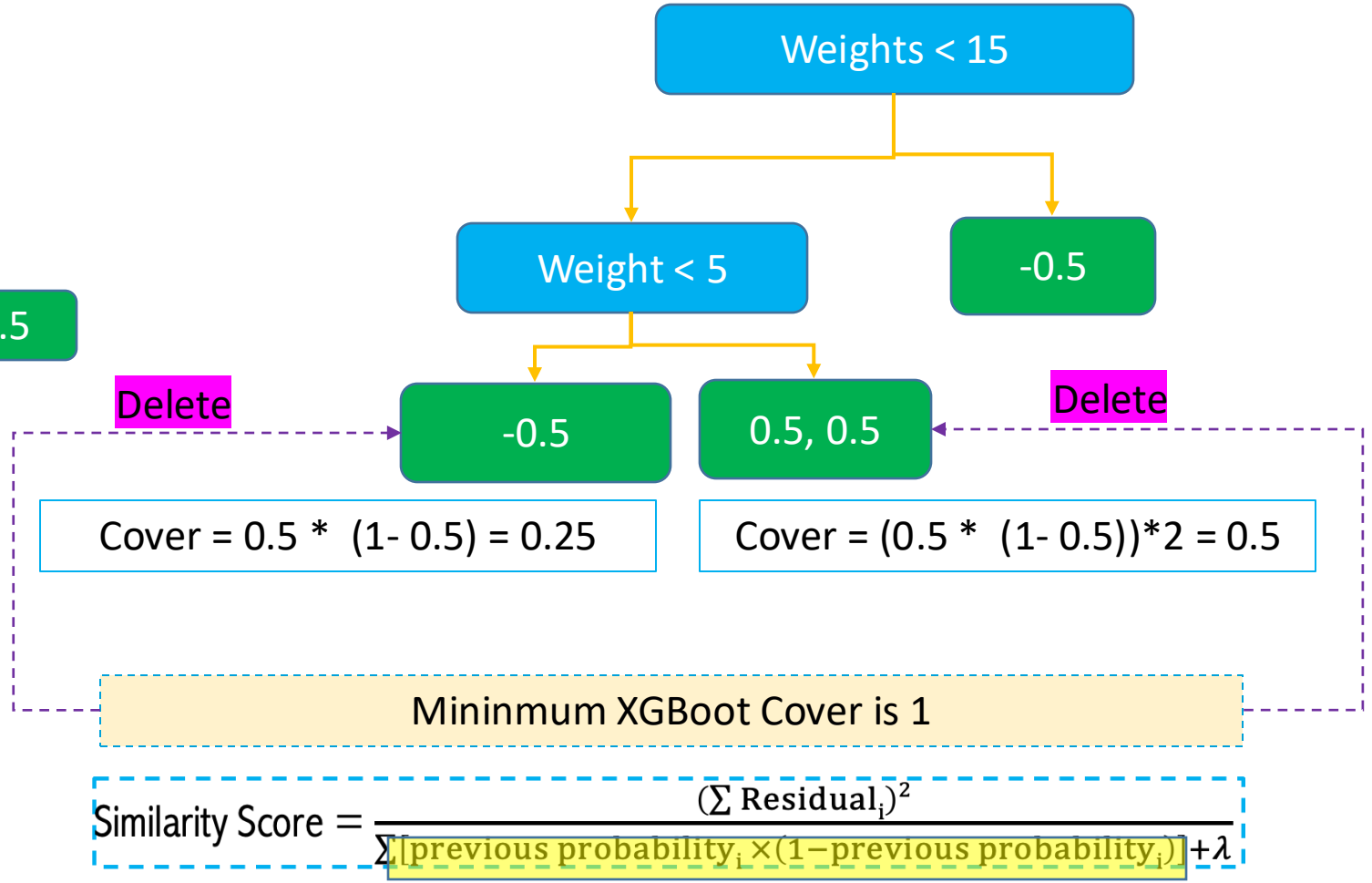
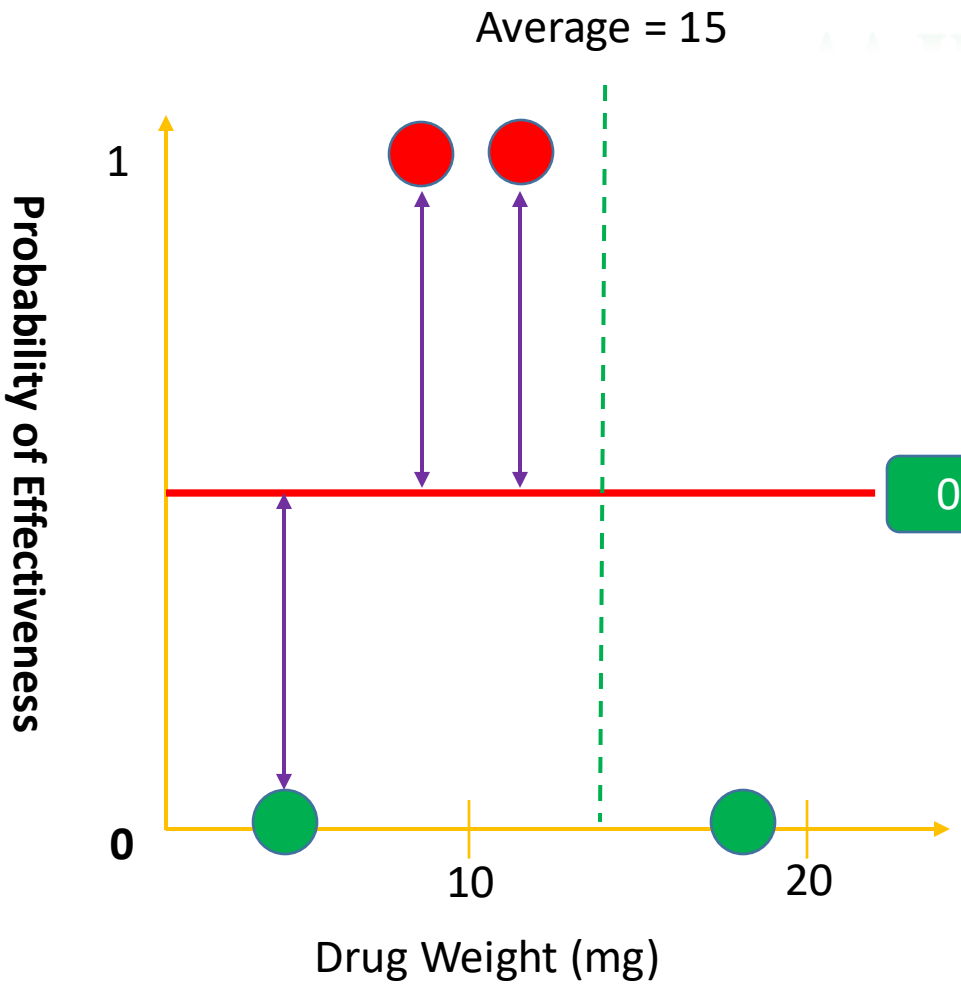
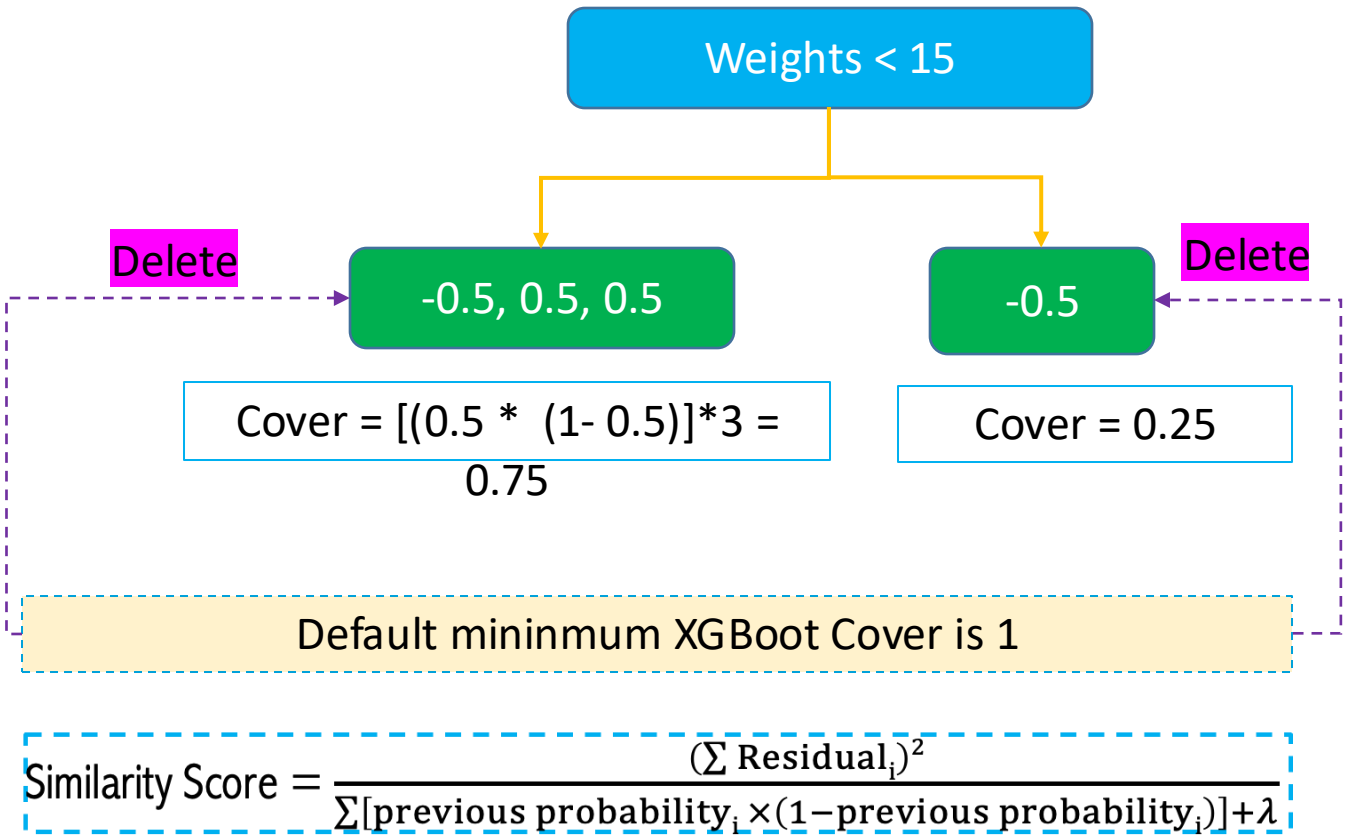
What is a Cover



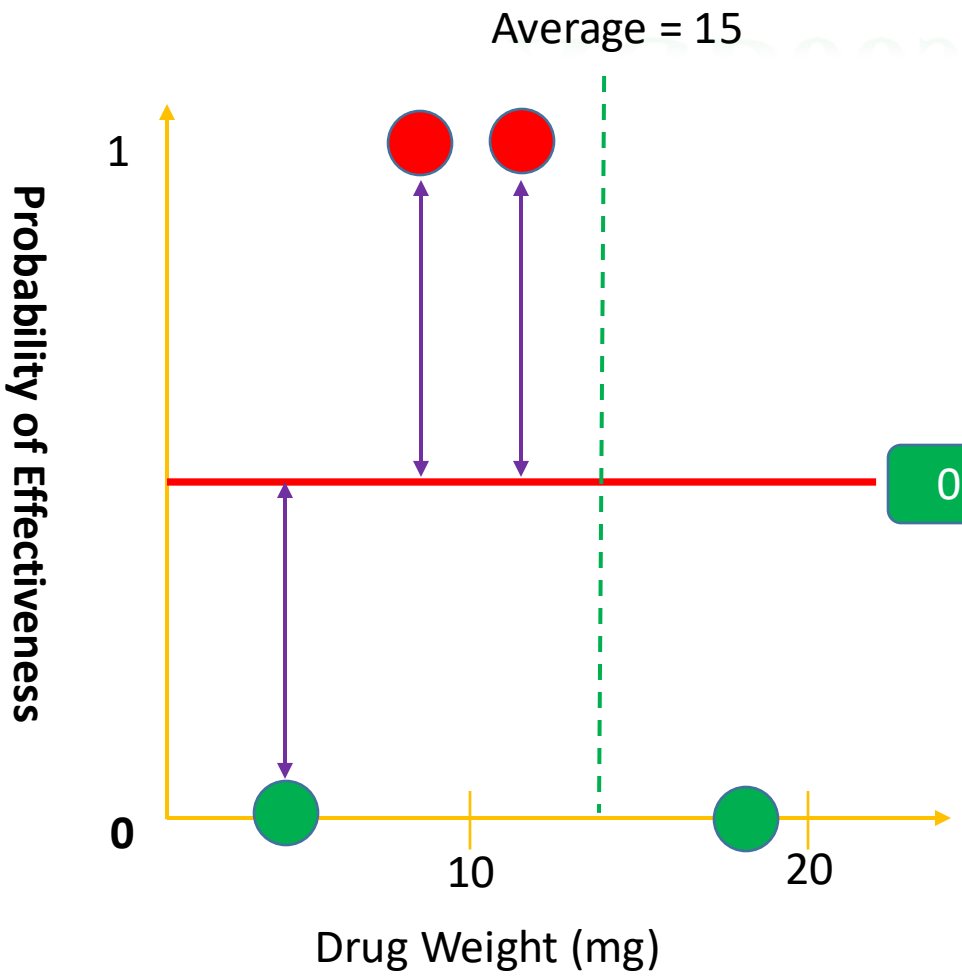


What is a Cover





XGBoost For Classification



Keep this node

-0.5, 0.5, 0.5, -0.5

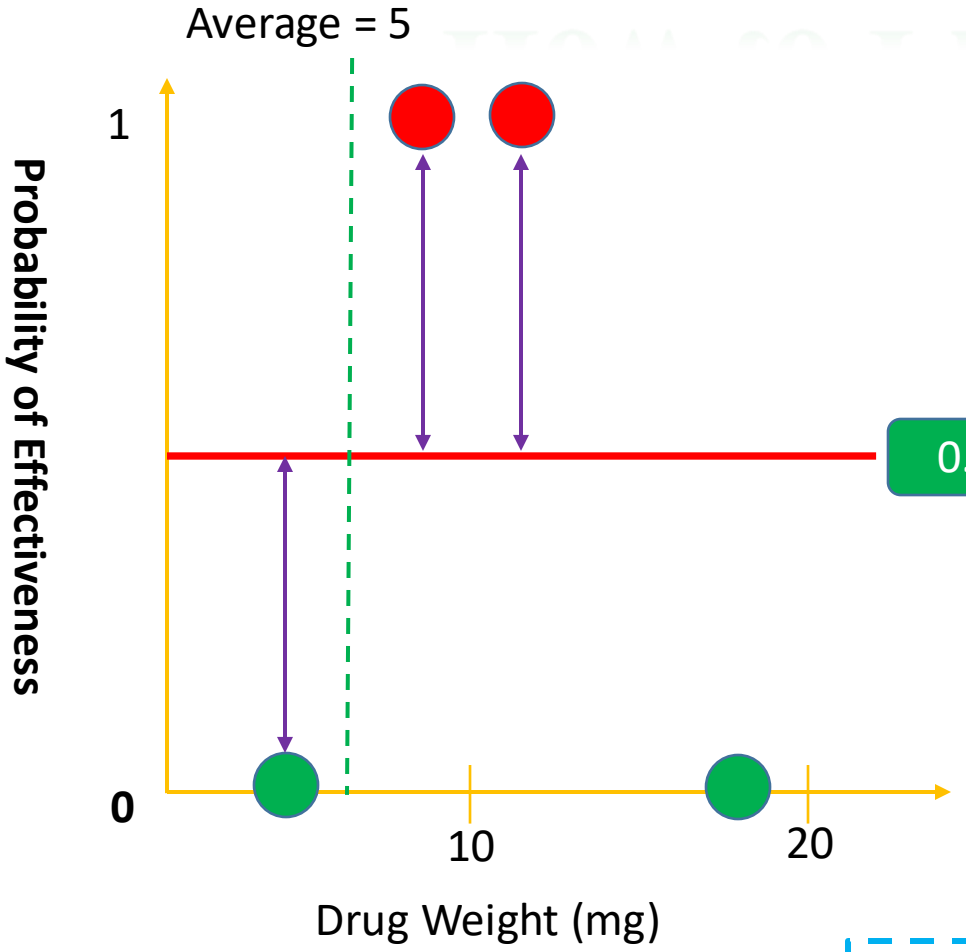
Cover = [(0.5 * (1- 0.5)]*4 = 1

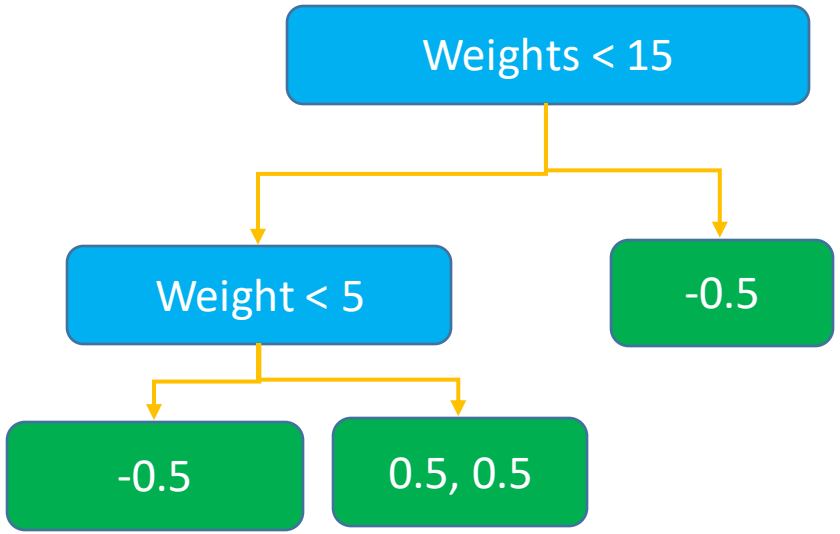
Default mininmum XGBoot Cover is 1

$$\text{Similarity Score} = \frac{(\sum \text{Residual}_i)^2}{\sum[\text{previous probability}_i \times (1-\text{previous probability}_i)]+\lambda}$$

How to Predict the Value

How to predict the value





$$\text{Output Value} = \frac{(\sum \text{Residual}_i)}{\sum[\text{previous probability}_i \times (1 - \text{previous probability}_i)] + \lambda}$$

How to Predict the Value

Drug Weight	Drug Effectiveness
	No
	Yes
	Yes
	No

Initial prediction is that the probability of drug effective is 50%

2 Yes and 2 No => Probablity Yes = 2/4 = 1/2 = 0.5

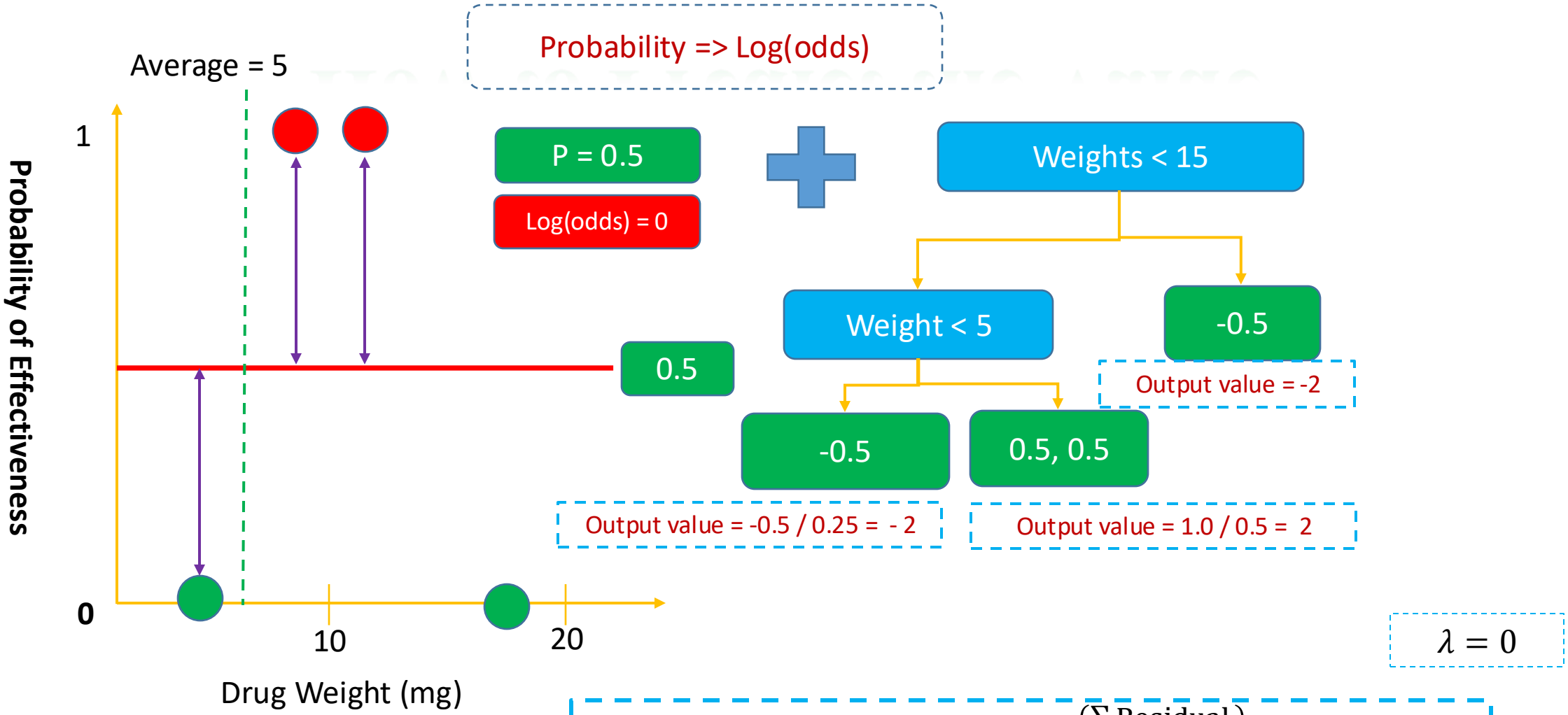
$$\text{Log(odds)} = \log\left(\frac{\text{Probablity Yes}}{\text{Probablity No}}\right) = 0$$

In XGBoost (or Gradient Boost), the initial prediction is that the log(odds)

$$\text{Probability of Drug Effectiveness} = \frac{e^{\log(\text{odds})}}{1 + e^{\log(\text{odds})}}$$

$$\text{Probability of Drug Effectiveness} = \frac{e^{0}}{1+e^{0}} = 0.5$$

How to Predict the Value



$$\lambda = 0$$

$$\text{Output Value} = \frac{(\sum \text{Residual}_i)}{\sum[\text{previous probability}_i \times (1 - \text{previous probability}_i)] + \lambda}$$

Tranformation formular for getting value at a leaf.

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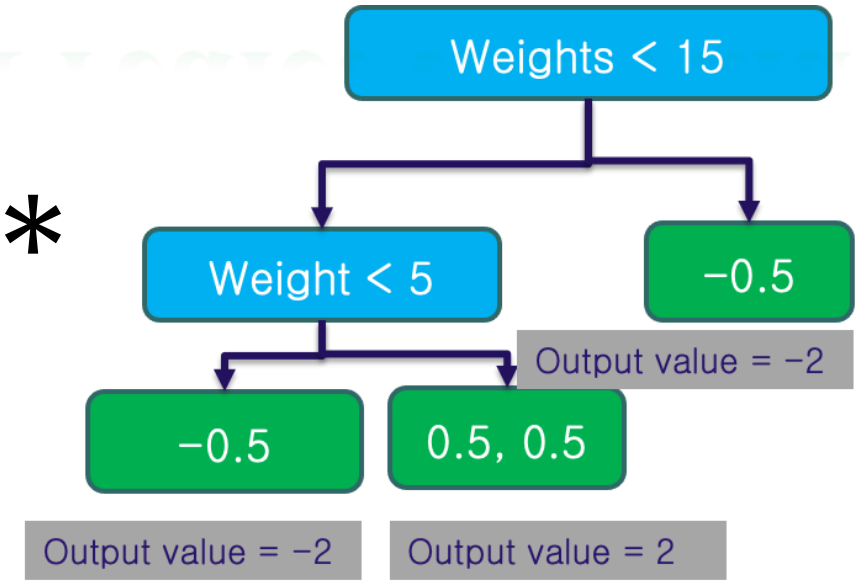
How to Predict the Value

Probability => Log(odds)

P = 0.5

Log(odds) = 0

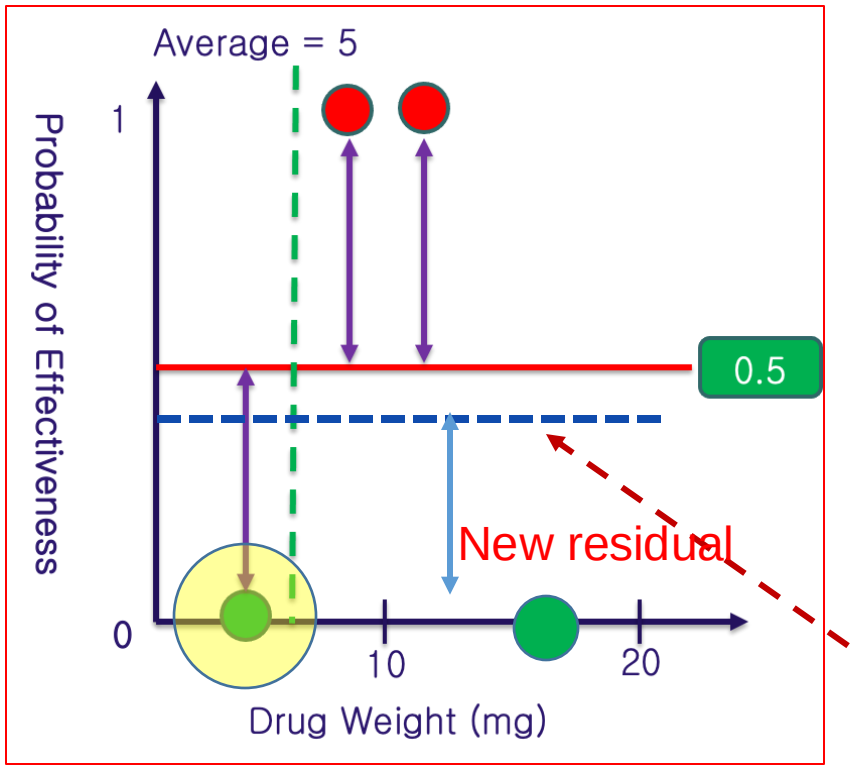
α *



$\frac{p}{1-p}$ = odds

$\text{Log}(\frac{p}{1-p})$ = log(odds)

α = 0.3



Prediction = 0 + 0.3 * (-2) = -0.6

$$\text{Probability} = \frac{e^{\log(odds)}}{1 + e^{\log(odds)}}$$

$$\text{Probability} = \frac{e^{-0.6}}{1+e^{-0.6}} = 0.35$$

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How to Predict the Value

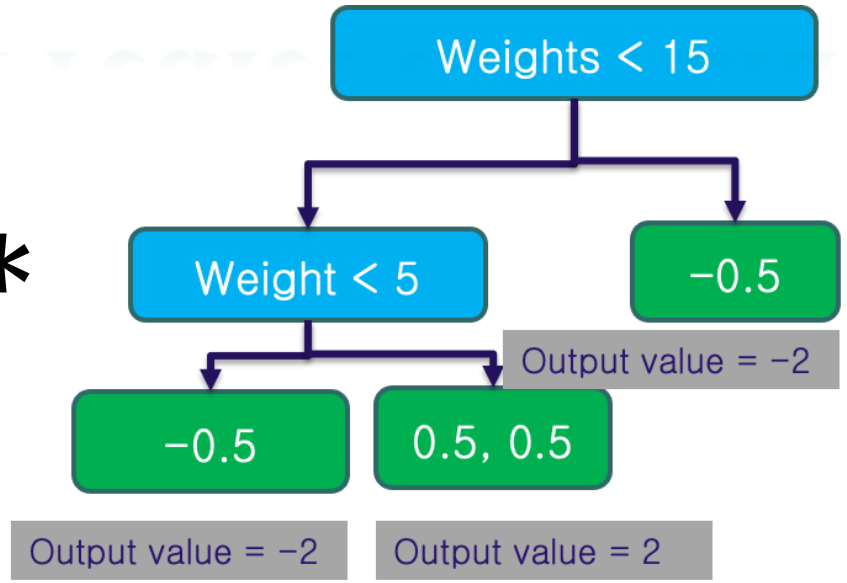
Probability => Log(odds)

Can we change P?

P = 0.5

Log(odds) = 0

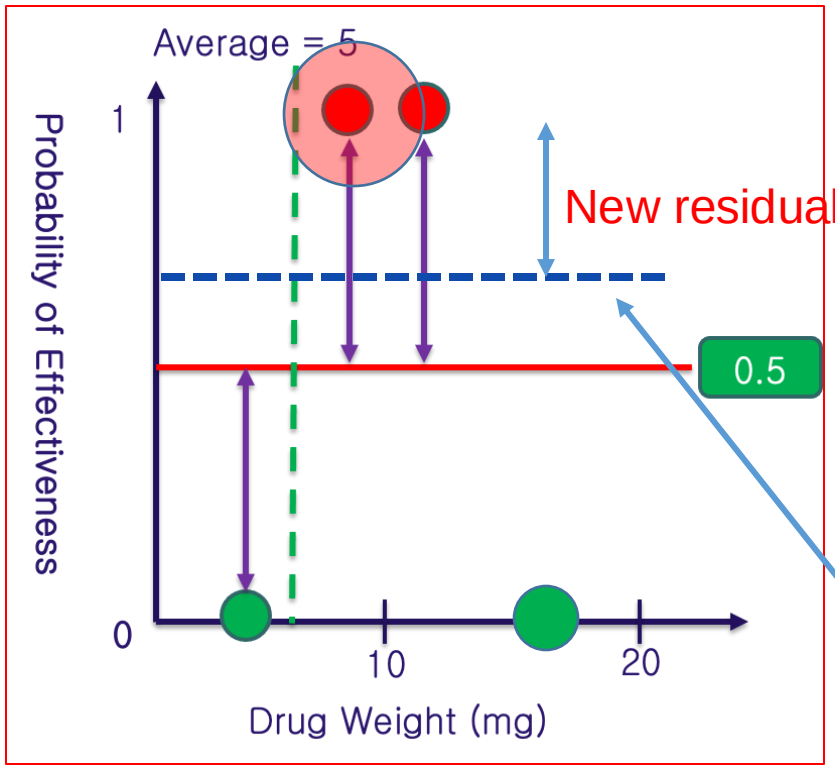
α *



$\frac{p}{1-p}$ = odds

$\text{Log}(\frac{p}{1-p})$ = log(odds)

α = 0.3



Log(odds) = Prediction = 0 + 0.3 * (2) = 0.6

$$\text{Probability} = \frac{e^{\log(odds)}}{1 + e^{\log(odds)}}$$

$$\text{Probability} = \frac{e^{0.6}}{1+e^{0.6}} = 0.65$$

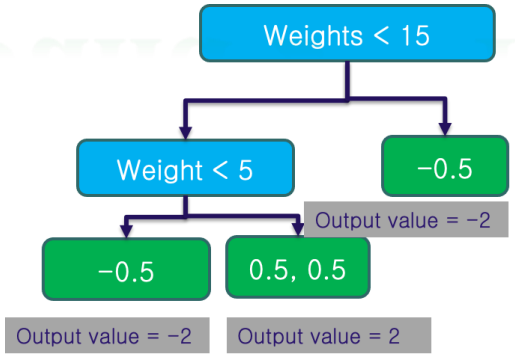
Build 2nd Tree

Probability => Log(odds)

P = 0.5

Log(odds) = 0

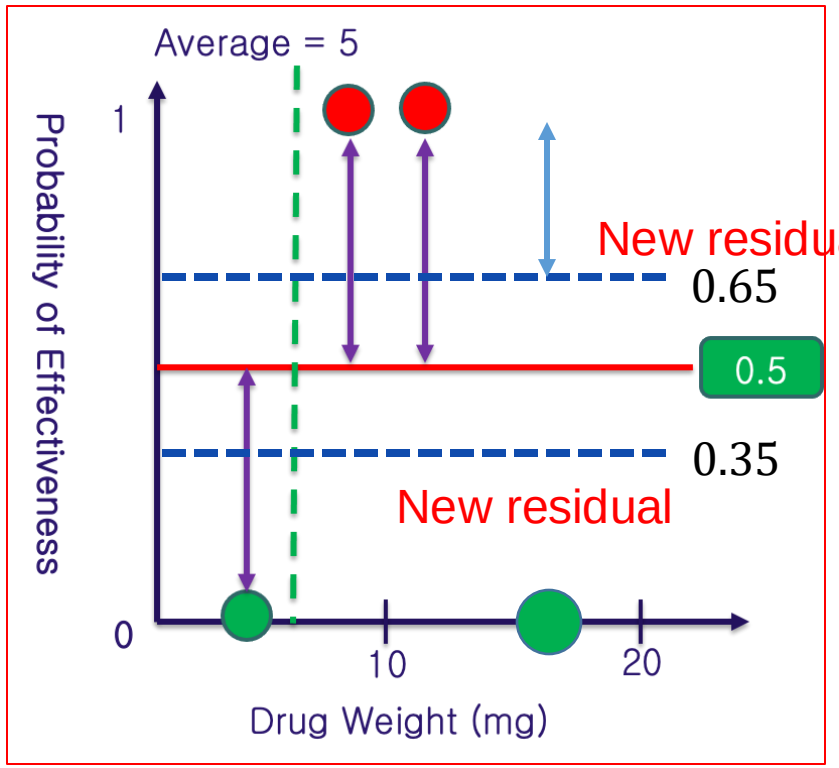
$\propto *$



$+$

$\propto *$

-0.35, 0.35, 0.35, -0.35



Similarity Score = $\dfrac{(-0.35 + 0.35 + 0.35 - 0.35)^2}{0.35\times(1-0.35)+0.65\times(1-0.65)+0.65\times(1-0.65)+0.35\times(1-0.35)}$

Similarity Score = $\dfrac{(\sum \text{Residual}_i)^2}{\sum[\text{previous probability}_i \times(1-\text{previous probability}_i)]+\lambda}$

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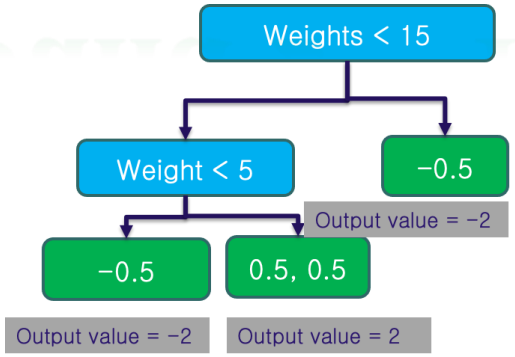
Build 2nd Tree

Probability => Log(odds)

P = 0.5

Log(odds) = 0

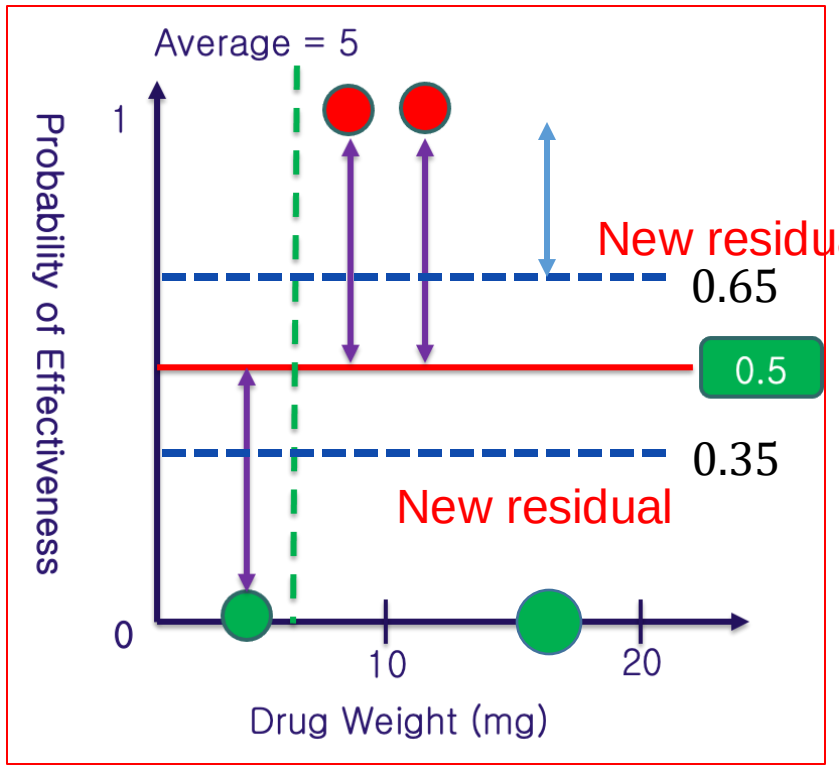
$\propto *$



$+$

$\propto *$

-0.35, 0.35, 0.35, -0.35



$$\text{Output Score} = \frac{(-0.35 + 0.35 + 0.35 - 0.35)}{0.35\times(1-0.35)+0.65\times(1-0.65)+0.65\times(1-0.65)+0.35\times(1-0.35)+\lambda}$$

$$\text{Output Score} = \frac{(\sum \text{Residual}_i)}{\sum[\text{previous probability}_i \times(1-\text{previous probability}_i)]+\lambda}$$

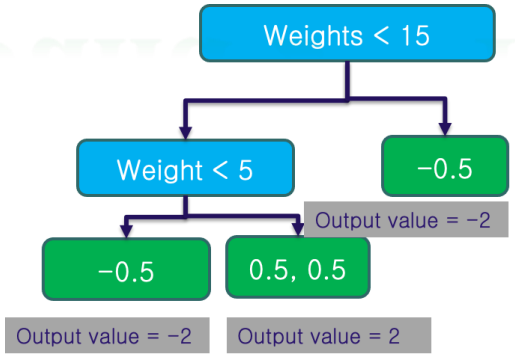
Build 2nd Tree

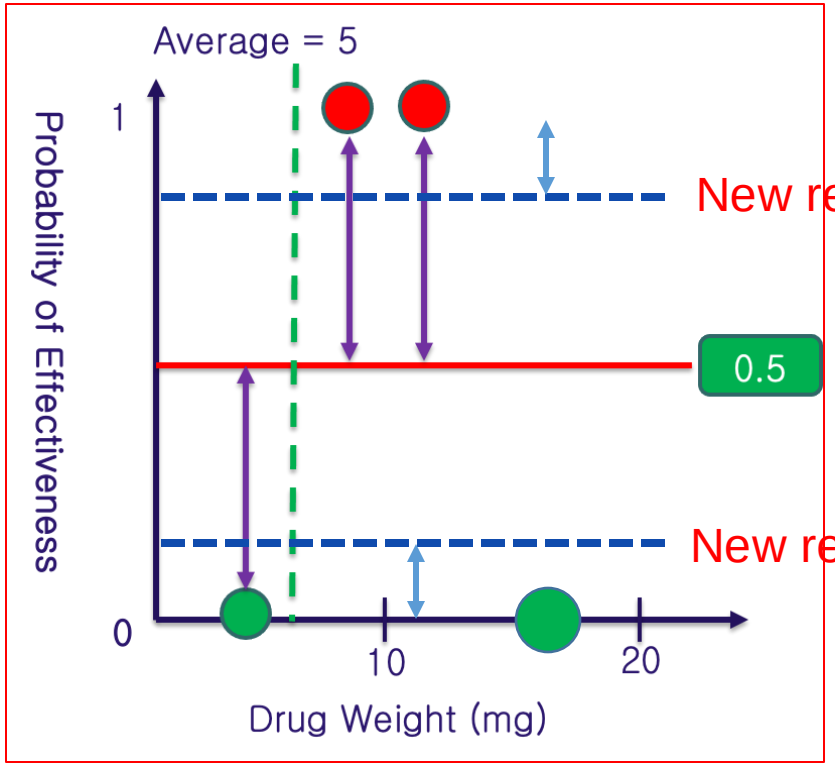
Probability => Log(odds)

P = 0.5

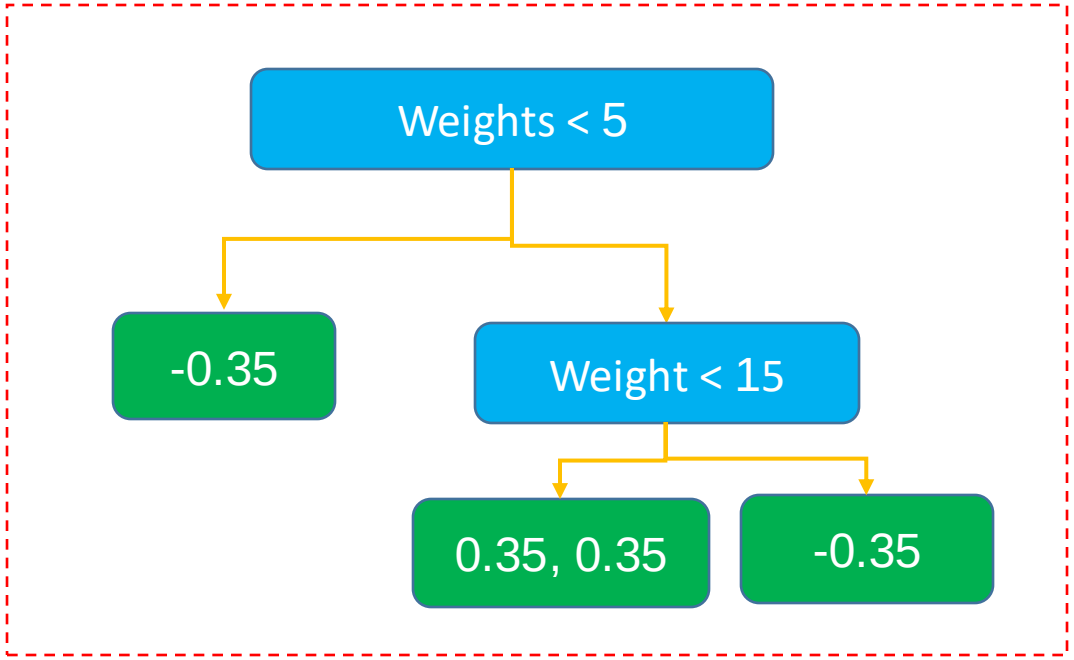
Log(odds) = 0

$\propto *$





$\propto *$



Q & A

1. When do you stop to build the Tree

2. What's happen when $\lambda > 0$

$$\text{Similarity Score} = \frac{(\sum \text{Residual}_i)^2}{\sum[\text{previous probability}_i \times (1 - \text{previous probability}_i)] + \lambda}$$

Outline

- Regularization
- Regression XGBoost
- Classification XGBoost
- XGBoost: Clearly Explain
- Time Series Example
- Summary

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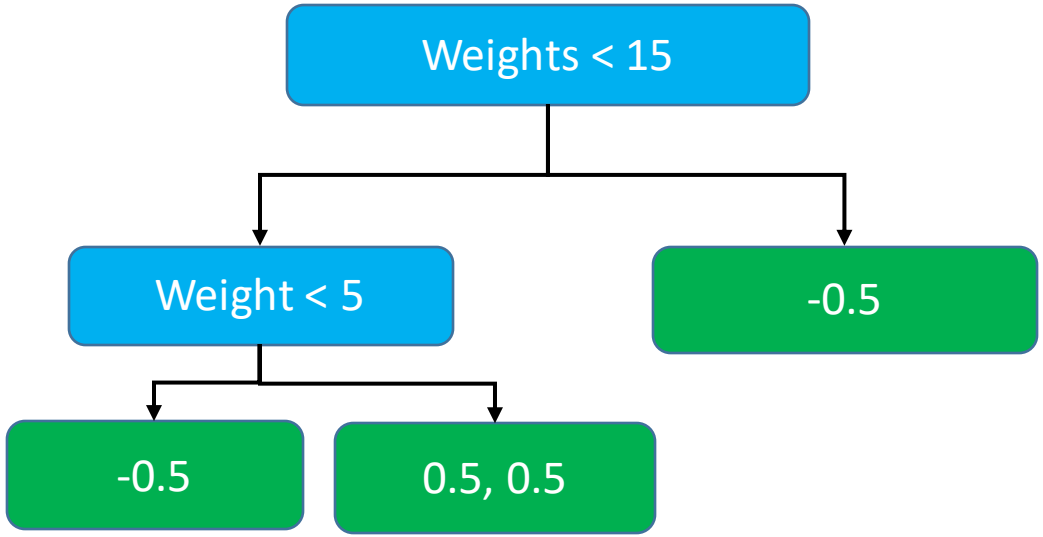
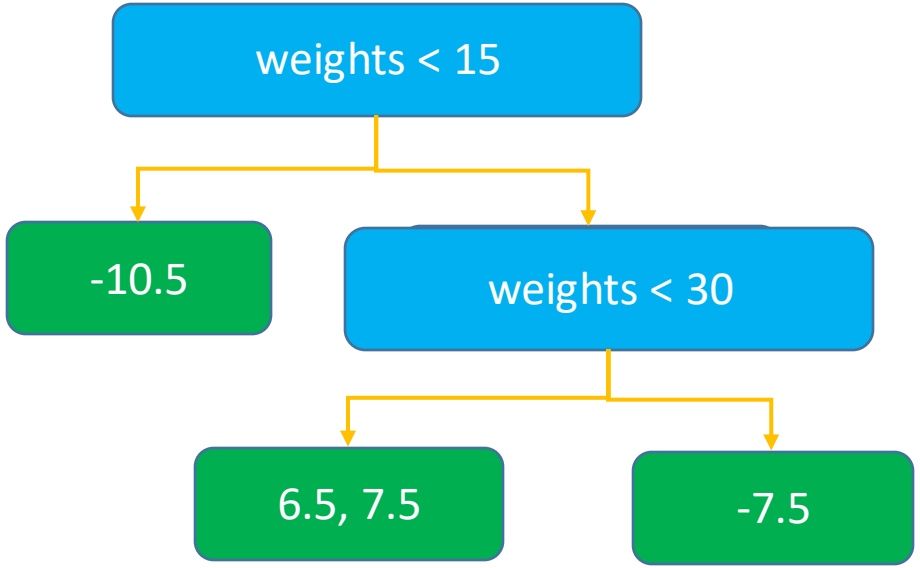
XGBoost: Behind The Scenes

Classification



Regression



$$\text{Similarity Score} = \frac{(\sum \text{Residual})^2}{\sum \bar{y}_i \times (1 - \bar{y}_i) + \lambda}$$

$$\text{Ouput value} = \frac{(\sum \text{Residual})}{\sum \bar{y}_i \times (1 - \bar{y}_i) + \lambda}$$

$$\text{Similarity Score} = \frac{(\sum \text{Residual})^2}{\text{Number of Residual} + \lambda}$$

$$\text{Output Value} = \frac{(\sum \text{Residual})}{\text{Number of Residual} + \lambda}$$

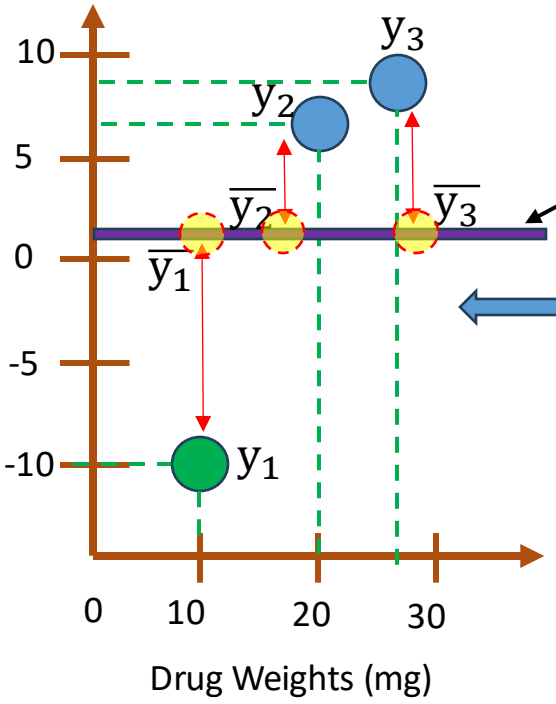
XGBoost: Behind The Scenes

Regression

Dự đoán ban đầu hiệu quả thuốc

0.5

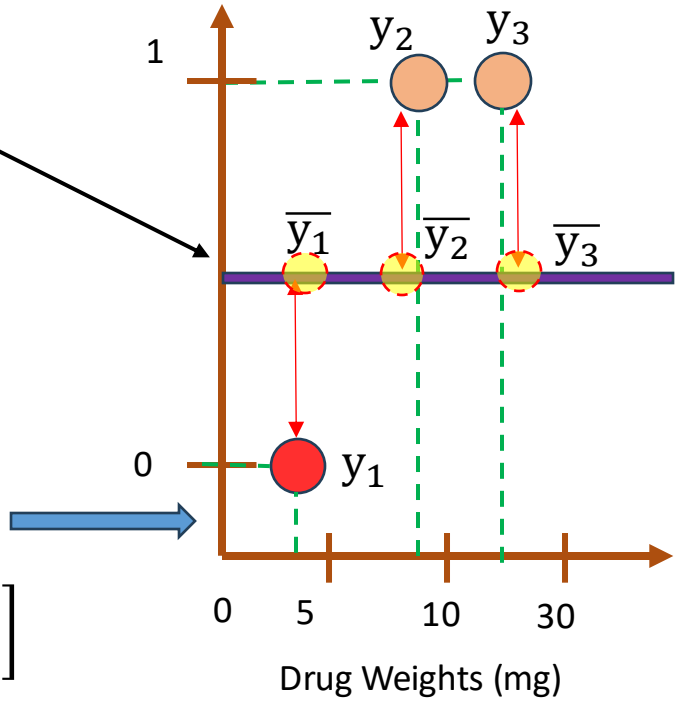
Classificati on



$$\sum_{i=1}^{3} \mathcal{L}(y_i, \overline{y_i}) \quad \text{Where} \quad \mathcal{L}(y_i, \overline{y_i}) = \frac{1}{2}(y_i - \overline{y_i})^2$$

$$\sum_{i=1}^{3} \mathcal{L}(y_i, \overline{y_i}) \quad \text{Where}$$

$$\mathcal{L}(y_i, \overline{y_i}) = -\begin{bmatrix} y_i \log(\overline{y_i}) + \\ (1 - y_i)\log(1 - \overline{y_i}) \end{bmatrix}$$



Sử dụng loss functions xây dựng cây

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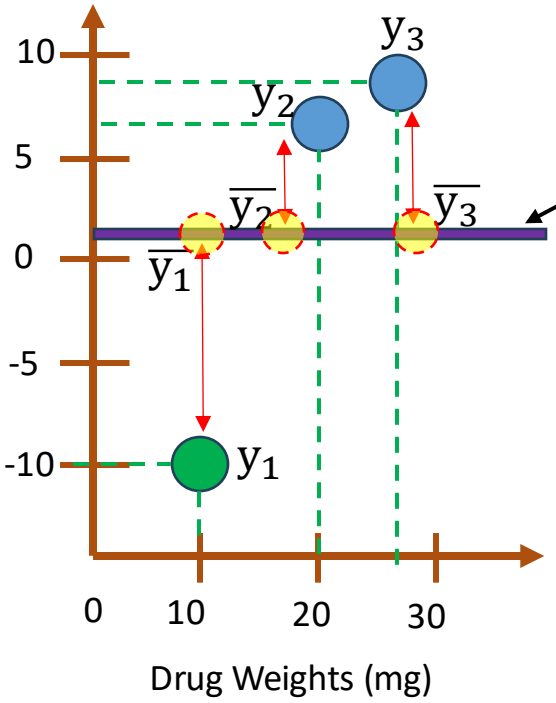
XGBoost: Behind The Scenes

Regression

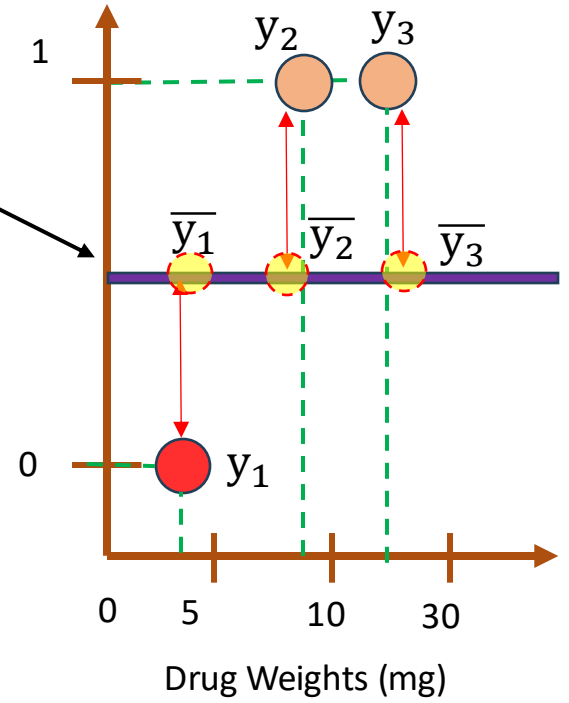
Dự đoán ban đầu hiệu quả thuốc

0.5

Classification



$$\sum_{i=1}^{n} \mathcal{L}(y_i, \overline{y_i}) + \gamma T + \lambda P^2$$



γ is a user definable penalty to encourse pruning

XGBoost can prune even when $\gamma = 0$

Pruning is excuted after the full tree built
=> It plays no role in deriving the Optimal

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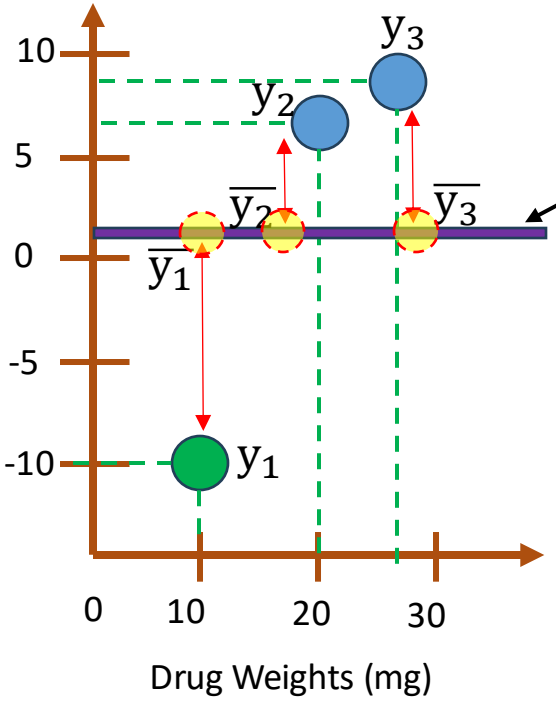
XGBoost: Behind The Scenes

Regression

Dự đoán ban đầu hiệu quả thuốc

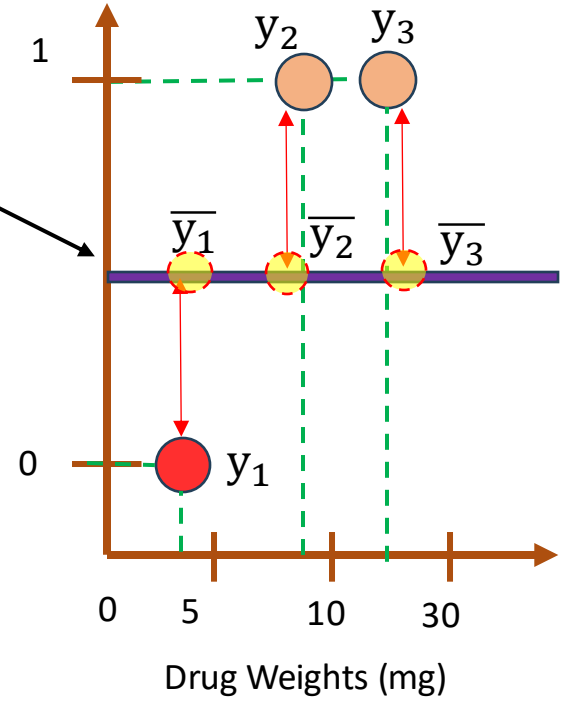
0.5

Classificati on



Bỏ qua γ

$$\sum_{i=1}^{n} \mathcal{L}(y_i, \overline{y_i}) + \gamma T + \lambda P^2$$



γ is a user definable penalty to encourse pruning

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Pruning is excuted after the full tree built => It plays no role in deriving the Optimal

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XGBoost: Behind The Scenes

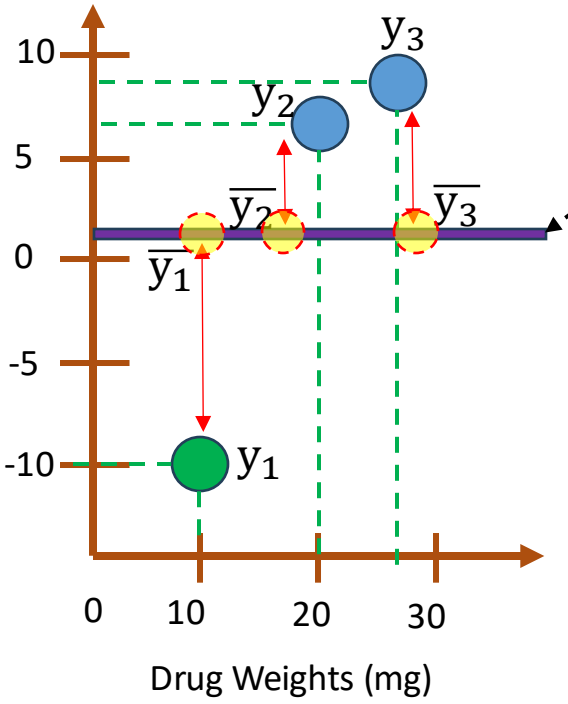
Regression

Dự đoán ban đầu hiệu quả thuốc

0.5

1st New Tree

Classification

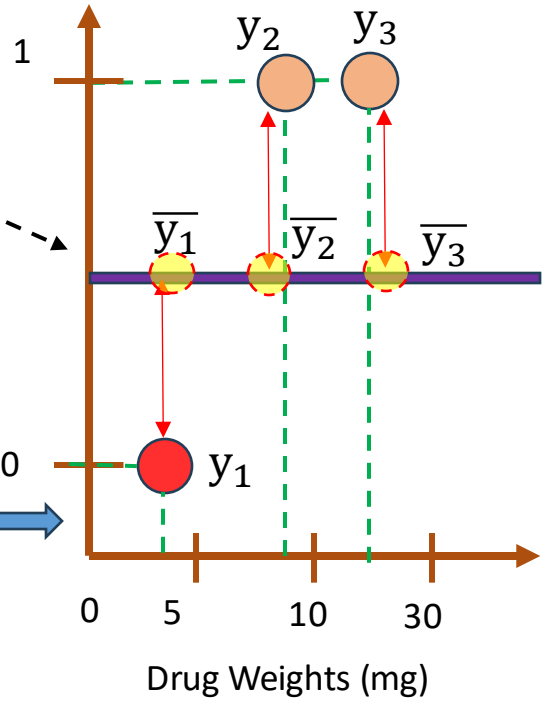


XGBoost builds the new tree based on the loss function:

$$\sum_{i=1}^{n} \mathcal{L}\left(y_i, \overline{y_i^0} + P\right) + \frac{1}{2}\lambda P^2$$

Rigde Regression Regularization term

Mục tiêu: tìm giá trị dự đoán cho mỗi leaf (P) của cây mới nhằm minimize hàm loss.



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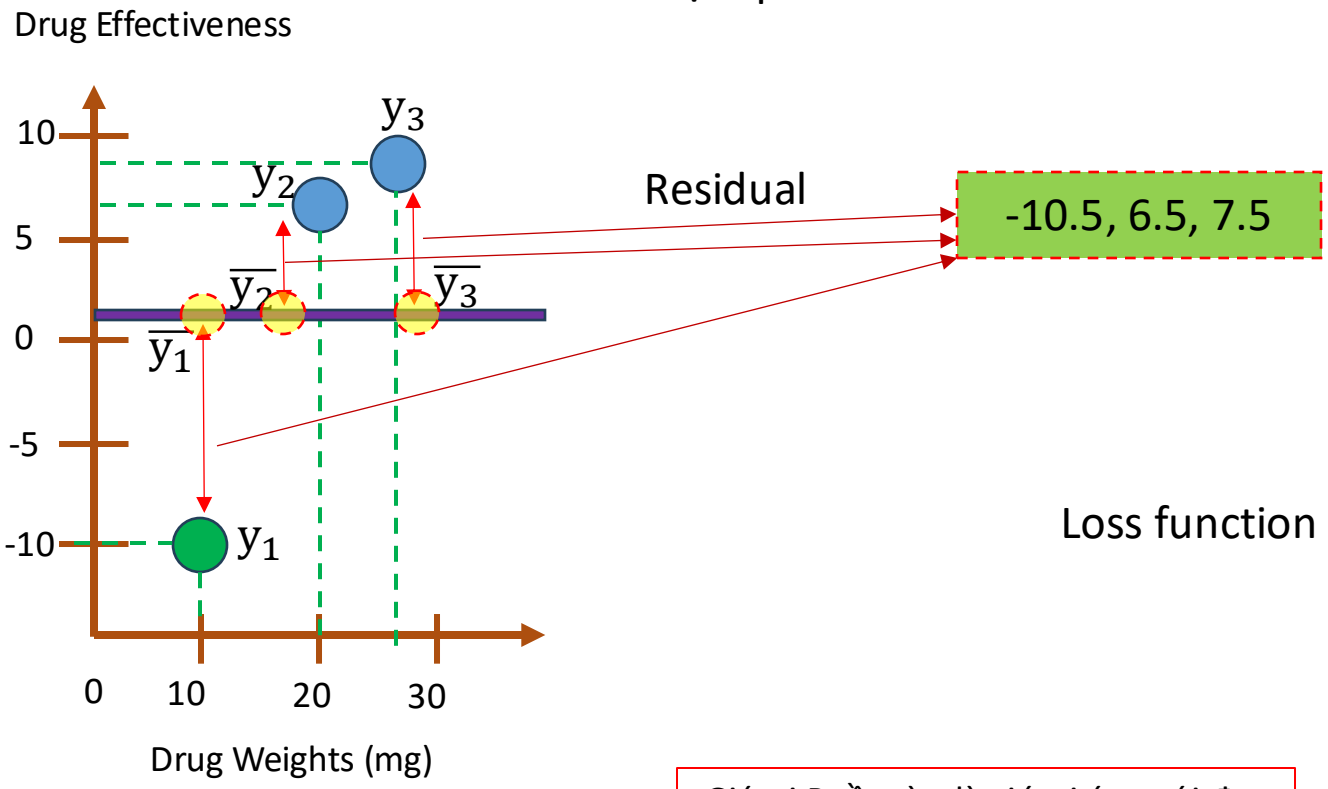
XGBoost: Behind The Scenes

Regression

Dự đoán ban đầu hiệu quả thuốc: 0.5

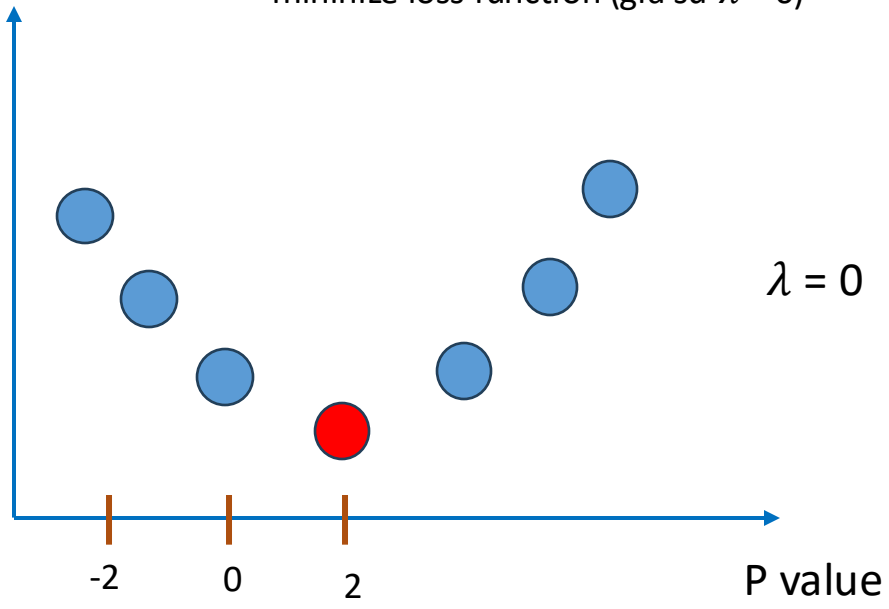


Residual: -10.5, 6.5, 7.5

Loss function

$$\sum_{i=1}^{n} \mathcal{L}\left(y_i, \overline{y_i^0} + P\right) + \frac{1}{2}\lambda P^2$$

Chúng cần tìm giá trị đầu ra của nút lá này (giá trị P) bằng cách mininize loss function (giả sử $\lambda = 0$)



Giá trị P cần tìm là giá trị ứng với đạo hàm của loss theo P bằng 0

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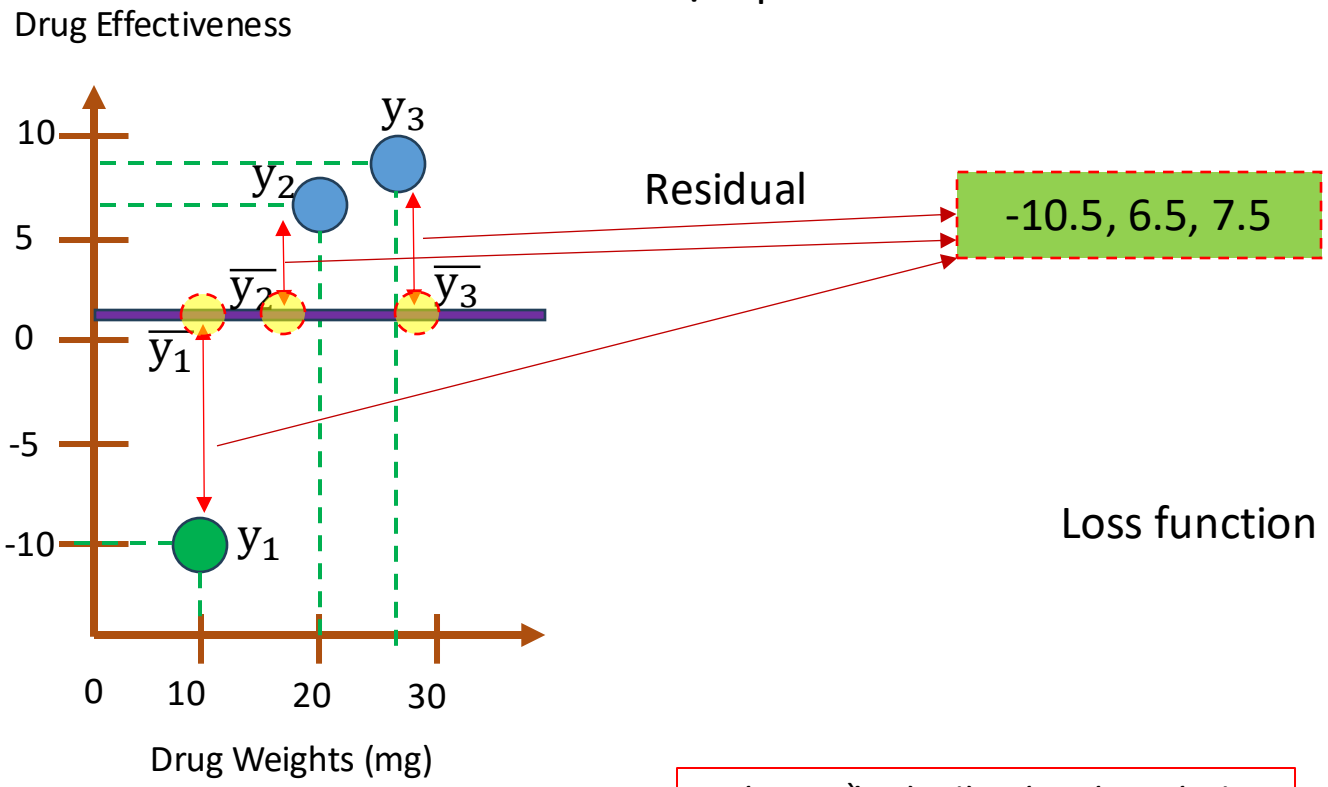
XGBoost: Behind The Scenes

Regression

Dự đoán ban đầu hiệu quả thuốc: 0.5

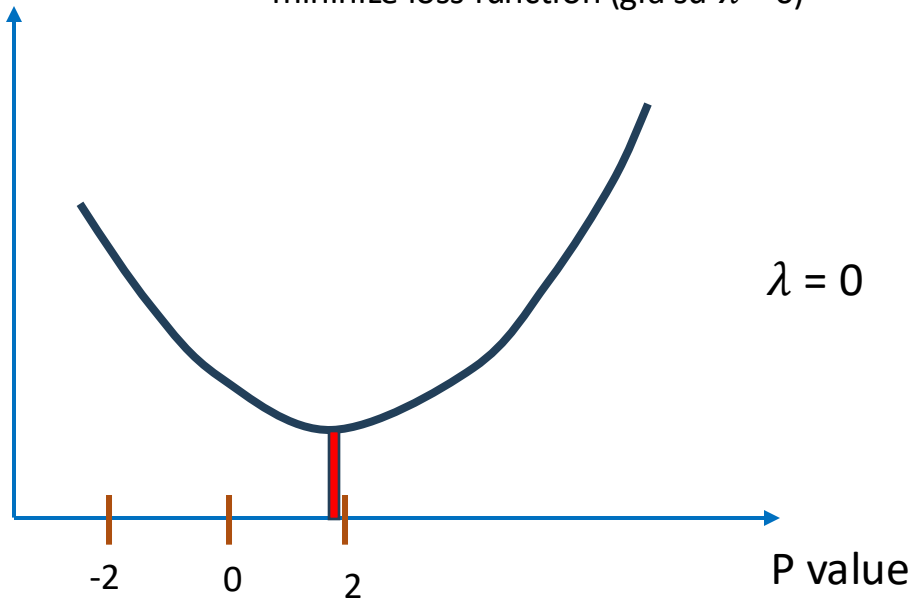


Residual: -10.5, 6.5, 7.5

Loss function

$$\sum_{i=1}^{n} \mathcal{L}\left(y_i, \overline{y_i^0} + P\right) + \frac{1}{2}\lambda P^2$$

Chúng cần tìm giá trị đầu ra của nút lá này (giá trị P) bằng cách mininize loss function (giả sử $\lambda = 0$)



Giá trị P cần tìm là giá trị ứng với đạo hàm của loss theo P bằng 0

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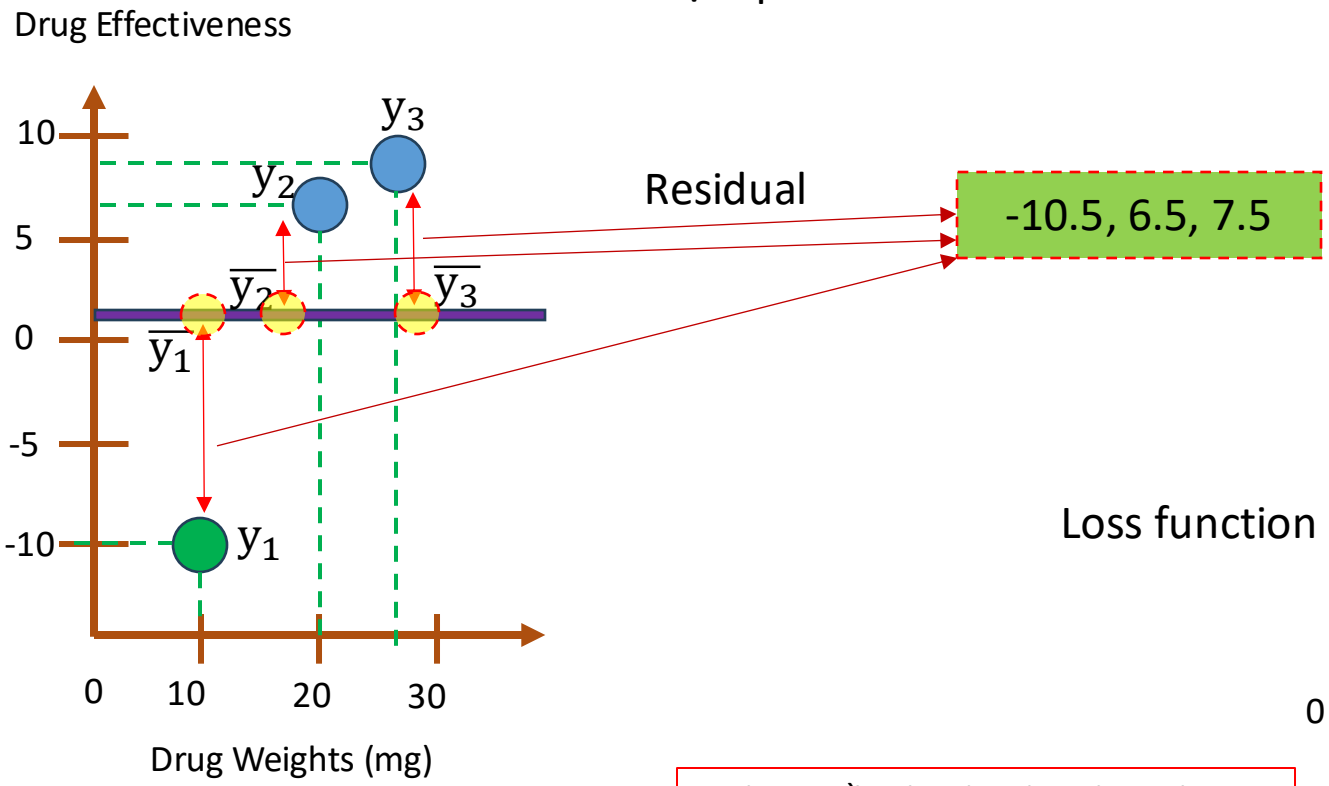
XGBoost: Behind The Scenes

Regression

Dự đoán ban đầu hiệu quả thuốc: 0.5

Whats happen if λ is very large?

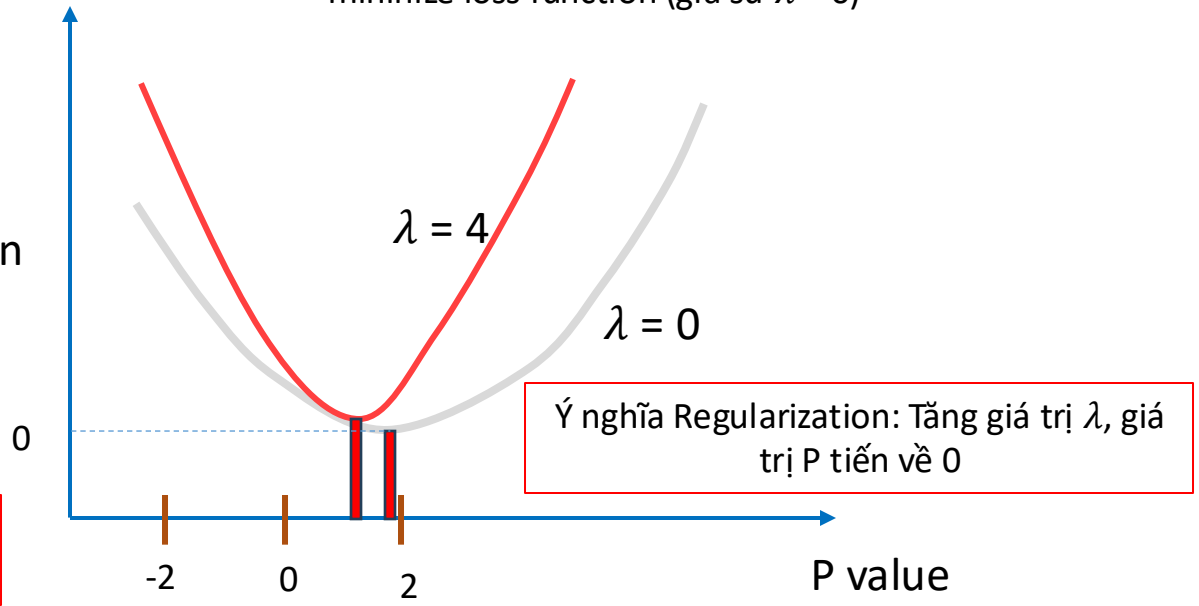


Residual: -10.5, 6.5, 7.5

$$\sum_{i=1}^{n} \mathcal{L}\left(y_i, \overline{y_i^0} + P\right) + \frac{1}{2}\lambda P^2$$

Chúng cần tìm giá trị đầu ra của nút lá này (giá trị P) bằng cách mininize loss function (giả sử $\lambda = 0$)

Loss function



Giá trị P cần tìm là giá trị ứng với đạo hàm của loss theo P bằng 0

Ý nghĩa Regularization: Tăng giá trị λ, giá trị P tiến về 0

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XGBoost: Behind The Scenes

$$\sum_{i=1}^{n} \mathcal{L}\left(y_i, \overline{y_i^0} + P\right) + \frac{1}{2}\lambda P^2$$

Rất khó để tìm optimalization, nên cũng ta sẽ sắp sỉ hàm loss bằng Second Order Tayler Appriximation

$$\mathcal{L}(y_i, \overline{y_i} + P) \approx \mathcal{L}(y_i, \overline{y_i}) + \left[\frac{d}{d\overline{y_i}}\mathcal{L}(y_i, \overline{y_i})\right] P + \frac{1}{2}\left[\frac{d}{d\overline{y_i^2}}\mathcal{L}(y_i, \overline{y_i})\right] P^2$$

$$\mathcal{L}(y_i, \overline{y_i} + P) \approx \mathcal{L}(y_i, \overline{y_i}) + gP + \frac{1}{2}hP^2$$

g (gradient) presents the first derivative of the loss function

h (hessian) presents the second derivative of the loss function

$$\mathcal{L}\left(y_1, \overline{y_1^0}\right) + g_1 P + \frac{1}{2}h_1 P^2 + \mathcal{L}\left(y_2, \overline{y_2^0}\right) + g_2 P + \frac{1}{2}h_2 P^2 + \cdots + \mathcal{L}\left(y_n, y_n^0\right) + g_n P + \frac{1}{2}h_n P^2 + \frac{1}{2}\lambda P^2$$

Tìm giá trị P cần tìm sao cho đạo hàm của loss function theo P bằng 0

$$\frac{d}{dP}\left[(g_1 + g_2 + \cdots + g_n)P + \frac{1}{2}(h_1 + h + \cdots + h_n + \lambda)P^2\right] = 0$$

XGBoost Regression: Output Value

$$\frac{d}{dP}\left[(g_1 + g_2 + \cdots + g_n)P + \frac{1}{2}(h_1 + h + \cdots + h_n + \lambda)P^2\right] = 0$$

$$(g_1 + g_2 + \cdots + g_n) + (h_1 + h + \cdots + h_n\lambda)P \quad = 0$$

1

$$g_i = \frac{d}{d\bar{y}_i}\frac{1}{2}(y_i - \bar{y}_i)^2 = -(y_i - \bar{y}_i)$$

2

$$h_i = \frac{d^2}{d\bar{y}_i^2}\frac{1}{2}(y_i - \bar{y}_i)^2 = 1$$

$$P = \frac{-(g_1+g_2+\cdots+g_n)}{h_1+h_2+\cdots+h_n+\lambda} = \frac{-(-(y_1-\bar{y_1})+-(y_2-\bar{y_2})+\cdots+-(y_n-\bar{y_n}))}{1+1+\cdots+1+\lambda} = \frac{\text{sum of residual}}{\text{numer of sum residual}+\lambda}$$

Output value af the leaf (or terminal node)

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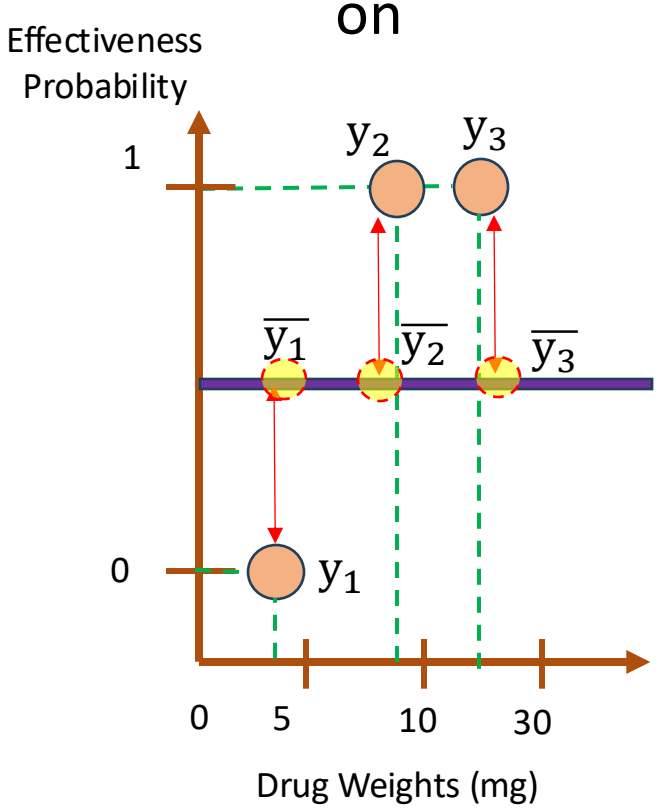
XGBoost For Classification: Output Value

Classificati
on



$$\mathcal{L}(y_i, \overline{y_i}) = -[y_i \log(\overline{y_i}) + (1 - y_i)\log(1 - \overline{y_i})]$$

Convert probability to Log(odds)

$$\mathcal{L}(y_i, \log(\text{odds})) = -y_i \log(odds) + log(1 + e^{\log(odds)})$$

$$g_i = \frac{d}{dlog(odds)} \mathcal{L}(y_i, \log(\text{odds})) = -y_i + \frac{e^{\log(odds)})}{1+e^{\log(odds)}} = -(y - \overline{y_i})$$

$$h_i = \frac{d^2}{dlog(odds)^2} \mathcal{L}(y_i, \log(\text{odds})) = \frac{e^{\log(odds)})}{1 + e^{\log(odds)}} \times \frac{1}{1 + e^{\log(odds)}} = \overline{y_i} \times (1 - \overline{y_i})$$

YOU ARE HERE

$$P = \frac{-(g_1+g_2+\cdots+g_n)}{h_1+h_2+\cdots+h_n+\lambda} = \frac{\text{sum of residual}}{\overline{y_1}\times(1-\overline{y_1})+\overline{y_2}\times(1-\overline{y_2})+\cdots+\overline{y_n}\times(1-\overline{y_n})+\lambda} = \frac{(\sum \text{Residual})}{\sum \overline{y_i}\times(1-\overline{y_i})+\lambda}$$

XGBoost: Similarity Score

1

$$\sum_{i=1}^{n} \mathcal{L}\left(y_i, \overline{y_i^0} + P\right) + \frac{1}{2}\lambda P^2$$

Rất khó để tìm optimalization, nên cũng ta sẽ sắp sỉ hàm loss bằng Second Order Tayler Appriximation

$$\mathcal{L}(y_i, \overline{y_i} + P) \approx \mathcal{L}(y_i, \overline{y_i}) + \left[\frac{d}{d\overline{y_i}}\mathcal{L}(y_i, \overline{y_i})\right] P + \frac{1}{2}\left[\frac{d}{d\overline{y_i^2}}\mathcal{L}(y_i, \overline{y_i})\right] P^2$$

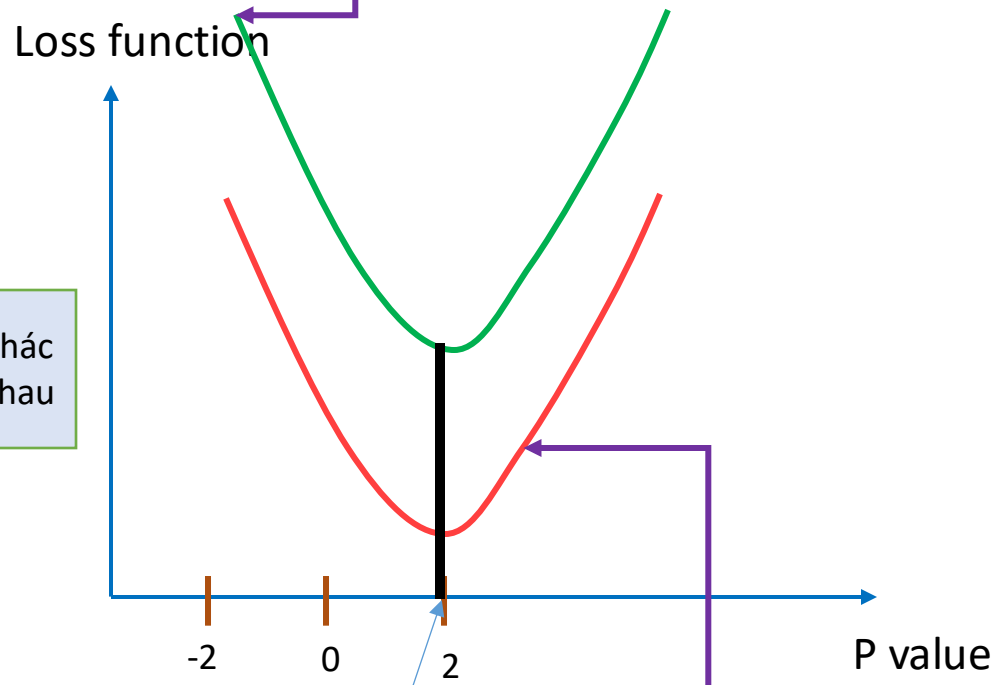
$$\mathcal{L}(y_i, \overline{y_i} + P) \approx \mathcal{L}(y_i, \overline{y_i}) + gP + \frac{1}{2}hP^2$$

2

$$(g_1 + g_2 + \cdots + g_n)P + \frac{1}{2}(h_1 + h + \cdots + h_n + \lambda)P^2$$

Cả (1) và (2) đều có cùng optimization point P

Khác nhau



$$P = \frac{-(g_1 + g_2 + \cdots + g_n)}{h_1 + h_2 + \cdots + h_n + \lambda}$$

XGBoost: Similarity Score

1

$$\sum_{i=1}^{n} \mathcal{L}(y_i, \overline{y_i^0} + P) + \frac{1}{2}\lambda P^2$$

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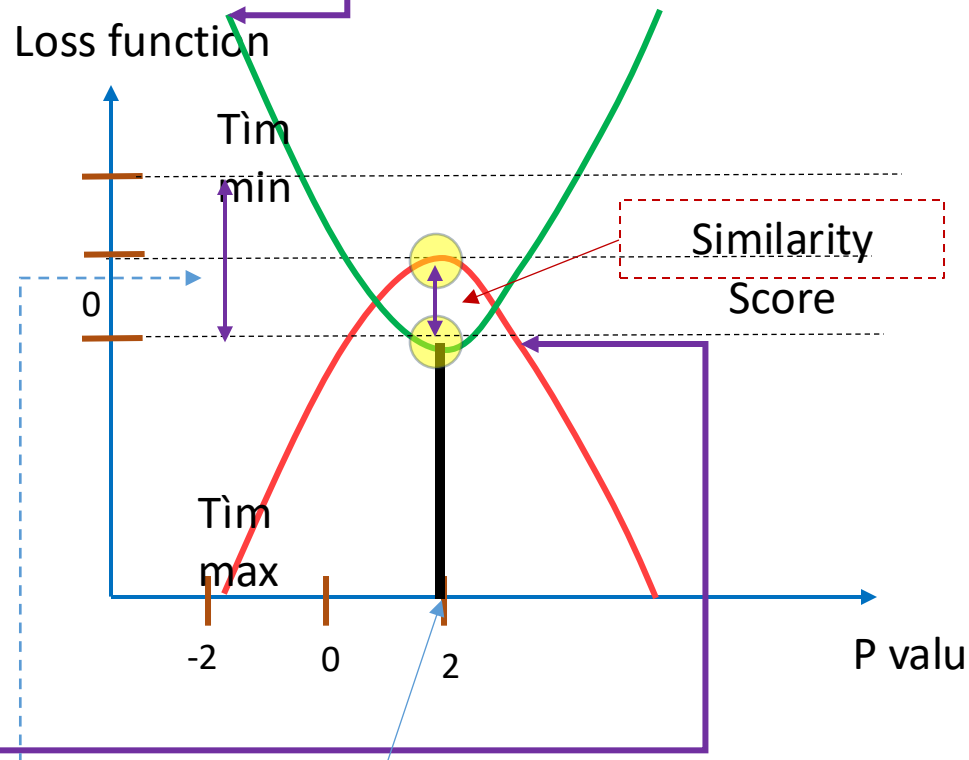
$$\mathcal{L}(y_i, \overline{y_i} + P) \approx \mathcal{L}(y_i, \overline{y_i}) + \left[\frac{d}{d\overline{y_i}}\mathcal{L}(y_i, \overline{y_i})\right] P + \frac{1}{2}\left[\frac{d}{d\overline{y_i^2}}\mathcal{L}(y_i, \overline{y_i})\right] P^2$$

$$\mathcal{L}(y_i, \overline{y_i} + P) \approx \mathcal{L}(y_i, \overline{y_i}) + gP + \frac{1}{2}hP^2$$

2

$$\text{-1 X } (g_1 + g_2 + \cdots + g_n)P + \text{-1 X } \frac{1}{2}(h_1 + h + \cdots + h_n + \lambda)P^2$$

Cả (1) và (2) đều có cùng optimization point P



Implementation Similarity Score

$$P = \frac{-(g_1 + g_2 + \cdots + g_n)}{h_1 + h_2 + \cdots + h_n + \lambda}$$

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XGBoost: Similarity Score

1

$$\sum_{i=1}^{n} \mathcal{L}\left(y_i, \overline{y_i^0} + P\right) + \frac{1}{2}\lambda P^2$$

Rất khó để tìm optimalization, nên cũng ta sẽ sắp sỉ hàm loss bằng Second Order Tayler Appriximation

$$\mathcal{L}(y_i, \overline{y_i} + P) \approx \mathcal{L}(y_i, \overline{y_i}) + \left[\frac{d}{d\overline{y_i}}\mathcal{L}(y_i, \overline{y_i})\right] P + \frac{1}{2}\left[\frac{d}{d\overline{y_i^2}}\mathcal{L}(y_i, \overline{y_i})\right] P^2$$

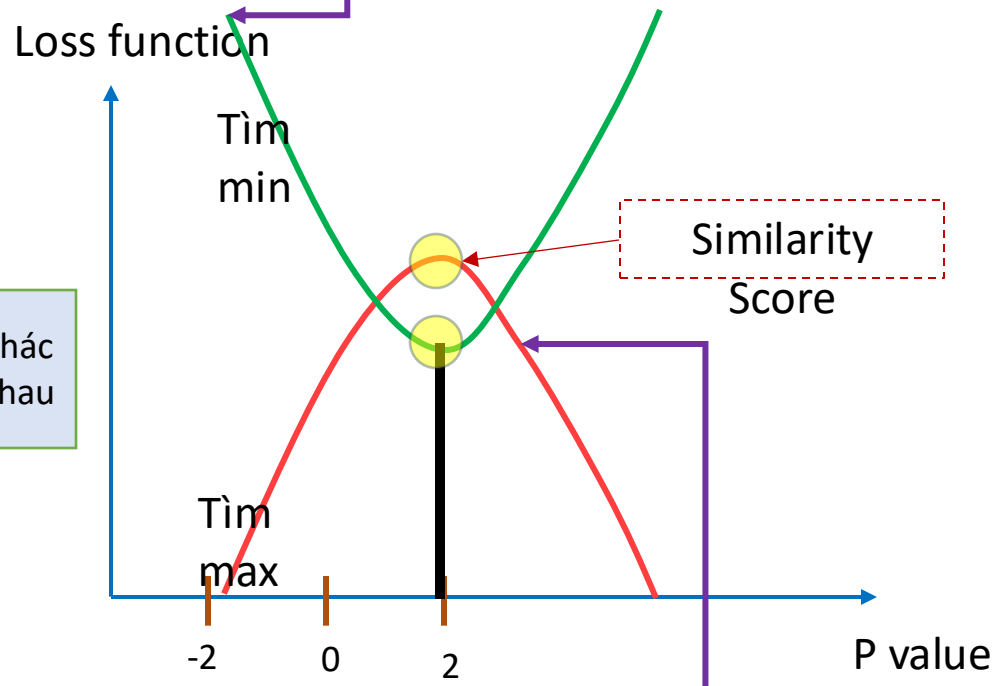
$$\mathcal{L}(y_i, \overline{y_i} + P) \approx \mathcal{L}(y_i, \overline{y_i}) + gP + \frac{1}{2}hP^2$$

2

$$\text{-1 X } (g_1 + g_2 + \cdots + g_n)P + \text{-1 X } \frac{1}{2}(h_1 + h + \cdots + h_n + \lambda)P^2$$

Khác nhau

Cả (1) và (2) đều có cùng optimization point P



$$\text{Similarity Score} = \frac{1}{2}\frac{(g_1 + g_2 + \cdots + g_n)^2}{(h_1 + h_2 + \cdots + h_n + \lambda)}$$

$$P = \frac{-(g_1 + g_2 + \cdots + g_n)}{h_1 + h_2 + \cdots + h_n + \lambda}$$

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XGBoost Regression: Similarity Score

$$P = \frac{-(g_1 + g_2 + \cdots + g_n)}{h_1 + h_2 + \cdots + h_n + \lambda}$$

Cả (1) và (2) đều có cùng optimization point P

1

$$\sum_{i=1}^{n} \mathcal{L}(y_i, \overline{y_i^0} + P) + \frac{1}{2}\lambda P^2$$

Rất khó để tìm optimalization, nên cũng ta sẽ sắp sỉ hàm loss bằng Second Order Tayler Appriximation

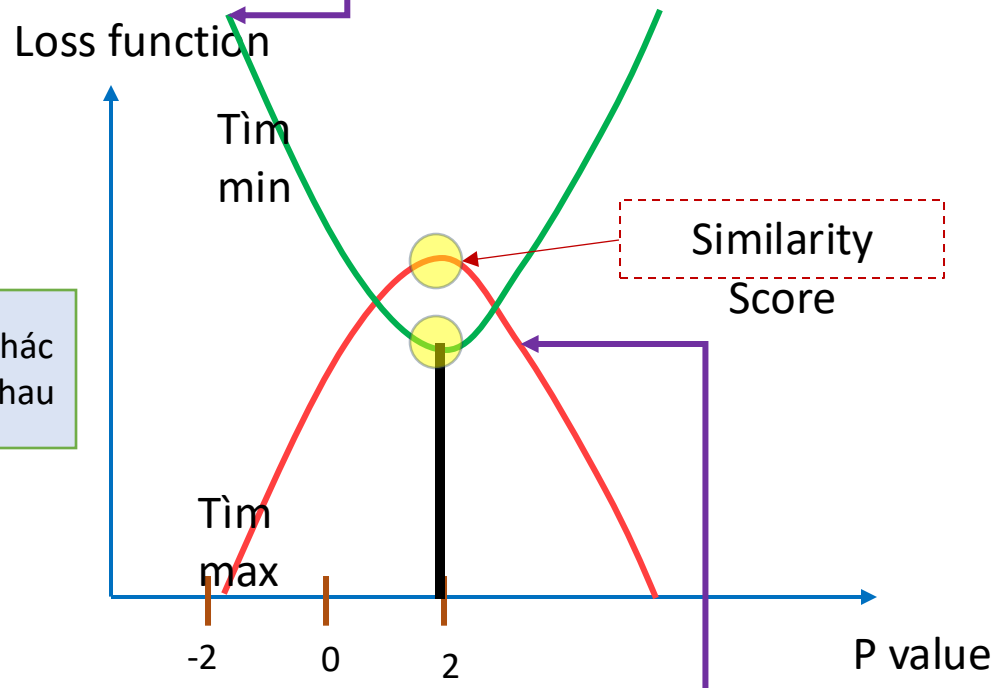
$$\mathcal{L}(y_i, \overline{y_i} + P) \approx \mathcal{L}(y_i, \overline{y_i}) + \left[\frac{d}{d\overline{y_i}}\mathcal{L}(y_i, \overline{y_i})\right] P + \frac{1}{2}\left[\frac{d}{d\overline{y_i^2}}\mathcal{L}(y_i, \overline{y_i})\right] P^2$$

$$\mathcal{L}(y_i, \overline{y_i} + P) \approx \mathcal{L}(y_i, \overline{y_i}) + gP + \frac{1}{2}hP^2$$

2

$$\text{-1 X } (g_1 + g_2 + \cdots + g_n)P + \text{-1 X } \frac{1}{2}(h_1 + h + \cdots + h_n + \lambda)P^2$$

Khác nhau



$$h_i = \frac{d}{d\overline{y_i^2}}\frac{1}{2}(y_i - \overline{y_i})^2 = 1$$

$$g_i = \frac{d}{d\overline{y_i}}\frac{1}{2}(y_i - \overline{y_i})^2 = -(y_i - \overline{y_i})$$

$$\text{Similarity Score} = \frac{1}{2}\frac{(g_1 + g_2 + \cdots + g_n)^2}{(h_1 + h_2 + \cdots + h_n + \lambda)}$$

$$\text{Similarity Score} = \frac{(\sum \text{Residual})^2}{\text{Number of Residual} + \lambda}$$

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XGBoost Classification: Similarity Score

1

$$\sum_{i=1}^{n} \mathcal{L}\left(y_i, \overline{y_i^0} + P\right) + \frac{1}{2}\lambda P^2$$

Rất khó để tìm optimalization, nên cũng ta sẽ sắp sỉ hàm loss bằng Second Order Tayler Appriximation

$$\mathcal{L}(y_i, \bar{y}_i + P) \approx \mathcal{L}(y_i, \bar{y}_i) + \left[\frac{d}{d\bar{y}_i}\mathcal{L}(y_i, \bar{y}_i)\right] P + \frac{1}{2}\left[\frac{d}{d\bar{y}_i^2}\mathcal{L}(y_i, \bar{y}_i)\right] P^2$$

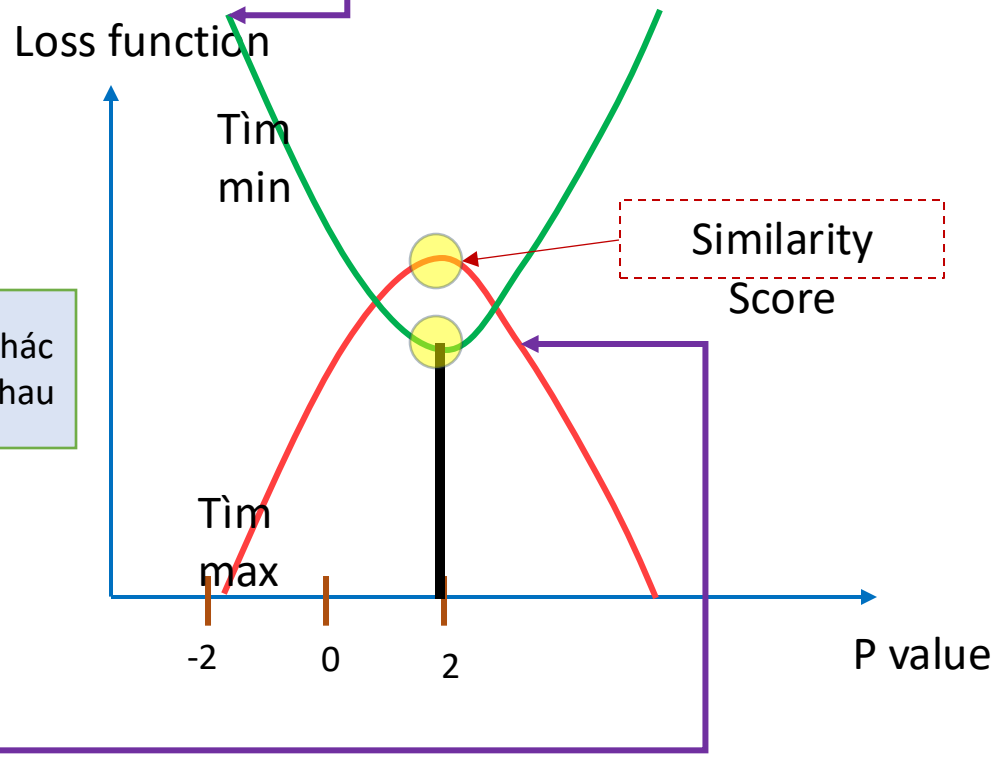
$$\mathcal{L}\left(y_i, \overline{y_i^0} + P\right) \approx \mathcal{L}(y_i, \bar{y}_i) + gP + \frac{1}{2}hP^2$$

2

$$\text{-1 X } (g_1 + g_2 + \cdots + g_n)P + \text{-1 X } \frac{1}{2}(h_1 + h + \cdots + h_n + \lambda)P^2$$

Khác nhau

Cả (1) và (2) đều có cùng optimization point P



$g_i = -(y_i - \bar{y}_i)$

$h_i = \bar{y}_i \times (1 - \bar{y}_i)$

$$\text{Similarity Score} = \frac{1}{2}\frac{(g_1 + g_2 + \cdots + g_n)^2}{(h_1 + h_2 + \cdots + h_n + \lambda)}$$

$$\text{Similarity Score} = \frac{(\sum \text{Residual})^2}{\sum \bar{y}_i \times (1 - \bar{y}_i) + \lambda}$$

Outline

- Regularization
- Regression XGBoost
- Classification XGBoost
- XGBoost: Clearly Explain
- Time Series Example
- Summary

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Time Series Forecasting

We will focus on the energy consumption problem, where given a sufficiently large dataset of the daily energy consumption of different households in a city, we are tasked to predict as accurately as possible the future energy demands.

london_energy

LCLid	Date	KWH
MAC000002	2012-10-12	7.098
MAC000002	2012-10-13	11.087
MAC000002	2012-10-14	13.223
MAC000002	2012-10-15	10.257
MAC000002	2012-10-16	9.769
MAC000002	2012-10-17	10.885
MAC000002	2012-10-18	10.751
MAC000002	2012-10-19	8.431
MAC000002	2012-10-20	17.578
MAC000002	2012-10-21	24.49
MAC000002	2012-10-22	18.885
MAC000002	2012-10-23	10.485
MAC000002	2012-10-24	15.537

Preprocessing

```
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_csv("/content/drive/MyDrive/AIO2024/london_energy.csv")
print(df.isna().sum())
df.head()
```

```
LCLid    0
Date     0
KWH      0
dtype: int64
```

	LCLid	Date	KWH
0	MAC000002	2012-10-12	7.098
1	MAC000002	2012-10-13	11.087
2	MAC000002	2012-10-14	13.223
3	MAC000002	2012-10-15	10.257
4	MAC000002	2012-10-16	9.769

Time Series Forecasting

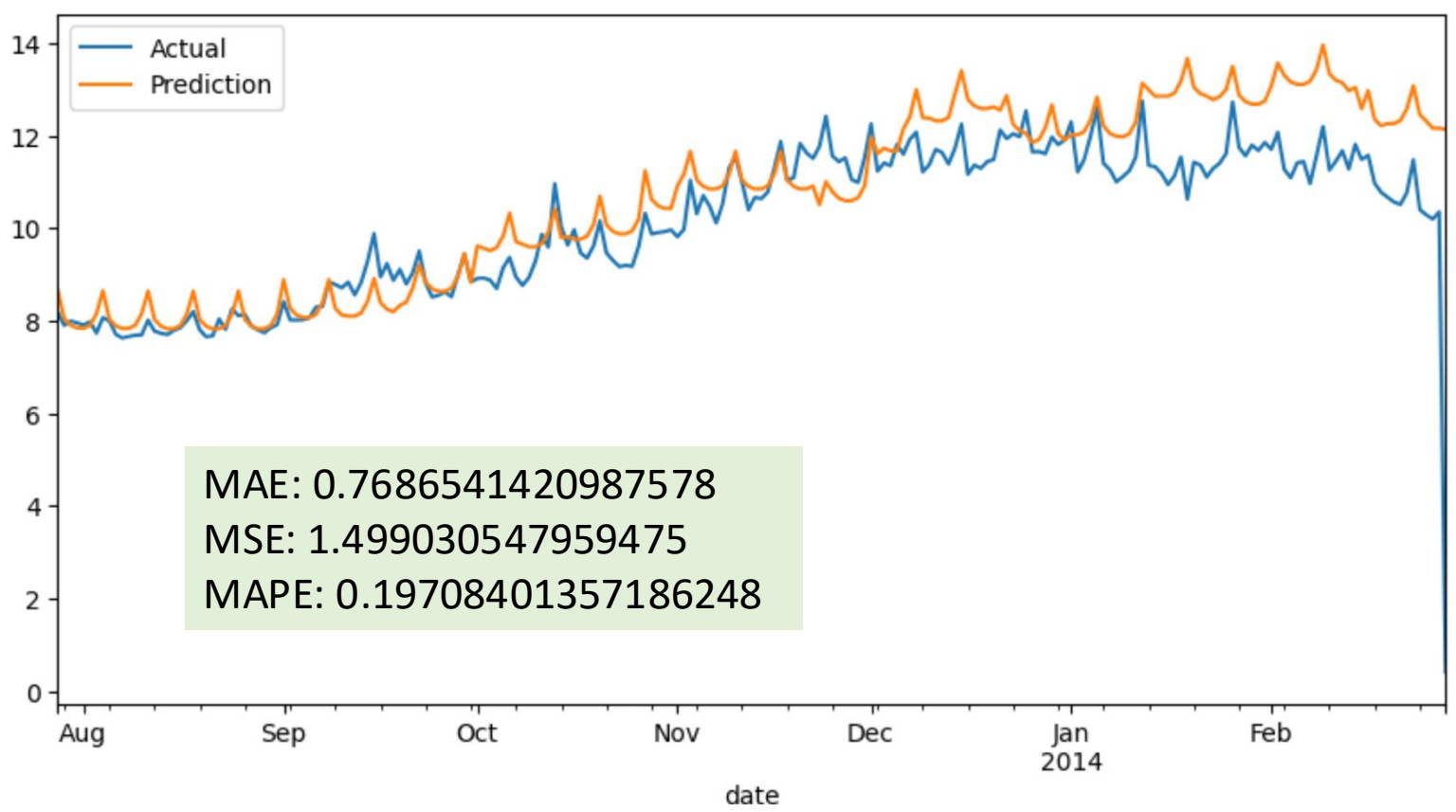
XGBoost For Training

```
from xgboost import XGBRegressor
import lightgbm as lgb
from sklearn.model_selection import TimeSeriesSplit, GridSearchCV

# XGBoost
cv_split = TimeSeriesSplit(n_splits=4, test_size=100)
model = XGBRegressor()
parameters = {
    "max_depth": [3, 4, 5],
    "learning_rate": [0.01, 0.05],
    "n_estimators": [100, 300],
    "colsample_bytree": [0.3]
}


grid_search = GridSearchCV(estimator=model, cv=cv_split, param_grid=parameters)
grid_search.fit(X_train, y_train)
```

Time Series Forecasting

XGBoost For Predicting

```
# Evaluating GridSearch results
prediction = grid_search.predict(X_test)
plot_predictions(testing_dates, y_test, prediction)
evaluate_model(y_test, prediction)
```



MAE: 0.7686541420987578
MSE: 1.499030547959475
MAPE: 0.19708401357186248

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Time Series Forecasting

The model performs relatively well, but is there a way to improve it even further?
The answer is yes

Metric	XGBoost	GDBoost
MAE	0.768	0.753
MSE	1.499	1.344
MAPE	0.197	0.195

Enhance our dataset with weather data from the London Weather Dataset

london_weather

date	cloud_cover	sunshine	global_radiation	max_temp	mean_temp	min_temp	precipitation	pressure	snow_depth
19790101	2.0	7.0	52.0	2.3	-4.1	-7.5	0.4	101900.0	9.0
19790102	6.0	1.7	27.0	1.6	-2.6	-7.5	0.0	102530.0	8.0
19790103	5.0	0.0	13.0	1.3	-2.8	-7.2	0.0	102050.0	4.0
19790104	8.0	0.0	13.0	-0.3	-2.6	-6.5	0.0	100840.0	2.0
19790105	6.0	2.0	29.0	5.6	-0.8	-1.4	0.0	102250.0	1.0
19790106	5.0	3.8	39.0	8.3	-0.5	-6.6	0.7	102780.0	1.0
19790107	8.0	0.0	13.0	8.5	1.5	-5.3	5.2	102520.0	0.0
19790108	8.0	0.1	15.0	5.8	6.9	5.3	0.8	101870.0	0.0
19790109	4.0	5.8	50.0	5.2	3.7	1.6	7.2	101170.0	0.0
19790110	7.0	1.9	30.0	4.9	3.3	1.4	2.1	98700.0	0.0
19790111	1.0	6.8	55.0	2.9	2.6	0.3	2.3	98960.0	0.0
19790112	3.0	6.4	54.0	2.0	0.4	-2.0	0.0	100650.0	1.0

Time Series Forecasting

Data Analysis: Filling missing value

```
[22] df_weather = pd.read_csv("/content/drive/MyDrive/AIO2024/london_weather.csv")
     print(df_weather.isna().sum())
     df_weather.head()
```

```
date                0
cloud_cover        19
sunshine            0
global_radiation   19
max_temp            6
mean_temp          36
min_temp            2
precipitation       6
pressure            4
snow_depth       1441
dtype: int64
```

```
# Parsing dates
df_weather["date"] = pd.to_datetime(df_weather["date"], format="%Y%m%d")

# Filling missing values through interpolation
df_weather = df_weather.interpolate(method="ffill")

# Enhancing consumption dataset with weather information
df_avg_consumption = df_avg_consumption.merge(df_weather, how="inner", on="date")
df_avg_consumption.head()
```

Time Series Forecasting

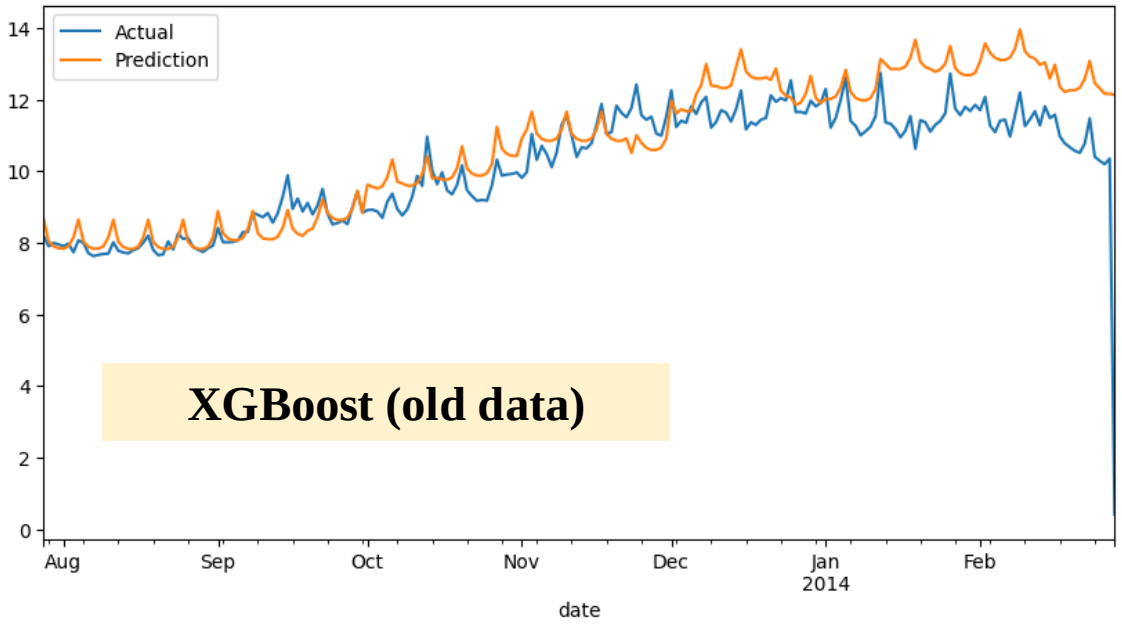
Prepare New Dataset

```
# Dropping unnecessary `date` column
training_data = training_data.drop(columns=["date"])
testing_dates = testing_data["date"]
testing_data = testing_data.drop(columns=["date"])

X_train = training_data[["day_of_week", "day_of_year", "month", "quarter", "year",\
                         "cloud_cover", "sunshine", "global_radiation", "max_temp",\
                         "mean_temp", "min_temp", "precipitation", "pressure",\
                         "snow_depth"]]
y_train = training_data["consumption"]


X_test = testing_data[["day_of_week", "day_of_year", "month", "quarter", "year",\
                         "cloud_cover", "sunshine", "global_radiation", "max_temp",\
                         "mean_temp", "min_temp", "precipitation", "pressure",\
                         "snow_depth"]]
y_test = testing_data["consumption"]
```

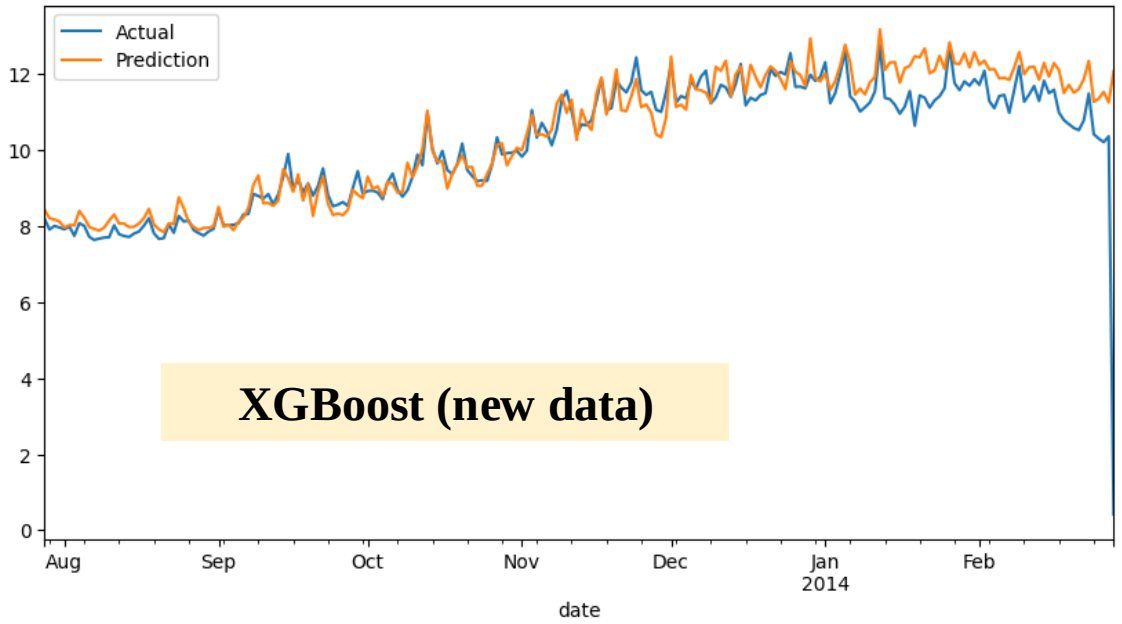
Time Series Forecasting

Performance Evaluation

Metric	XGBoost (old data)	XGBoost (new data)
MAE	0.768	0.423
MSE	1.499	0.864
MAPE	0.197	0.164



XGBoost (old data)



XGBoost (new data)

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